Time Series Analysis

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Contents

Chapter 1

Introduction

1.1 Practical

Practical information

- Problem sheets, solutions, slides and lecture notes will be uploaded on Moodle.
- There is an exam in the summer for the course.

Course topics

- ARIMA models
- Frequency domain analysis
- Forecasting
- ARCH models
- Multivariate Time Series

Useful resources

- Brockwell, P. J. and Davis, R. A. (2002). *Introduction to time series and forecasting.* Springer
- Shumway, R. H. and Stoffer, D. S. (2000). *Time series analysis and its applications*, volume 3. Springer
- Tsay, R. S. (2005). *Analysis of financial time series.* John Wiley & Sons
- Percival, D. B. and Walden, A. T. (1993). *Spectral analysis for physical applications.* Cambridge University Press

1.2 Motivation

Dependence

In basic data analysis we assume that observations are independent or even independent and identically distributed,

$$X_1, \ldots, X_n \overset{\text{ind}}{\sim} F_1, \ldots, F_n, \quad X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}\left(\mu, \sigma^2\right).$$

- Time series is the study of observations that arise in some order (usually time) and so are dependent.
- There are many more ways to be dependent than to be independent, and almost all data are collected in time order, so time series arise in a vast range of disciplines.
- Many of these disciplines have developed special techniques to deal with their specific problems, and we will only scratch the surface of them in this course, by surveying some main ideas.

Unobserved Components Models

- In econometrics for example, the notion that a time series is an aggregation of different phenomenological behaviours is common, see Figure 1.1.

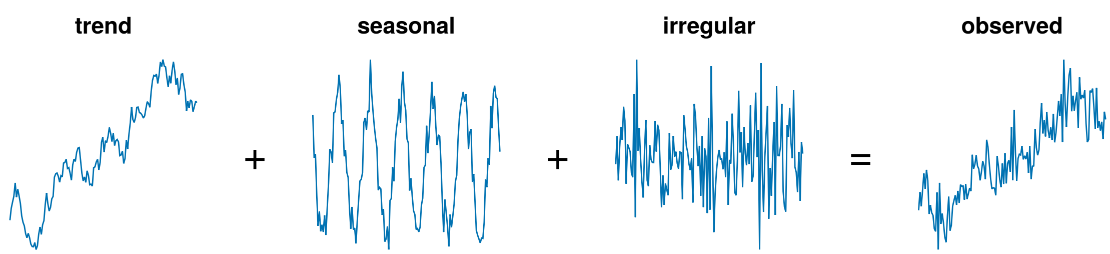


Figure 1.1: Decomposition of temporal series

1.3 Basic Notation

What is a time series?

- Informally, a time series X_t is just data recorded over time.

- We shall use the word 'time series' to mean both the data, and the process from which the data is a realisation.
- More formally, we think of X_t as a stochastic process, i.e. as a family of random variables $\{X_t : t \in \mathcal{T}\}$ defined on a probability space $(\Omega, \mathscr{F}, P)$.
- In time series analysis the index (or parameter) set $\mathcal{T}$ is a set of time points, very often $\mathbb{R}$ or $\Delta\mathbb{Z}$ (or a subset of them).
- Here $\Delta \in \mathbb{R}$ is the time step between observations.

What is a time series practically?

- Whilst it is mathematically useful to think of processes with infinite index sets, in practice we can only make finitely many observations.
- Therefore, the set of observations X_t we actually record are in some set of time points $\mathcal{S} \subset \mathcal{T}$.
- Normally $\mathcal{S}$ is a discrete set (often with a regular sampling interval) $\{\Delta, \ldots, n\Delta\}$.
- The time series may also be recorded over an interval $\mathcal{S} = [0, T_0]$ (though it obviously cannot be stored digitally in this way directly).

Kolmogorov's theorem

Let $\mathscr{F} := \{\mathbf{t} = (t_1, \ldots, t_n)^T \in \mathcal{T}^n : \ t_1 < \cdots < t_n, \ n = 1, 2, \ldots\}$. Then the (finite-dimensional) distribution functions of $\{X_t\}_{t \in \mathcal{T}}$ are the functions $\{F_{\mathbf{t}}(\cdot), \ \mathbf{t} \in \mathcal{T}^n\}$ defined by

$$F_{\mathbf{t}}(\mathbf{x}) = \Pr\left(X_{t_1} \le x_1, \ldots, X_{t_n} \le x_n\right), \quad \mathbf{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n.$$

Theorem 1.1 (Kolmogorov's theorem)**.** *The probability distribution functions $\{F_{\mathbf{t}}(\cdot), \ \mathbf{t} \in \mathscr{F}\}$ are the distribution functions of a given stochastic process if and only if for any natural number n and $\mathbf{t} \in \mathscr{F}$ and $1 \le i \le n$ we have*

$$\lim_{x_i \to \infty} F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}(i)}(\mathbf{x}(i)),$$

where we have defined $\mathbf{t}(i)$ and $\mathbf{x}(i)$ as the $(n-1)$-component vectors obtained by deleting the ith component of $\mathbf{t}$ and $\mathbf{x}$ respectively.

1.4 Stationarity

Weak stationarity

Definition 1.1 ((Weak) Stationarity)**.** The time series $\{X_t\}$ is said to be second-order/weak or covariance stationary if for all $n \ge 1$ for any $t_1, \ldots, t_n \in \mathcal{T}$ and for all τ such that $t_1 + \tau, \ldots, t_n + \tau \in \mathcal{T}$ all the joint moments of order 1 and 2 of $X_{t_1}, \ldots, X_{t_n}$ exist, are all finite and equal to the corresponding joint moments of $X_{t_1+\tau}, \ldots, X_{t_n+\tau}$.

In fact this corresponds to: $\forall t, s \in \mathcal{T}$, $\forall \tau$ such that $t + \tau, s + \tau \in \mathcal{T}$

1. $\mathbb{E}\left[X_t\right] = \mu$,

2. $\mathrm{Var}(X_t) = \sigma^2 < \infty$,
3. $\mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[X_s X_{s+\tau}]$.

One may deduce from this that $\mathbb{E}[X_t X_{t+\tau}]$ can be written as a function of τ only.

Strong stationarity

We can go beyond the first two moments and define strong stationarity.

Definition 1.2 (Strong Stationarity)**.** The time series $\{X_t\}$ is said to be completely/strong or strictly stationary if for all $n \geq 1$ for any $t_1, \dots, t_n \in \mathcal{T}$ and for all τ such that $t_1 + \tau, \dots, t_n + \tau \in \mathcal{T}$ the joint distribution of $X_{t_1}, \dots, X_{t_n}$ is the same as $X_{t_1+\tau}, \dots, X_{t_n+\tau}$.

In general:

- second order stationary $\nRightarrow$ strictly stationary,
- strict stationarity $\nRightarrow$ 2nd order stationarity.

For example iid student t with non-finite variance.

Gaussian processes

Definition 1.3 (Gaussian processes)**.** A stochastic process is called a Gaussian process if, for all $n \geq 1$ and for any $t_1, \dots t_n \in \mathcal{T}$, the joint distribution of $X_{t_1}, \dots, X_{t_n}$ is multivariate normal.

- In other words, the distribution of the process at any finite collection of time points is multivariate normal.
- Gaussian processes are then completely characterised by their first two moments (which are finite).
- Thus for Gaussian processes, strong stationarity and weak stationarity are the same!

1.5 Dependence

Measures of dependence

- As we discussed earlier, often in statistics one works with collections of independent observations.
- Here, we do not have independent data.
- To understand dependence better for finite collections of random variables, we often compute their covariance matrix.
- For a time series $\{X_t\}$ the extension of the covariance matrix corresponds to the autocovariance function.
- We usually only have one time series, but if the process is stationary we might hope to still be able to do some kind of averaging.
- For notational simplicity, assume that $\Delta = 1$ for the remainder of this lecture.

Autocovariance Sequence (ACVS)

Definition 1.4 (ACVS)**.** For a discrete time second-order stationary process $\{X_t\}$ we define the autocovariance sequence (ACVS) by

$$\gamma_\tau = \mathrm{Cov}\left(X_0, X_\tau\right), \tag{1.1}$$

where $\tau \in \mathbb{Z}$ is the lag.

- By stationarity, we have $\gamma_\tau = \mathrm{Cov}\left(X_t, X_{t+\tau}\right)$ for any $t \in \mathbb{Z}$.
- Clearly $\gamma_0 = \mathrm{Var}\left(X_t\right)$ and $\gamma_{-\tau} = \gamma_\tau$.
- But how do we estimate this from a single realisation?

Estimating the ACVS: moment matching

- A natural estimator for the acvs is based on matching moments, i.e.

$$\tilde{\gamma}_\tau = \begin{cases} \frac{1}{n-|\tau|}\sum_{t=1}^{n-|\tau|}(X_t - \bar{X})(X_{t+|\tau|} - \bar{X}) & \text{if } |\tau| \leq n-1, \\ 0 & \text{otherwise.} \end{cases} \tag{1.2}$$

- If we knew the true mean μ, and replace the sample mean by this we obtain for $|\tau| \leq n-1$

$$\mathbb{E}\left[\tilde{\gamma}_\tau\right] = \frac{1}{n-|\tau|}\sum_{t=1}^{n-|\tau|}\mathbb{E}\left[(X_t - \mu)(X_{t+|\tau|} - \mu)\right] = \gamma_\tau. \tag{1.3}$$

- However, $\tilde{\gamma}_\tau$ is not always positive semi-definite, motivating a slightly different estimator.[1]

Estimating the ACVS: a positive semi-definite estimate

Definition 1.5 (The sample autocovariance)**.** Given observations of a time series $X_1, \ldots, X_n$. We define the sample autocovariance to be

$$\hat{\gamma}_\tau = \begin{cases} \frac{1}{n}\sum_{t=1}^{n-|\tau|}(X_t - \bar{X})(X_{t+|\tau|} - \bar{X}) & \text{if } |\tau| \leq n-1, \\ 0 & \text{otherwise,} \end{cases} \tag{1.4}$$

where $\bar{X}$ is the sample mean.

- Again in the known mean case

$$\mathbb{E}\left[\hat{\gamma}_\tau\right] = \frac{n-|\tau|}{n}\gamma_\tau, \tag{1.5}$$

  for $|\tau| \leq n-1$, which is biased.
- But, $\hat{\gamma}_\tau$ is positive semi-definite.

[1] Consider the case $X_1 = -1, X_2 = 0, X_3 = -1$, and look at the eigenvalues of the covariance matrix.

Autocorrelation sequence (ACF)

Definition 1.6 (ACF)**.** The autocorrelation sequence, usually called autocorrelation function, (ACF) is defined as

$$\rho_\tau = \mathrm{Corr}\left(X_t, X_{t+\tau}\right), \tag{1.6}$$

for $\tau \in \mathbb{Z}$.

- We have $\rho_\tau = \gamma_\tau/\gamma_0$.
- Even though it is a sequence, it is still usually referred to as the autocorrelation function.
- Estimation uses the plug in estimator

$$\hat{\rho}_\tau = \hat{\gamma}_\tau/\hat{\gamma}_0. \tag{1.7}$$

Example time series vs sample ACF

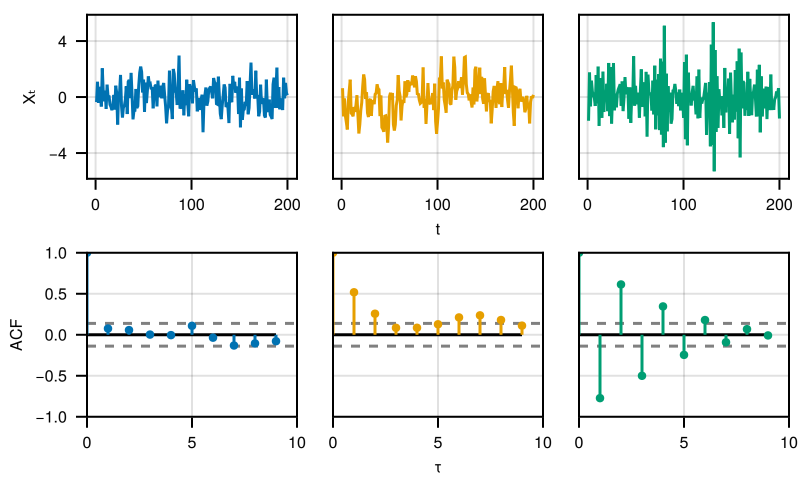


Figure 1.2: Simulated time series and their sample ACF.

ACVS is positive semi-definite

Theorem 1.2. *The sequence* $\{\gamma_\tau\}$ *is positive semi-definite, that is for all* $n \geq 1$ *for any* $t_1, \ldots, t_n \in \mathbb{Z}$ *and for any* $a_1, \ldots, a_n \in \mathbb{R}$ *we have*

$$\sum_{j=1}^{n} \sum_{k=1}^{n} a_j a_k \gamma_{j-k} \geq 0. \tag{1.8}$$

Proof. Consider the random variable $W = \sum_{j=1}^{n} a_j X_j$. Now we have

$$0 \leq \mathrm{Var}\,(W) = \sum_{j=1}^{n} \sum_{k=1}^{n} a_j a_k \,\mathrm{Cov}\,(X_j, X_k) = \sum_{j=1}^{n} \sum_{k=1}^{n} a_j a_k \gamma_{j-k}$$

as required. □

The impact of dependence on estimation

- In estimation we will only have a single realisation available.
- We will use a time-average to give time replication, as we saw in the estimation of the autocovariance.
- However, we have yet to examine the effect of the dependence on our estimates.
- Assume that the autocovariance satisfies $\sum\limits_{\tau=-\infty}^{\infty} |\gamma_\tau| < \infty$.
- Define
$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$
- What are the properties of this estimator?
$$\mathbb{E}\left[\bar{X}\right] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[X_i\right] = \mu.$$
- So $\bar{X}$ is unbiased as an estimator of μ.
- What about the variance? We say that $\bar{X}$ will converge to μ in mean square if
$$\lim_{n\to\infty} \mathrm{Var}\left(\bar{X}\right) = 0.$$
- How do we figure this out?
- We calculate
$$\begin{aligned}\mathrm{Var}\left(\bar{X}\right) &= \mathbb{E}\left[(\bar{X} - \mu)^2\right] \\ &= \mathbb{E}\left[\left(\frac{1}{n} \sum_{i=1}^{n} X_i - \mu\right)^2\right] \\ &= \frac{1}{n^2} \sum_{i,j=1}^{n} \mathbb{E}\left[(X_i - \mu)(X_j - \mu)\right].\end{aligned}$$

  How can we simply this?

- We need to acknowledge the correlation in the process. If the covariance was σ^2 everywhere then we could not have mean square convergence.
- We find that

$$\begin{aligned}
\operatorname{Var}\left(\bar{X}\right) &= \frac{1}{n^2}\sum_{i,j=1}^{n}\mathbb{E}\left[(X_i-\mu)(X_j-\mu)\right] \\
&= \frac{1}{n^2}\sum_{i,j=1}^{n}\gamma_{j-i} \\
&= \frac{1}{n^2}\sum_{\tau=-(n-1)}^{n-1}(n-|\tau|)\gamma_\tau
\end{aligned}$$

- We now need the Césaro summability theorem which says that if $\sum_{\tau=-(n-1)}^{n-1}\gamma_\tau$ converges to a limit then $\sum_{\tau=-(n-1)}^{n-1}\frac{(n-|\tau|)}{n}\gamma_\tau$ converges to the same limit.
- Thus

$$\begin{aligned}
\lim_{n\to\infty} n\operatorname{Var}\left(\bar{X}\right) &= \lim_{n\to\infty}\sum_{\tau=-(n-1)}^{n-1}\frac{(n-|\tau|)}{n}\gamma_\tau \\
&= \lim_{n\to\infty}\sum_{\tau=-(n-1)}^{n-1}\gamma_\tau \\
&= \sum_{\tau=-\infty}^{\infty}\gamma_\tau \\
&= C^{(\gamma)}
\end{aligned}$$

  say.
- Now we know that $C^{(\gamma)} < \infty$ by assumption, therefore

$$\lim_{n\to\infty}\operatorname{Var}\left(\bar{X}\right) = 0.$$

- We just showed that the sample mean was consistent, e.g. $\bar{X} \xrightarrow{P} \mu$, if the autocovariance satisfies $\sum_\tau |\gamma_\tau| < \infty$.
- Seems like a general idea: "when can we replace a sample average by a population average"? But what about the correlation? Does it matter? Does it change things?
- For example, consider the AR(1) process:

$$X_t = \phi X_{t-1} + \varepsilon_t,$$

  for $X_0 \sim \mathcal{N}\left(0, \frac{1}{1-\phi^2}\right)$. Can we always average this? Do other statistical operations? What happens as ϕ changes?
- (Note we will see more on the AR process next week!)

Effect of correlation on mean estimates

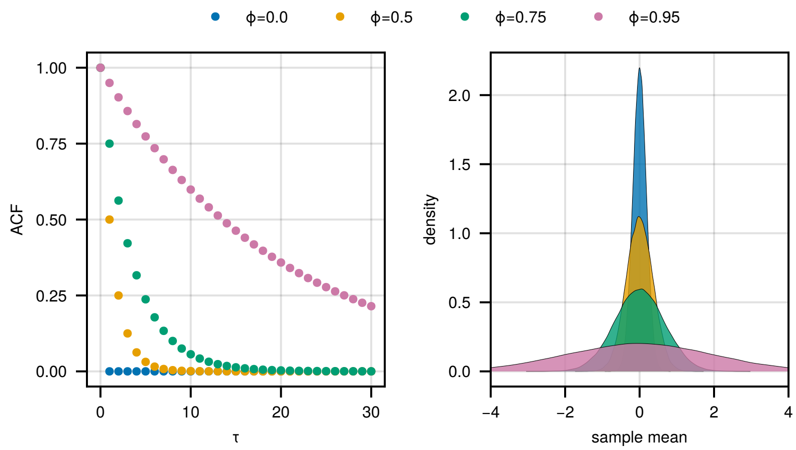


Figure 1.3: Density estimates of the distribution of the sample mean for different AR(1) processes. Time series of length 30.

Definition 1.7 ((Mean) Ergodic)**.** The time series $\{X_t\}$ is said to be mean ergodic if its first and second moments are finite and

$$\lim_{n\to\infty} \bar{X} \xrightarrow{P} \mathbb{E}\left[X_t\right].$$

- The funny squiggle $\xrightarrow{P}$ means "converges in probability" and informally it implies that the mean stabilises as the expectation tends to a constant and the variance goes to zero.
- The concept can be generalised to the d^{th} moment for $d \geq 2$, not just for the mean.
- The informal understanding of this is "sample means converge to population means", or "temporal averages are equivalent to population averages".
- Ergodicity and stationarity are not equivalent. The former concept is popular in econometrics.

Chapter 2

ARMA

2.1 ARMA models

2.1.1 White noise

Example of stationary process: white noise

Definition 2.1. An example of a stationary process is a white noise, also known as a purely random process. This corresponds to a sequence $\{X_t\}$ of uncorrelated RVs such that for all $t \in \mathbb{Z}$

$$\mathbb{E}[X_t] = 0, \quad \mathrm{Var}(X_t) = \sigma^2 < \infty.$$

In this case

$$\gamma_\tau = \begin{cases} \sigma^2 & \text{if } \tau = 0, \\ 0 & \text{otherwise,} \end{cases}$$

or equivalently

$$\rho_\tau = \begin{cases} 1 & \text{if } \tau = 0, \\ 0 & \text{otherwise.} \end{cases}$$

White noise is a building block for other time series models. Figure 2.1 shows an example of a white noise process.

2.1.2 Moving average models

Moving average processes (MA)

Definition 2.2 (MA(q))**.** Let $\{\varepsilon_t\}$ be a mean-zero white noise process. Then we define the q-th order moving average process, denoted MA(q), as

$$X_t = \mu - \sum_{j=0}^{q} \theta_j \varepsilon_{t-j}, \tag{2.1}$$

where θ_j are constants such that $\theta_0 = -1$ and $\theta_q \neq 0$.

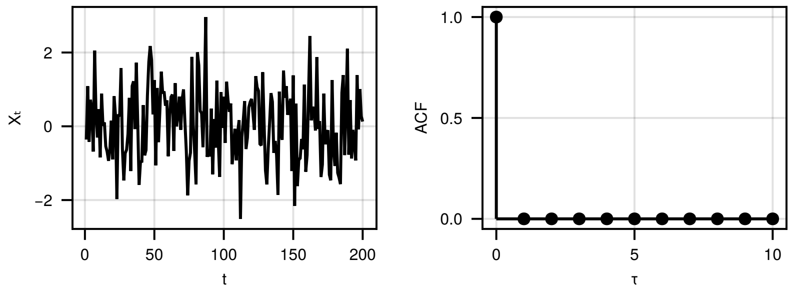


Figure 2.1: A simulated realisation of a white noise process (left) and true ACF (right)

- All we have done is taken a noisy process and averaged consecutive values.
- This isn't necessarily smoother than the original!

Moments of MA(q)

We can calculate the moments of this process.

- We find for the expectation that

$$\mathbb{E}\left[X_t\right] = \mu. \tag{2.2}$$

- Now for the covariance, let $\theta_j = 0$ for $j > q$, then

$$\operatorname{Cov}\left(X_{t+\tau}, X_t\right) = \begin{cases} \sigma_\varepsilon^2 \sum_{j=0}^{q} \theta_j \theta_{j+|\tau|} & \text{if } |\tau| \leq q, \\ 0 & \text{otherwise.} \end{cases} \tag{2.3}$$

Therefore, the only constraint for stationarity is that $|\theta_j| < \infty$.

Moments of MA(q): computation

$$\begin{aligned} \operatorname{Cov}\left(X_{t+\tau}, X_t\right) &= \operatorname{Cov}\left(\mu - \sum_{k=0}^{q} \theta_k \varepsilon_{t+\tau-k}, \mu - \sum_{j=0}^{q} \theta_j \varepsilon_{t-j}\right) \\ &= \sum_{j=0}^{q} \sum_{k=0}^{q} \theta_j \theta_k \operatorname{Cov}\left(\varepsilon_{t+\tau-k}, \varepsilon_{t-j}\right) \\ &= \sigma_\varepsilon^2 \sum_{j=0}^{q} \sum_{k=0}^{q} \theta_j \theta_k \mathbb{1}_{\tau=k-j} \end{aligned}$$

To find the terms which survive consider $0 \leq \tau \leq q$, we see that for a given j only terms $k = j + \tau$ survive. The full result follows from symmetry.

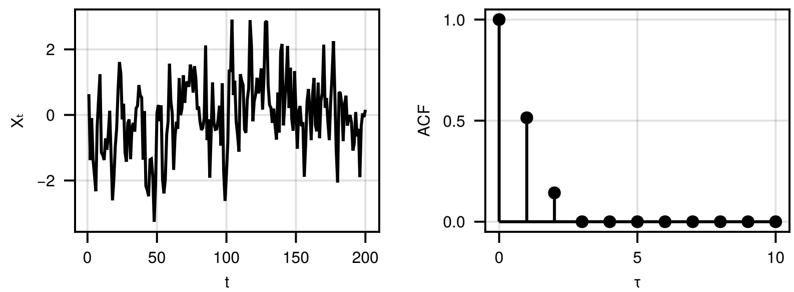


Figure 2.2: A simulated realisation of an MA(2) process (left) and true ACF (right)

An MA(1) example

Consider an MA(1) with zero mean, with ε a mean-zero white noise with variance σ_ε^2

$$X_t = \varepsilon_t - \theta\varepsilon_{t-1}.$$

The autocovariance of $\{X_t\}$ is given by

$$\gamma_\tau = \begin{cases} \sigma_\varepsilon^2\left(1+\theta^2\right) & \text{if } \tau = 0, \\ -\sigma_\varepsilon^2\theta & \text{if } |\tau| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Now consider a different white noise process $\{\eta_t\}$ with variance $\sigma_\eta^2 = \sigma_\varepsilon^2\theta^2$ and define

$$Y_t = \eta_t - \frac{1}{\theta}\eta_{t-1}.$$

Write $\tilde{\gamma}_\tau$ for the ACVS of $\{Y_t\}$, then

$$\begin{aligned} \tilde{\gamma}_0 &= \sigma_\varepsilon^2\theta^2\left(1 + \frac{1}{\theta}^2\right) \\ &= \sigma_\varepsilon^2\left(1+\theta^2\right) \\ \tilde{\gamma}_1 &= -\sigma_\varepsilon^2\theta^2/\theta \\ &= -\sigma_\varepsilon^2\theta \end{aligned}$$

and $\tilde{\gamma}_{-1} = \tilde{\gamma}_1$ and the remainder being zero.

So $\{\tilde{\gamma}_\tau\} = \{\gamma_\tau\}$ and we cannot identify the two processes from the autocovariance sequence.

2.1.3 Autoregressive models

Autoregressive process of order p, AR(p)

Definition 2.3. Let $\{\varepsilon_t\}$ be a mean-zero white noise process. Then we define the <u>p-th order autoregressive process</u>, denoted AR(p), as

$$X_t = \sum_{j=1}^{p} \phi_j X_{t-j} + \varepsilon_t, \tag{2.4}$$

where ϕ_j are constants such that $\phi_p \neq 0$.

- In contrast with the moving average process, we have constraints on $\{\phi_j\}$ to obtain a stationary process (see lecture 3).
- We also cannot give a nice closed form for the autocovariance in general.

AR(1) example

Say that we have an AR(1) process, so that

$$X_t = \phi X_{t-1} + \varepsilon_t.$$

Assume that $|\phi| < 1$. Then we may iterate and informally find that:

$$\begin{aligned} X_t &= \phi\left(\phi X_{t-2} + \varepsilon_{t-1}\right) + \varepsilon_t \\ &= \phi^2 X_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t \\ &= \phi^3 X_{t-3} + \phi^2\varepsilon_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t \\ &= \dots \\ &= \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}. \end{aligned}$$

Note that the mean and covariance is given by

$$\mathbb{E}\left[X_t\right] = \sum_{k=0}^{\infty} \phi^k \cdot 0 = 0. \tag{2.5}$$

$$\begin{aligned} \operatorname{Cov}\left(X_{t+\tau}, X_t\right) &= \sum_{k=0}^{\infty}\sum_{j=0}^{\infty} \phi^{k+j} \operatorname{Cov}\left(\varepsilon_{j+\tau}, \varepsilon_k\right) \\ &= \sigma_\varepsilon^2 \sum_{k=0}^{\infty} \phi^{2k+|\tau|} \\ &= \frac{\sigma_\varepsilon^2}{1-\phi^2}\phi^{|\tau|}. \end{aligned} \tag{2.6}$$

2.1.4 ARMA

Auto-regressive Moving Average Process ARMA(p, q)

Definition 2.4 (ARMA(p, q))**.** A time series $\{X_t\}$ is an autoregressive moving average process of order p and q, denoted ARMA(p, q), if

$$X_t = \sum_{j=1}^{p} \phi_j X_{t-j} - \sum_{k=0}^{q} \theta_k \varepsilon_{t-k}$$

This statement can be made formally correct under certain conditions, where the equality can be seen to hold with probability one.

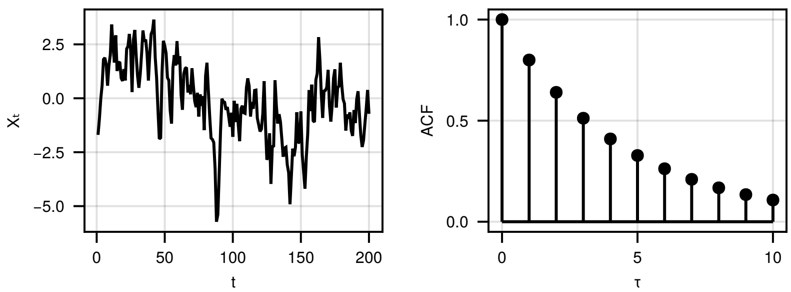


Figure 2.3: A simulated realisation of a AR(1) (left) and true ACF (right)

where $\{\varepsilon_t\}$ is a mean-zero white noise process, and ϕ_j, θ_k are the same as in the AR and MA cases respectively.

- Again we will have conditions on the AR parameters in order for this to be a valid model.
- Again general closed forms for the autocovariance aren't obtainable (though techniques to compute the autocovariance exist).

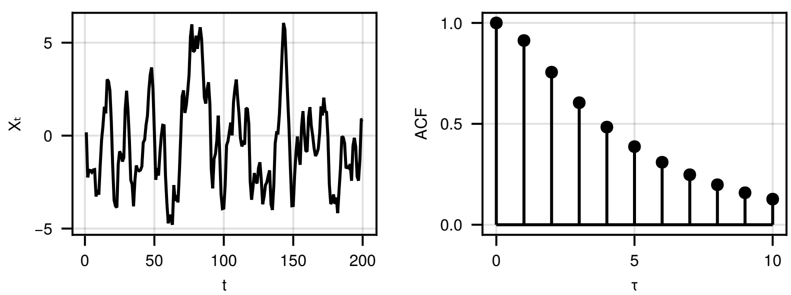


Figure 2.4: A simulated realisation of a ARMA process (left) and true ACF (right)

Causal Process

Definition 2.5 (Causal ARMA(p, q))**.** An ARMA(p, q) process is said to be causal (or more specifically to be a causal function of $\{\varepsilon_t\}$) if there exists a sequence of constants $\{\psi_j\}$ such that $\sum_{j=0}^{\infty} |\psi_j| < \infty$ and

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad t = 0, \pm 1, \dots$$

- A mean-zero $MA(q)$ process is always causal.

2.2 Partial autocorrelation

2.2.1 Motivation

AR & Dependence

Consider an AR(1) model:

$$Y_t = \phi Y_{t-1} + \epsilon_t, \quad t \in \mathbb{Z}$$

We have $\rho_0 = 1, \rho_1 = \phi, \rho_2 = \phi^2, \dots$

- Thus Y_t and Y_{t-2} are correlated even if we do not write Y_t in terms of Y_{t-2}.
- This follows because Y_t is specified in terms of Y_{t-1} and as Y_{t-1} is given in terms of Y_{t-2} there is correlation.
- We would therefore like to calculate the covariance of Y_t and Y_{t-2} given the effect of the intervening variable Y_{t-1}.

2.2.2 Partial correlation

Partial correlation

Say that we have two random variables X and Y.

- We might want to understand their correlation in order to understand something about their dependence.
- However, if they both depend on some other random variables, say $\boldsymbol{Z} = (Z_1, \dots, Z_n)^T$, then this correlation may be spuriously generated by these confounders.
- One simple way to try and avoid this is partial correlation, which aims to remove this effect.
- Partial correlation is computed by fitting the best linear model for X given $\boldsymbol{Z}$, and also for Y given $\boldsymbol{Z}$, and then looking at the correlation of the residuals.

Linear prediction

In order to define partial correlation, we need to define the best linear predictor.

Definition 2.6. Consider a collection of mean-zero random variables $X_1, \dots, X_n, Y$. The best linear predictor of Y from $X_1, \dots, X_n$ is

$$\mathcal{P}_{X_1,\dots,X_n}(Y) = \sum_{j=1}^{n} \beta_j X_j \tag{2.7}$$

such that the β_js minimise

$$\mathbb{E}\left[(Y - \mathcal{P}_{X_1,\dots,X_n}(Y))^2\right].$$

Partial correlation: formal definition

Definition 2.7 (Partial correlation)**.** Consider two random variables X and Y, and some confounding random variables $\boldsymbol{Z} = (Z_1, \ldots, Z_n)^T$, then the partial correlation of X and Y given Z is

$$\rho_{XY\bullet Z} = \operatorname{Corr}\left(X - \mathcal{P}_{\boldsymbol{Z}}(X), Y - \mathcal{P}_{\boldsymbol{Z}}(Y)\right) \tag{2.8}$$

where $\mathcal{P}_{\boldsymbol{Z}}(X)$ denotes the best linear predictor of X from $\boldsymbol{Z}$.

- If $X, Y, \boldsymbol{Z}$ are jointly Gaussian then

$$\rho_{XY\bullet Z} = \mathbb{E}\left[\operatorname{Corr}\left(X, Y \mid \boldsymbol{Z}\right)\right].$$

- In fact, in the Gaussian case,

$$\rho_{XY\bullet Z} \stackrel{a.s.}{=} \operatorname{Corr}\left(X, Y \mid \boldsymbol{Z}\right).$$

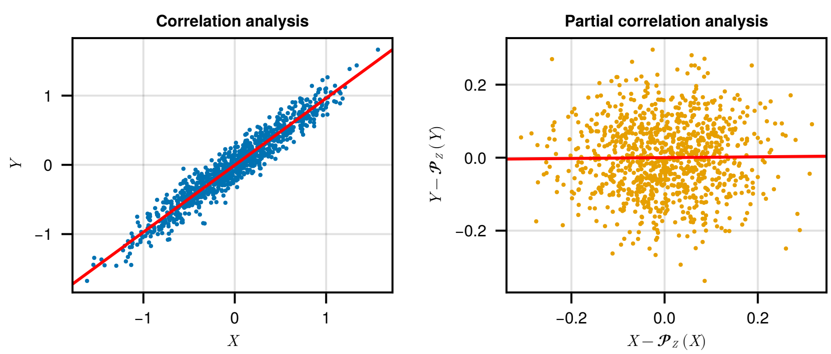


Figure 2.5: Comparison of correlation and partial correlation for some variables X and Y with some other variable Z acting as a confounder. Accounting for Z removes all correlation between X and Y, so standard correlation is misleading here!

2.2.3 Partial autocorrelation

Partial autocorrelation

- Partial autocorrelation is simply the partial correlation of the time series at time t with the time series at time $t + \tau$ accounting for all values in between, i.e.

$$\alpha_k = \rho_{XY\bullet\boldsymbol{Z}} \tag{2.9}$$

  if we set $X = X_t$, $Y = X_{t+\tau}$ and $\boldsymbol{Z} = (X_{t+1}, \ldots, X_{t+\tau-1})^T$.

- Pictorially

$$X_t,\ \underbrace{X_{t+1}, \ldots, X_{t+\tau-1}}_{\text{confounders}},\ X_{t+\tau}.$$

Partial autocorrelation function

Definition 2.8. Consider a mean-zero stationary time series $\{X_t\}$. Denote the best linear predictors of X_t and $X_{t+\tau}$ from the intervening values

$$\hat{X}_t = \mathcal{P}_{X_{t+1},\ldots,X_{t+\tau-1}}(X_t),$$
$$\hat{X}_{t+\tau} = \mathcal{P}_{X_{t+1},\ldots,X_{t+\tau-1}}(X_{t+\tau}).$$

The partial autocorrelation function α_τ is given by

$$\alpha_\tau = \operatorname{Corr}\left(X_{t+\tau} - \hat{X}_{t+\tau}, X_t - \hat{X}_t\right) \tag{2.10}$$

- Here we are measuring the correlation after removing the linear effects of the intervening values.

Reformulation as linear regression

Consider a mean-zero stationary time series $\{X_t\}$. Fix $t = 0$ and let $\tau \in \mathbb{Z}$, $\tau \geq 0$, the best linear predictor of $X_{t+\tau}$ from $X_t, \ldots, X_{t+\tau-1}$ takes the form

$$\mathcal{P}_{X_0,\ldots,X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}. \tag{2.11}$$

- One can show that $\alpha_{\tau,\tau} = \alpha_\tau$.
- This representation is useful as we can construct a system of equations to solve for $\alpha_{\tau,\tau}$.

Relation to the ACF

For any $\tau > 0$, for $k \in \{1, \ldots, \tau\}$, write $\hat{X}_\tau = \mathcal{P}_{X_0,\ldots,X_{\tau-1}}(X_\tau)$

$$\begin{aligned}\gamma_k = \operatorname{Cov}(X_{\tau-k}, X_\tau) &= \operatorname{Cov}\left(X_{\tau-k}, \hat{X}_\tau\right) + \operatorname{Cov}\left(X_{\tau-k}, X_\tau - \hat{X}_\tau\right) \\ &= \sum_{j=1}^{\tau} \alpha_{\tau,j} \operatorname{Cov}(X_{\tau-k}, X_{\tau-j}) \\ &= \sum_{j=1}^{\tau} \alpha_{\tau,j} \gamma_{k-j}\end{aligned}$$

and so

$$\rho_k = \sum_{j=1}^{\tau} \alpha_{\tau,j} \rho_{k-j}. \tag{2.12}$$

Thus we have equations to relate the ACF and the PACF.

ACF and PACF of ARMA models

For causal and invertible ARMA(p, q) models, the ACF and PACF have the properties in Table 2.1.

	AR(p)	MA(q)	ARMA(p,q)
ACF	Tails off	Cuts off after lag q	Tails off
PACF	Cuts off after lag p	Tails off	Tails off

Table 2.1: ACF and PACF properties of ARMA models.

Computation via Durbin-Levinson recursions

One can show that the Durbin-Levinson recursions can be used to solve for $\alpha_{\tau,\tau}$ given the ACF:

$$\begin{aligned}
\alpha_{1,1} &= \rho_1, \\
\alpha_{\tau,\tau} &= \frac{\rho_\tau - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j}\rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j}\rho_j}, \\
\alpha_{\tau,j} &= \alpha_{\tau-1,j} - \alpha_{\tau,\tau}\alpha_{\tau-1,\tau-j}.
\end{aligned}$$

- This can be used either with the theoretical or estimated ACF.

Example time series

- A time series model has $\rho_1 = 2/5, \rho_2 = -1/20$ and $\rho_3 = -1/8$.
- Find the PACF at lags 1,2 and 3.
- We use the Durbin-Levinson recursions. These give

$$\begin{aligned}
\alpha_{1,1} &= \rho_1 = \frac{2}{5} \\
\alpha_{2,2} &= \frac{\rho_2 - \alpha_{1,1}\rho_1}{1 - \alpha_{1,1}\rho_1} = -\frac{1}{4} \\
\alpha_{2,1} &= \alpha_{1,1} - \alpha_{2,2}\alpha_{1,1} = 1/2 \\
\alpha_{3,3} &= \ldots = 0
\end{aligned}$$

We don't know about $\alpha_{k,k}$ but the latter indicates that this may be an AR(2) model.

Example time series continued

- In fact the AR(2) of

$$Y_t = \frac{1}{2}Y_{t-1} - \frac{1}{4}Y_{t-2} + \varepsilon_t$$

has the same PACF to the process that we noted (for the first three lags).

- Note that α_{22} is equal to the coefficient of Y_{t-2} in the model.

Partial autocorrelation of an AR model

Proposition 2.1. *For an AR(p), we have*

$$\alpha_\tau = 0, \qquad \forall \tau > p.$$

- Furthermore, it can be shown that asymptotically the estimated partial autocorrelation at lags greater than p have mean 0 and variance $1/n$, where n is the sample size.

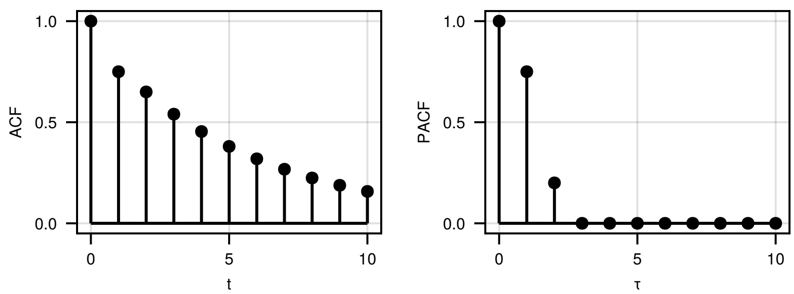


Figure 2.6: The ACF (left) and PACF (right) of model 1.

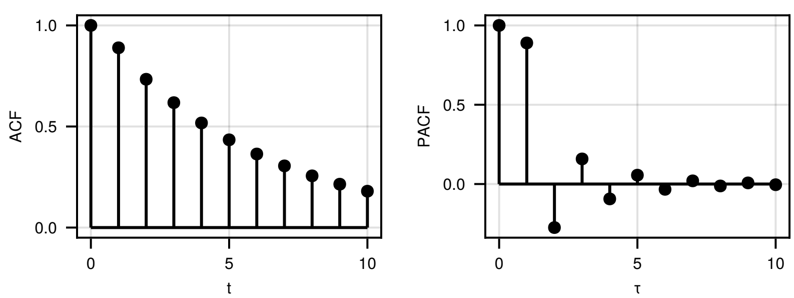


Figure 2.7: The ACF (left) and PACF (right) of model 2.

Solutions

The true models were as follows:

- model 1: AR(2)

  $\rightarrow$ $\phi_1 = 0.6$ and $\phi_2 = 0.2$

- model 2: ARMA$(1, 2)$

  $\rightarrow$ $\phi_1 = 0.6$, $\phi_2 = 0.2$ and $\theta_1 = -0.6$

- model 3: MA(1)

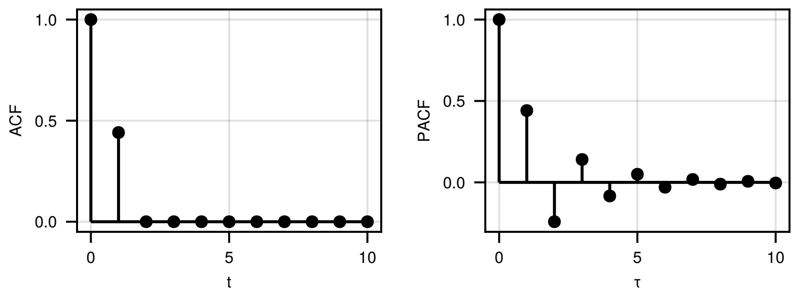


Figure 2.8: The ACF (left) and PACF (right) of model 3.

$\rightarrow\ \theta_1 = -0.6$

in all cases the noise had variance 1.

Chapter 3

ARMA revisited

3.1 ARMA polynomial notation

ARMA recap

- Recall that an ARMA process has the following form

$$X_t = \sum_{j=1}^{p} \phi_j X_{t-j} - \sum_{k=0}^{q} \theta_k \varepsilon_{t-k}$$

  where $\{\varepsilon_t\}$ is a mean-zero white noise process and $\theta_0 = -1$.

- We could rewrite this as

$$\sum_{j=0}^{p} \phi_j X_{t-j} = \sum_{k=0}^{q} \theta_k \varepsilon_{t-k} \tag{3.1}$$

  where $\phi_0 = -1$.

- This is elegant, but we can simplify things a little further with the help of the backshift operator.

The backshift operator

Definition 3.1 (The backshift operator)**.** Define the backshift operator as a map from one time series $\{X_t\}$ to another time series, say $\{y_t\}$ which simply shifts the first series back one step in time. Formally write

$$\{Y_t\} = \mathrm{B}[\{X_t\}]$$

then for all $t \in \mathbb{Z}$ we have

$$Y_t = X_{t-1}.$$

- Often we will use the informal notation $\mathrm{B}X_t$ to refer to $\mathrm{B}[\{X_t\}]$.
- Note that applying the backshift multiple times will shift the series by multiple lags (i.e. the operator B^k shifts by k lags).

MA and the backshift operator

Take the MA(q) example, we can write this as

$$X_t = \sum_{k=0}^{q} \theta_k \varepsilon_{t-k} \tag{3.2}$$

$$= \sum_{k=0}^{q} \theta_k \mathrm{B}^k \varepsilon_t \tag{3.3}$$

$$= \Theta(\mathrm{B})\varepsilon_t \tag{3.4}$$

where Θ is the polynomial given by

$$\Theta(z) = \sum_{k=0}^{q} \theta_k z^k. \tag{3.5}$$

ARMA and the backshift operator

Now we can apply the same trick for the general ARMA(p, q) process, and write

$$\Phi(\mathrm{B})X_t = \Theta(\mathrm{B})\varepsilon_t \tag{3.6}$$

to specify the ARMA process, where

$$\Theta(z) = \sum_{k=0}^{q} \theta_k z^k, \tag{3.7}$$

$$\Phi(z) = \sum_{j=0}^{p} \phi_j z^j. \tag{3.8}$$

Later in the lecture, we shall explore how properties of the polynomials Θ and Φ relate to properties of the resultant process.

3.2 Wold decomposition

Linear combinations of white noise

So far, we have seen finite combinations of white noise processes, but we may want to consider something more general:

$$X_t = \sum_{k=-\infty}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 < \infty. \tag{3.9}$$

where $\{g_k\}$ is a real valued sequence and $\{\varepsilon_t\}$ is mean-zero white noise.

- We have for all $t, \tau \in \mathbb{Z}$

$$\mathbb{E}\left[X_t\right] = 0, \tag{3.10}$$
$$\operatorname{Var}\left(X_t\right) = \|g\|_2^2 \operatorname{Var}\left(\epsilon_t\right) < \infty, \tag{3.11}$$
$$\operatorname{Cov}\left(X_t, X_{t+\tau}\right) = \sum_{k=-\infty}^{\infty} g_k g_{k+\tau} \operatorname{Var}\left(\varepsilon_t\right). \tag{3.12}$$

Definition 3.2 (General linear process)**.** If in (3.9) we set $g_{-1},\ g_{-2},\ \cdots = 0$, then we obtain a general linear process:

$$X_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 < \infty, \tag{3.13}$$

- Now X_t depends only on the past and the present, making this into a causal process.
- The same more general equations for mean and autocovariance apply, so the process is stationary.

Theorem 3.1 (Wold Decomposition Theorem)**.** *Any stationary process $\{X_t\}$ can be expressed in the form:*

$$X_t = U_t + V_t,$$

with U_t and V_t uncorrelated such that

- *U_t has a one–sided linear representation*

$$U_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k},$$

  with $g_0 = 1$, $\|g\|_2^2 < \infty$ and ε_t a mean-zero white noise process uncorrelated with V_t so that $\forall s, t$ $\mathbb{E}\left[\varepsilon_s V_t\right] = 0$. The sequences $\{g_u\}$ and $\{\varepsilon_t\}$ are then uniquely determined.
- *V_t is singular (can be predicted from its own past with no error).*

- To study such processes, we introduce the function $G(z)$:

$$G(z) = \sum_{k=0}^{\infty} g_k z^k,$$

  so that $X_t = G(\mathrm{B})\varepsilon_t$.
- We represent $G(z)$ via its Laurent series (a fancy Taylor series) but will first study it as a ratio:

$$G(z) = \frac{G_1(z)}{G_2(z)}.$$

- Note that the roots of $G_1(z)$ are the roots or zeros of $G(z)$ and the roots of $G_2(z)$ are the poles of $G(z)$.
- Call the zeros of $G_2(z)$ $z_1, z_2, \ldots z_p$ ordered so that $z_1, z_2, \ldots z_k$ are inside the unit circle $|z| = 1$ and $z_{k+1}, \ldots, z_p$ are outside the unit circle $|z| = 1$.

- With this specification the Laurent expansion gives us

$$\begin{aligned}\frac{1}{G_2(z)} &= \sum_{j=1}^{p} \frac{A_j}{z - z_j} \\ &= \sum_{j=1}^{k} \frac{A_j}{z} \sum_{l=0}^{\infty} \left(\frac{z_j}{z}\right)^l - \sum_{j=k+1}^{p} \frac{A_j}{z_j} \sum_{l=0}^{\infty} \left(\frac{z_j}{z}\right)^{-l}.\end{aligned}$$

This expansion is convergence on $|z| = 1$. We can therefore replace z by B and arrive at

$$\begin{aligned}\frac{1}{G_2(\mathrm{B})}\varepsilon_t &= \left\{\sum_{j=1}^{k} A_j \mathrm{B}^{-1} \sum_{l=0}^{\infty} z_j^l \mathrm{B}^{-l} - \sum_{j=k+1}^{p} A_j \sum_{l=0}^{\infty} z_j^{-(l+1)} \mathrm{B}^l\right\}\varepsilon_t \\ &= \left\{\sum_{j=1}^{k} A_j \sum_{l=0}^{\infty} z_j^l \mathrm{B}^{-l-1} - \sum_{j=k+1}^{p} A_j \sum_{l=0}^{\infty} z_j^{-(l+1)} \mathrm{B}^l\right\}\varepsilon_t.\end{aligned}$$

- It therefore follows that

$$\frac{1}{G_2(\mathrm{B})}\varepsilon_t = \sum_{j=1}^{k} A_j \sum_{l=0}^{\infty} z_j^l \underbrace{\varepsilon_{t+1+l}}_{\text{future}} - \sum_{j=k+1}^{p} A_j \sum_{l=0}^{\infty} z_j^{-(l+1)} \underbrace{\varepsilon_{t-l}}_{\text{past+now}}.$$

Hence if all the roots of $G_2(z)$ are outside the unit circle then only past and present values of X_t are involved. Then the general linear process exists.

- Another way of stating this is that $|G(z)| < \infty$ for $|z| \leq 1$. This means that $G(z)$ is analytic inside and on the unit circle.

- If a particular value of ε_t affects X_t and all subsequent X_t then we say this is an innovations outlier.

3.3 Stationarity and invertibility of ARMA processes

- Consider the $MA(q)$ model in this setting. Then

$$X_t = \Theta(\mathrm{B})\varepsilon_t = \varepsilon_t - \theta_{1,q}\varepsilon_{t-1} - \cdots - \theta_{q,q}\varepsilon_{t-q}.$$

Thus we have in the general linear process representation:

$$X_t = \Theta(\mathrm{B})\varepsilon_t \;\Leftrightarrow \Theta^{-1}(\mathrm{B})X_t = \varepsilon_t.$$

- Similarly for the AR model we may write

$$\Phi(\mathrm{B})X_t = \varepsilon_t.$$

Here $\Phi(\mathrm{B})$ has a finite order but $\Phi^{-1}(\mathrm{B})$ has an infinite order.

- Invertibility: Consider inverting the general linear process

$$X_t = G(\mathrm{B})\varepsilon_t \Rightarrow G^{-1}(\mathrm{B})X_t = \varepsilon_t.$$

- The expansion of $G^{-1}(\mathrm{B})$ in powers of B gives its AR form provided that $G^{-1}(\mathrm{B})$ admits a power expansion

$$G^{-1}(z) = \sum_{k=0}^{\infty} h_k z^k,$$

  and that must be analytic on $|z| \leq 1$.

- For a general linear process the model is invertible if $|G^{-1}(z)| < \infty$ for $|z| \leq 1$.
- This means all the poles of $G^{-1}(z)$ are outside the unit circle.
- $X_t = G(\mathrm{B})\varepsilon_t$ is the general linear model. If the poles of $G(z)$ are outside the unit circle, then the zeros of $G^{-1}(z)$ are inside the unit circle.
- For the MA(q) process we have $G(\mathrm{B}) = \Theta(\mathrm{B})$.
- For the AR(p) process we have $\Phi(\mathrm{B})X_t = \varepsilon_t$.
- Thus

$$\begin{aligned} X_t &= \Phi^{-1}(\mathrm{B})\varepsilon_t = G(\mathrm{B})\varepsilon_t \\ &\Rightarrow G(z) = \Phi^{-1}(z). \end{aligned}$$

- Thus in this scenario (AR) the requirement for stationarity is that the roots of $\Phi(z)$ are outside the unit disc.
- For the MA(q) process we have

$$X_t = \Theta(\mathrm{B})\varepsilon_t = G(\mathrm{B})\varepsilon_t.$$

  Thus since $\Theta(\mathrm{B}) = G(\mathrm{B})$ is a polynomial of finite order with have $|G(z)| < \infty$ as long as all parameters are finite.

Summary of stationarity and invertibility of ARMA models

We can therefore summarise our understanding as follows:

- An AR(p) must have the roots of $\Phi(z)$ outside $|z| = 1$ to be stationary. It is always invertible.
- An MA(q) is always stationary but must have the roots of $\Theta(z)$ outside $|z| = 1$ to be invertible.
- An ARMA(p, q) must have the roots of $\Phi(z)$ outside $|z| = 1$ to be stationary, and must have the roots of $\Theta(z)$ outside $|z| = 1$ to be invertible.

Characteristic polynomials

Definition 3.3. Recall we can write an ARMA model as

$$\Phi(\mathrm{B})X_t = \Theta(\mathrm{B})\varepsilon_t.$$

We call

- $\Phi(z)$ the characteristic polynomial of the autoregressive part,
- $\Theta(z)$ the characteristic polynomial of the moving average part.

In the specific cases of MA and AR models, this will be shortened to characteristic polynomial, i.e.

- For an AR, $\Phi(\mathrm{B})X_t = \varepsilon_t$, the characteristic polynomial is $\Phi(z)$.
- For an MA, $X_t = \Theta(\mathrm{B})\varepsilon_t$ the characteristic polynomial is $\Theta(z)$.

ARMA Example

Consider the following example, let

$$\{\mathrm{I} - \mathrm{B} + \frac{1}{4}\mathrm{B}^2\}X_t = \{\mathrm{I} + \mathrm{B}\}\varepsilon_t. \tag{3.14}$$

Determine the auto-covariance of X_t assuming $\{\varepsilon_t\}$ is white noise.

- We can cross–multiply equation (3.14) by $X_{t-\tau}$ for $\tau \geq 2$. We then arrive at

  $$X_tX_{t-\tau} - X_{t-1}X_{t-\tau} + \frac{1}{4}X_{t-2}X_{t-\tau} = \varepsilon_tX_{t-\tau} + \varepsilon_{t-1}X_{t-\tau}.$$

  Taking expectations we arrive at

  $$\gamma_\tau - \gamma_{\tau-1} + \frac{1}{4}\gamma_{\tau-2} = 0.$$

- This leaves $\tau = 0, 1$ to figure out. This must be done separately.
- If we set $\tau = 0$ then we get

  $$X_t^2 - X_{t-1}X_t + \frac{1}{4}X_{t-2}X_t = \varepsilon_tX_t + \varepsilon_{t-1}X_t.$$

  Then taking expectations we get that

  $$\gamma_0 - \gamma_1 + \frac{1}{4}\gamma_2 = \sigma^2 + 2\sigma^2.$$

  Similarly for $\tau = 1$

  $$\gamma_1 - \gamma_0 + \frac{1}{4}\gamma_1 = \sigma^2.$$

  A general solution will be of the format

  $$\gamma_\tau = \{\beta_{10} + \beta_{11}\tau\}2^{-\tau}, \quad \tau > 0,$$

  and by using the initial conditions we may recover the constants.

Chapter 4

Parametric estimation

4.1 Yule-Walker (method-of-moments) for AR models

Setup

For the first two sections of this lecture, assume that X_t is a mean-zero causal AR process of order p:

$$
\begin{aligned}
X_t &= \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t \\
&= \sum_{k=1}^{p} \phi_k X_{t-k} + \epsilon_t,
\end{aligned}
$$

where $\epsilon_t \overset{iid}{\sim} \mathcal{N}\left(0, \sigma_\epsilon^2\right)$ is a Gaussian white noise process. We suppose that we have observed $X_1, \ldots, X_n$ and we want to estimate the parameters $\phi_1, \ldots, \phi_p$ and σ_ϵ^2.

Yule-Walker derivation

To derive the estimation method we start by multiplying through the AR equation by X_{t-k} for $k > 0$

$$
\begin{aligned}
X_t X_{t-k} &= \sum_{i=1}^{p} \phi_i X_{t-i} X_{t-k} + \epsilon_t X_{t-k} \\
\Rightarrow \mathbb{E}\left[X_t X_{t-k}\right] &= \sum_{i=1}^{p} \phi_i \mathbb{E}\left[X_{t-i} X_{t-k}\right] + \mathbb{E}\left[\epsilon_t X_{t-k}\right]
\end{aligned}
$$

- Remember the process is causal, so $\mathbb{E}\left[\epsilon_t X_{t-k}\right] = 0$.
- Furthermore, we have assumed X_t is mean-zero, so

$$
\mathbb{E}\left[X_{t-j} X_{t-k}\right] = \gamma_{k-j}.
$$

- Therefore

$$\gamma_k = \sum_{j=1}^{p} \phi_j \gamma_{k-j} \tag{4.1}$$

Therefore using (4.1) and symmetry of $\gamma_\tau = \gamma_{-\tau}$

$$\begin{aligned} \gamma_1 &= \phi_1\gamma_0 + \phi_2\gamma_1 + \cdots + \phi_p\gamma_{p-1}, \\ \gamma_2 &= \phi_1\gamma_1 + \phi_2\gamma_0 + \cdots + \phi_p\gamma_{p-2}, \\ &\vdots \\ \gamma_p &= \phi_1\gamma_{p-1} + \phi_2\gamma_{p-2} + \cdots + \phi_p\gamma_0. \end{aligned} \tag{4.2}$$

- In matrix form we have

$$\boldsymbol{\gamma} = \boldsymbol{\Gamma}\boldsymbol{\phi}$$

where $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_p)^T$, and $\boldsymbol{\phi} = (\phi_1, \dots, \phi_p)^T$ and

$$\boldsymbol{\Gamma} = \begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots & \gamma_0 \end{pmatrix}.$$

- If the mean is zero take

$$\hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} X_t X_{t+|\tau|}$$

- We estimate $\boldsymbol{\phi}$ via

$$\widehat{\boldsymbol{\phi}} = \widehat{\boldsymbol{\Gamma}}^{-1}\widehat{\boldsymbol{\gamma}}$$

- Now we just need to estimate σ_ϵ^2. We have

$$\mathbb{E}\left[X_t^2\right] = \phi_1 \mathbb{E}\left[X_t X_{t-1}\right] + \cdots + \phi_p \mathbb{E}\left[X_t X_{t-p}\right] + \mathbb{E}\left[\epsilon_t X_t\right].$$

- Therefore

$$\gamma_0 = \sum_{j=1}^{p} \phi_j \gamma_j + \mathbb{E}\left[\epsilon_t X_t\right]$$

- Furthermore,

$$\epsilon_t X_t = \phi_1 \epsilon_t X_{t-1} + \cdots + \phi_p \epsilon_t X_{t-p} + \epsilon_t^2$$

- We take expectations of the left and right hand side and get that

$$\begin{aligned} \mathbb{E}\left[\epsilon_t X_t\right] &= \phi_1 \mathbb{E}\left[\epsilon_t X_{t-1}\right] + \cdots + \phi_p \mathbb{E}\left[\epsilon_t X_{t-p}\right] + \mathbb{E}\left[\epsilon_t^2\right] \\ &= \phi_1 \cdot 0 + \cdots + \phi_p \cdot 0 + \sigma_\epsilon^2 \\ \Rightarrow \gamma_0 &= \sum_{j=1}^{p} \phi_j \gamma_j + \sigma_\epsilon^2. \end{aligned}$$

- Thus the estimator is $\widehat{\sigma}_\epsilon^2 = \hat{\gamma}_0 - \sum_{j=1}^p \widehat{\phi}_j \hat{\gamma}_j$.

Definition 4.1 (Yule-Walker estimators)**.** The estimators $\widehat{\phi}_p$ and $\widehat{\sigma}_\epsilon^2$ are the Yule-Walker estimators.

$$\widehat{\boldsymbol{\phi}}_p = \widehat{\boldsymbol{\Gamma}}_p^{-1} \widehat{\boldsymbol{\gamma}}_p \tag{4.3}$$

$$\widehat{\sigma}_\epsilon^2 = \hat{\gamma}_0 - \sum_{j=1}^p \widehat{\phi}_j \hat{\gamma}_j \tag{4.4}$$

- Inverting $\boldsymbol{\Gamma}_p$ naively is $O\left(p^3\right)$, but the Levinson-Durbin algorithm is $O\left(p^2\right)$ by using the structure of $\boldsymbol{\Gamma}_p$ (symmetric Toeplitz matrix).

4.2 Least squares for Gaussian mean-zero AR models

Forward least squares: formulation

In order to formulate a least squares approach we begin from the definition of an AR process

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t.$$

Specifically, we can write

$$\begin{aligned} X_{p+1} &= \phi_1 X_p + \phi_2 X_{p-1} + \cdots + \phi_p X_1 + \epsilon_{p+1} \\ &\vdots \\ X_n &= \phi_1 X_{n-1} + \phi_2 X_{n-2} + \cdots + \phi_p X_{n-p} + \epsilon_n \end{aligned} \tag{4.5}$$

- Other terms $X_1, \ldots, X_p$ have to be discarded, as we do not observe times before X_1.

Forward least squares: matrix form

We can rewrite this in matrix form

$$\mathbf{X}_F = \mathbf{F}\boldsymbol{\phi} + \boldsymbol{\epsilon}$$

where

$$\mathbf{X}_F = \begin{pmatrix} X_{p+1} \\ \vdots \\ X_n \end{pmatrix}, \quad \boldsymbol{\phi} = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_{p+1} \\ \vdots \\ \epsilon_n \end{pmatrix},$$

and

$$\mathbf{F} = \begin{pmatrix} X_p & X_{p-1} & \cdots & X_1 \\ X_{p+1} & X_p & \cdots & X_2 \\ \vdots & \vdots & \ddots & \cdots \\ X_{n-1} & X_{n-2} & \cdots & X_{n-p} \end{pmatrix}.$$

Forward least squares: objective function

We can estimate $\boldsymbol{\phi}$ by finding the vector which minimises

$$\begin{aligned}\mathrm{SS}(\boldsymbol{\phi}) &= \|\mathbf{X}_F - \mathbf{F}\boldsymbol{\phi}\|^2 \\ &= \sum_{t=p+1}^{n}\left(X_t - \sum_{k=1}^{p}\phi_k X_{t-k}\right)^2 \\ &= \sum_{t=p+1}^{n}\epsilon_t^2.\end{aligned}$$

- This is a standard least squares problem.

Definition 4.2 (Forward least squares estimators)**.** We write the minimiser by $\widehat{\boldsymbol{\phi}}_F$, the estimator takes the form

$$\widehat{\boldsymbol{\phi}}_F = \left(\mathbf{F}^T\mathbf{F}\right)^{-1}\mathbf{F}^T\mathbf{X}_F \tag{4.6}$$

which is the usual least squares estimator. Finally we may estimate σ_ϵ^2 using standard least squares methods:

$$\widehat{\sigma}_F^2 = \left\|\mathbf{X}_F - \mathbf{F}\widehat{\boldsymbol{\phi}}_F\right\|^2/(n-p-p). \tag{4.7}$$

Together, $\widehat{\phi}_F, \widehat{\sigma}_F^2$ are known as the forward least squares estimators.

Backward least squares: time reversibility

The backward least squares approach is based on the time reversibility property of a Gaussian AR process.

- In particular, there exists a representation $Y_t = X_{-t}$ as an AR process with the same parameters.
- Specifically,

$$Y_t = \sum_{j=1}^{p}\phi_{j,p}Y_{t-j} + \tilde{\nu}_t \tag{4.8}$$

  where $\tilde{\nu}_t$ has the same distribution as ϵ_t. Rewriting in terms of X_t, with $\nu_t = \tilde{\nu}_{-t}$ we have

$$X_t = Y_{-t} = \sum_{j=1}^{p}\phi_{j,p}Y_{-t-j} + \tilde{\nu}_{-t} = \sum_{j=1}^{p}\phi_{j,p}X_{t+j} + \nu_t.$$

The proof of (4.8) will be given later, using frequency domain methods.

Backward least squares: formulation

Now perform the same trick as in the forwards case:

$$\begin{aligned} X_1 &= \phi_1 X_2 + \phi_2 X_3 + \cdots + \phi_p X_{p+1} + \nu_1 \\ &\vdots \\ X_{n-p} &= \phi_1 X_{n-p+1} + \phi_2 X_{n-p+2} + \cdots + \phi_p X_n + \nu_n \end{aligned} \tag{4.9}$$

- This time we are regressing the <u>first</u> $n-p$ values in the time series against the values at appropriate lags.
- In contrast, the forwards approach used the <u>last</u> $n-p$ values.

Backward least squares: matrix form

In matrix form

$$\mathbf{X}_B = \mathbf{B}\boldsymbol{\phi} + \boldsymbol{\nu}$$

where

$$\mathbf{X}_B = \begin{pmatrix} X_1 \\ \vdots \\ X_{n-p} \end{pmatrix}, \quad \boldsymbol{\phi} = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix}, \quad \boldsymbol{\nu} = \begin{pmatrix} \nu_1 \\ \vdots \\ \nu_{n-p} \end{pmatrix}$$

and

$$\mathbf{B} = \begin{pmatrix} X_2 & X_3 & \cdots & X_{p+1} \\ X_3 & X_4 & \cdots & X_{p+2} \\ \vdots & \vdots & \ddots & \cdots \\ X_{n-p+1} & X_{n-p+2} & \cdots & X_n \end{pmatrix}.$$

Definition 4.3 (Backward least squares estimators)**.** We write the minimiser by $\widehat{\boldsymbol{\phi}}_B$, the estimator takes the form

$$\widehat{\boldsymbol{\phi}}_B = \left(\mathbf{B}^T\mathbf{B}\right)^{-1} \mathbf{B}^T \mathbf{X}_B \tag{4.10}$$

which is again the least squares estimator. Finally we may estimate $\sigma_\epsilon^2 = \sigma_\nu^2$ using standard least squares methods:

$$\widehat{\sigma}_B^2 = \left\| \mathbf{X}_B - \mathbf{B}\widehat{\boldsymbol{\phi}}_B \right\|^2 / (n-p-p). \tag{4.11}$$

Together, $\widehat{\phi}_F, \widehat{\sigma}_B^2$ are known as the <u>backward least squares estimators</u>.

- Analogously to the forward case, $\widehat{\boldsymbol{\phi}}_B$ is the vector which minimises

$$\mathrm{SS}_B(\boldsymbol{\phi}) = \|\mathbf{X}_B - \mathbf{B}\boldsymbol{\phi}\|^2 = \sum_{t=1}^{n-p} \left(X_t - \sum_{k=1}^{p} \phi_k X_{t+k} \right)^2.$$

Forward-backward least squares

Definition 4.4 (Forward-backward least squares estimators)**.** The vector $\widehat{\boldsymbol{\phi}}_{FB}$ that minimises

$$\mathrm{SS}_B(\boldsymbol{\phi}) + \mathrm{SS}_F(\boldsymbol{\phi}) \tag{4.12}$$

is called the forward-backward least squares estimator.

- Simulation studies indicate that it performs better than forward least squares or backward least squares.

Summary of least squares for AR models

- We have seen three different methods to estimate AR parameters via least squares.
- $\widehat{\boldsymbol{\phi}}_{FB}, \widehat{\boldsymbol{\phi}}_B$ and $\widehat{\boldsymbol{\phi}}_F$ produce estimated models which need not be stationary:
  - → This may be a concern for prediction (see Lecture 10).
  - → But for other purposes, e.g. spectral estimation (see Lecture 8), the parameter values will still produce a valid estimates.

4.3 Least squares for general ARMA models

Least squares for regression type models

- We now consider estimation for ARMA in general.
- We shall estimate them by minimising sums of squares of the residuals.
- Example: say we have an AR(1) model with an unknown mean

$$Y_t = \mu + \phi\,(Y_{t-1} - \mu) + \epsilon_t. \tag{4.13}$$

If we were to know μ and ϕ then we can work out the random perturbation by

$$\epsilon_t = Y_t - \mu - \phi\,(Y_{t-1} - \mu)\,, \quad t = 2, \ldots, n. \tag{4.14}$$

Least Squares: score equations for AR(1)

We seek estimates of μ and ϕ. We thus minimise the least squares of

$$S(\mu, \phi) = \sum_{t=2}^{n} \epsilon_t^2 = \sum_{t=2}^{n} \left(Y_t - \mu - \phi\,(Y_{t-1} - \mu)\right)^2.$$

- How do we find estimates of the parameters?

We will see stationarity conditions for AR models in Lecture 4.

- We use calculus:

$$\frac{\partial S(\mu,\phi)}{\partial \mu} = -2(1-\phi)\sum_{t=2}^{n}\left(Y_t - \mu - \phi\left(Y_{t-1}-\mu\right)\right),$$

$$\frac{\partial S(\mu,\phi)}{\partial \phi} = -2\sum_{t=2}^{n}\left(Y_t - \mu - \phi\left(Y_{t-1}-\mu\right)\right)\left(Y_{t-1}-\mu\right).$$

  We see that there is one solution to this; $\phi = 1$. This is not allowed as it is non-stationary.

Least Squares: solving the score equations

Thus our estimates satisfy

$$0 = \sum_{t=2}^{n}\left(Y_t - \widehat{\mu} - \widehat{\phi}\left(Y_{t-1}-\widehat{\mu}\right)\right),$$

$$0 = \sum_{t=2}^{n}\left(Y_t - \widehat{\mu} - \widehat{\phi}\left(Y_{t-1}-\widehat{\mu}\right)\right)\left(Y_{t-1}-\widehat{\mu}\right).$$

- We can arrive at closed form expressions:

$$\widehat{\mu} = \frac{1}{(n-1)(1-\widehat{\phi})}\left(\sum_{t=2}^{n} Y_t - \widehat{\phi}\sum_{t=2}^{n} Y_{t-1}\right)$$

$$\widehat{\phi} = \frac{\sum_{t=2}^{n}\left\{Y_{t-1}-\widehat{\mu}\right\}\left\{Y_t-\widehat{\mu}\right\}}{\sum_{t=1}^{n-1}\left\{Y_t-\widehat{\mu}\right\}^2}$$

Summary of the AR case

- For n very large, this is approximately the same as $\widehat{\mu} \approx \bar{Y}$ and $\widehat{\phi} \approx \rho_1$.
- So in this case the least-squares estimation method gives similar results as the method of moments.
- This is true for most AR(p) models.

Least Squares for MA models

Now say we want to estimate θ in an MA(1) model

$$Y_t = \epsilon_t - \theta\epsilon_{t-1}.$$

Again we minimise over θ the sum of squares

$$S(\theta) = \sum_{t=1}^{n}\left(Y_t + \theta\epsilon_{t-1}\right)^2.$$

But ϵ_{t-1} is not observed!

- Solution: assume that $\epsilon_0 = 0$ and then, for every θ we calculate estimates of $\{\epsilon_t\}$ recursively using the formula

$$\epsilon_t = Y_t + \theta\epsilon_{t-1}.$$

Least Squares for MA models: estimating innovations

- Thus

$$
\begin{aligned}
\hat{\epsilon}_1 &= Y_1, \\
\hat{\epsilon}_2 &= Y_2 + \theta\hat{\epsilon}_1, \\
\hat{\epsilon}_3 &= Y_3 + \theta\hat{\epsilon}_2, \\
&\vdots \\
\hat{\epsilon}_n &= Y_n + \theta\hat{\epsilon}_{n-1}.
\end{aligned}
$$

- In order to calculate the sum of squares $S(\theta)$ we replace the unobserved errors ϵ_t with the estimates $\hat{\epsilon}_t$.

Least Squares for ARMA models

- Consider an ARMA$(1,1)$ model:

$$Y_t = \phi Y_{t-1} + \varepsilon_t - \theta\varepsilon_{t-1}.$$

- In this case we can estimate

$$S(\phi,\theta) = \sum_{t=1}^{n} \left(Y_t - \phi Y_{t-1} + \theta\varepsilon_{t-1}\right)^2.$$

- We can estimate the residuals one-by-one recursively

$$\varepsilon_t = Y_t - \phi Y_{t-1} + \theta\varepsilon_{t-1}.$$

- The recursive estimation is initiated at $Y_0 = \varepsilon_0 = 0$.

Least Squares: takeaways

- This method will work for any ARMA(p,q) model.
- For models that have an MA part, one needs to assume that q terms are zero.

Chapter 5

SARIMA

5.1 Mean trends

Trend definition

Often an observed signal exhibits a trend. This is a tendency to increase or decrease over time. There may also be fluctuations over time. This model is given by

$$X_t = \mu_t + Y_t,$$

where μ_t is a time-dependent mean, and Y_t is a stationary process, for example $\mu_t = a + bt$.

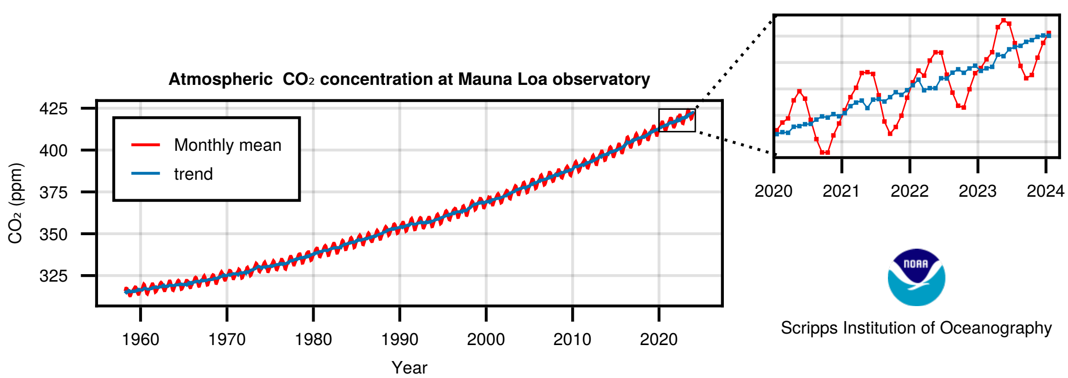


Figure 5.1: Example of a time series with a trend and seasonality.

Linear trend removal

There are two common approaches to trend adjustment:

1. Estimate a and b and remove the trend. We could use least squares regression, and then further analyse the residuals.

2. Take first differences, writing the difference operator $\nabla = (\mathrm{I} - \mathrm{B})$:

$$\begin{aligned}\nabla X_t &= X_t - X_{t-1} \\ &= a + bt + Y_t - (a + b(t-1) + Y_{t-1}) \\ &= b + Y_t - Y_{t-1} \\ &= b + \nabla Y_t.\end{aligned}$$

   Thus we have got rid of the trend but we are left with the constant b, and also ∇Y_t rather than Y_t.

- In fact, the first difference of a stationary process is stationary, so if Y_t was stationary then so is ∇Y_t.

- If we difference again then we arrive at

$$\begin{aligned}\nabla^2 X_t &= \nabla(b + \nabla Y_t) \\ &= \nabla Y_t - \nabla Y_{t-1} \\ &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{t-2}.\end{aligned}$$

  Thus the effect of μ_t has been completely removed from the observations.

- Let $X_t = Y_t + \mu_t$ where
  - Y_t is stationary,
  - μ_t is a $(q-1)^{\text{th}}$ degree polynomial.

- If μ_t is a $(q-1)^{\text{th}}$ degree polynomial in t then the q^{th} difference of μ_t will be zero. Therefore

$$\begin{aligned}\nabla^q X_t &= \nabla^q Y_t + \nabla^q \mu_t \\ &= \sum_{k=0}^{q} \binom{q}{k} (-1)^k Y_{t-k},\end{aligned}$$

Periodic trend removal

Sometimes we assume

$$X_t = s_t + Y_t,$$

and s_t is the periodic deterministic function. Y_t is assumed a zero-mean stationary process.

Because ∇ is a linear operator.

Periodic and linear trend removal

Let $\{Y_t\}$ be a stationary process with mean zero and let a and b be fixed constants. X_t is given by

$$X_t = a + bt + s_t + Y_t,$$

where s_t is a deterministic function with period 12, i.e. $s_{t+12} = s_t$. By forming the seasonal difference $\nabla^{(s)} X_t = X_t - X_{t-s}$ as well as the standard difference $\nabla X_t = X_t - X_{t-1} = \nabla^{(1)} X_t$, we can form $\nabla\nabla^{(12)} X_t$. Let us show that this is weakly stationary.

- First note that

$$\begin{aligned} \nabla^{(12)} X_t &= \nabla^{(12)} bt + \nabla^{(12)} Y_t \\ &= bt - b(t-12) + \nabla^{(12)} Y_t = 12b + \nabla^{(12)} Y_t \\ \nabla\nabla^{(12)} X_t &= \nabla\nabla^{(12)} Y_t \\ &= \nabla(Y_t - Y_{t-12}) = Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}. \end{aligned} \tag{5.1}$$

- We can calculate

$$\mathbb{E}\left[\nabla\nabla^{(12)} X_t\right] = 0$$

$$\begin{aligned} \operatorname{Cov}\left(\nabla\nabla^{(12)} X_t, \nabla\nabla^{(12)} X_{t-\tau}\right) = \gamma_\tau - \gamma_{\tau+1} - \gamma_{\tau+12} + \gamma_{\tau+13} \\ - \gamma_{\tau-1} + \gamma_\tau + \gamma_{\tau+11} - \gamma_{\tau+12} \\ - \gamma_{\tau-12} + \gamma_{\tau-11} + \gamma_\tau - \gamma_{\tau+1} \\ + \gamma_{\tau-13} - \gamma_{\tau-12} - \gamma_{\tau-1} + \gamma_\tau. \end{aligned}$$

  This is a function of τ only and so we have a weakly stationary process.

Interaction between periodic and linear trend components

- Now instead set

$$X_t = (a + bt)s_t + Y_t,$$

  where s_t is a deterministic function with period 12, i.e. $s_{t+12} = s_t$. Now instead show that $\nabla^{(12)2} X_t$ is stationary. We find that

$$\begin{aligned} \nabla^{(12)} X_t &= (a + bt)s_t + Y_t - (a + b(t-12))s_{t-12} + Y_{t-12} \\ &= 12bs_{t-12} + Y_t - Y_{t-12} \end{aligned}$$

$$\begin{aligned} \nabla^{(12)}\nabla^{(12)} X_t &= 12bs_{t-12} + Y_t - Y_{t-12} - (12bs_{t-24} + Y_{t-12} - Y_{t-24}) \\ &= Y_t - 2Y_{t-12} + Y_{t-24} \end{aligned}$$

  The expectation of this is clearly zero, and the covariance is clearly just a function of $|\tau|$.

5.2 Seasonal modelling: SARMA

Seasonality

Many econometric and financial processes have seasonal features ($s = 4$ for quarterly data, $s = 12$ for monthly data etc). Consider the following model:

$$X_t = \varepsilon_t - \theta\varepsilon_{t-s}$$

- We can derive the autocorrelation sequence for this process

$$\begin{aligned}\gamma_\tau &= \operatorname{Cov}(X_t, X_{t+\tau}) \\ &= \operatorname{Cov}(\varepsilon_t - \theta\varepsilon_{t-s}, \varepsilon_{t+\tau} - \theta\varepsilon_{t+\tau-s}) \\ &= \sigma_\varepsilon^2\delta_\tau - \theta\sigma_\varepsilon^2\delta_{\tau-s} - \theta\sigma_\varepsilon^2\delta_{\tau+s} + \theta^2\sigma_\varepsilon^2\delta_\tau \\ &= \sigma_\varepsilon^2\left(\left(1+\theta^2\right)\delta_\tau - \theta\left(\delta_{\tau-s} + \delta_{\tau+s}\right)\right)\end{aligned}$$

- This is a seasonal moving average of order 1.

Definition 5.1 (Seasonal moving average)**.** A general seasonal moving average process $\mathrm{SMA}(Q)_s$ takes the form

$$X_t = \varepsilon_t - \sum_{j=1}^{Q} \theta_j^{(s)} \varepsilon_{t-js}$$

where $\{\varepsilon_t\}$ is a mean-zero white noise process.

- We have the seasonal MA polynomial given by

$$\Theta^{(s)}(z) = 1 - \sum_{j=1}^{Q} \theta_j^{(s)} z^{sj}.$$

- So we could write

$$X_t = \Theta^{(s)}(\mathrm{B})\varepsilon_t.$$

Definition 5.2 (Seasonal autoregression)**.** A general seasonal autoregressive process $\mathrm{SAR}(P)_s$ takes the form

$$X_t = \varepsilon_t + \sum_{j=1}^{P} \phi_j^{(s)} X_{t-js}$$

where $\{\varepsilon_t\}$ is a mean-zero white noise process.

- We have the seasonal AR polynomial given by

$$\Phi^{(s)}(z) = 1 - \sum_{j=1}^{P} \phi_j^{(s)} z^{sj}.$$

- So we could write

$$\Phi^{(s)}(\mathrm{B})X_t = \varepsilon_t.$$

Definition 5.3 (SARMA)**.** We can combine the SAR and SMA with a standard ARMA to get a SARMA$(p, q) \times (P, Q)_s$, i.e.

$$\Phi(\mathrm{B})\Phi^{(s)}(\mathrm{B})X_t = \Theta(\mathrm{B})\Theta^{(s)}(\mathrm{B})\varepsilon_t$$

where $\{\varepsilon_t\}$ is a mean-zero white noise process.

- We have the following relations
  1. SMA$(Q)_s$ is SARMA$(0,0) \times (0,Q)_s$.
  2. SAR$(P)_s$ is SARMA$(0,0) \times (P,0)_s$.
- Example: SARMA$(0,1) \times (0,1)_{12}$:

$$X_t = (1 - \theta\mathrm{B})\left(1 - \theta^{(12)}\mathrm{B}^{12}\right)\varepsilon_t.$$

5.3 ARIMA

Motivation

Consider starting from the AR(1) process

$$X_t = \phi X_{t-1} + \varepsilon_t.$$

- Stationarity of this process depends on $|\phi| < 1$.
- If we take $\phi = 1$ then the process is a random walk (which is not stationary):

$$X_t = X_{t-1} + \varepsilon_t$$

- However, the difference $\nabla X_t = \varepsilon_t$ is stationary.

Definition 5.4. We say that a process is an ARIMA(p, d, q) if the d^{th} difference is an ARMA(p, q). In other words

$$\Phi(\mathrm{B})(1 - \mathrm{B})^d Y_t = \Theta(\mathrm{B})\varepsilon_t$$

- If we take $Z_t = \nabla^d Y_t$ then

$$Z_t = \sum_{j=1}^{p} \phi_j Z_{t-j} + \varepsilon_t - \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

- The first popular model we will study is the simple IMA $(1,1)$ model. We take Z_t as the difference $Y_t - Y_{t-1}$,

$$Z_t = \varepsilon_t - \theta\varepsilon_{t-1}$$

- Then recalling the link between Y_t and Z_t we may write this as

$$Y_t = Y_{t-1} + \varepsilon_t - \theta\varepsilon_{t-1}$$

- This is NOT a stationary process.

- Assuming the process starts at $t = 0$ at $Y_0 = 0$ then

$$\begin{aligned} Y_1 &= 0 + \varepsilon_1 - \theta\varepsilon_0 \\ Y_2 &= Y_1 + \varepsilon_2 - \theta\varepsilon_1 \\ &= 0 + \varepsilon_1 - \theta\varepsilon_0 + \varepsilon_2 - \theta\varepsilon_1 \\ &= \cdots \\ Y_t &= \varepsilon_t + (1-\theta)\sum_{j=1}^{t-1}\varepsilon_j - \theta\varepsilon_0. \end{aligned}$$

From this equation we can now find the variance of the process of interest:

$$\begin{aligned} \operatorname{Var}(Y_t) &= \sigma_\varepsilon^2 + (1-\theta)^2\sum_{j=1}^{t-1}\sigma_\varepsilon^2 + \theta^2\sigma_\varepsilon^2 \\ &= \sigma_\varepsilon^2\left\{1 + (1-\theta)^2(t-1) + \theta^2\right\} \end{aligned}$$

and the covariances (for $\tau > 0$) can also be calculated from this:

$$\operatorname{Cov}(Y_t, Y_{t+\tau}) = \sigma_\varepsilon^2\left\{1 - \theta + (1-\theta)^2(t-1) + \theta^2\right\}.$$

As $\tau/t \to 0$, $\operatorname{Corr}(Y_t, Y_{t+\tau})$ tends to 1.

- The ARI$(1,1)$ process is defined from a stationary AR process.
- The ARI$(1,1)$ process is defined as $(|\phi| < 1)$,

$$Z_t = \phi Z_{t-1} + \varepsilon_t$$

- Then recalling the link between Y_t and Z_t we may write this as

$$Y_t - Y_{t-1} = \phi\left(Y_{t-1} - Y_{t-2}\right) + \varepsilon_t$$

- Again assuming the process starts at $t = 0$ with $Y_0 = 0$ then

$$\begin{aligned} Y_1 &= \varepsilon_1 \\ Y_2 &= \varepsilon_2 + (1+\phi)\varepsilon_1 \\ \ldots &= \ldots \\ Y_t &= \sum_{j=1}^{t}\sum_{i=0}^{t-j}\phi^i\varepsilon_j = \sum_{j=1}^{t}\frac{1-\phi^{t-j+1}}{1-\phi}\varepsilon_j \end{aligned}$$

- We use this to determine the second order properties of the process.

$$\operatorname{Var}(Y_t) = \sigma_\varepsilon^2\sum_{j=1}^{t}\left(\frac{1-\phi^{t-j+1}}{1-\phi}\right)^2.$$

- We can also determine that when $\tau > 0$

$$\operatorname{Cov}(Y_t, Y_{t+\tau}) = \sigma_\varepsilon^2\sum_{j=1}^{t}\left(\frac{1-\phi^{t-j+1}}{1-\phi}\right)\left(\frac{1-\phi^{t+\tau-j+1}}{1-\phi}\right)$$

- Again the correlation will be near to unity.

Chapter 6

Fourier theory

6.1 Continuous time Fourier transform

The Fourier transform (continuous time)

Definition 6.1. Consider some function $g : \mathbb{R} \to \mathbb{C}$. Assume further that $g \in L^1$ then the Fourier transform of g, denoted by $\mathscr{G}$ is

$$\mathscr{G}(f) = \int_{-\infty}^{\infty} g(t)e^{-2\pi ift}\mathrm{d}t \tag{6.1}$$

for all $f \in \mathbb{R}$.

- Note the notational convention that upper case denotes the Fourier transform, other conventions exist such as $\hat{g}$.
- (This is not the only condition under which the Fourier transform can be defined, but it enough for our purposes.)

Inverse transform

Theorem 6.1. *Consider again the function $g : \mathbb{R} \to \mathbb{C}$, and assume that both $g, \mathscr{G} \in L^1$. Then the inverse Fourier transform of $\mathscr{G}$ recovers g, in that*

$$g(t) = \int_{-\infty}^{\infty} \mathscr{G}(f)e^{2\pi ift}\mathrm{d}f \tag{6.2}$$

for almost all $t \in \mathbb{R}$.

- Again, this result can be extended to a much broader class of functions, but this is not necessary for our purposes.
- The existence of such an inverse leads to the notion of a Fourier pair (a function and its Fourier transform), often denoted by

$$g \longleftrightarrow \mathscr{G}.$$

Basic properties of the Fourier transform (continuous time)

Let $g, h \in L^1$. The following table lists some common simple relationships.

Time	Frequency
$g(t)^*$	$\mathscr{G}(-f)^*$
$g(\alpha t) \quad (\alpha \neq 0)$	$\mathscr{G}(f/\alpha)/\lvert\alpha\rvert$
$g(t+\tau)$	$\mathscr{G}(f)e^{2\pi i\tau f}$
$\alpha g(t) + h(t)$	$\alpha\mathscr{G}(f) + \mathscr{H}(f)$

- Note that for notational convenience, we have committed a slight abuse of notation and used $g(t)$ and $\mathscr{G}(f)$ to refer to the functions.

Convolution (continuous time)

One of the most important results of Fourier theory we will make use of in this course is the convolution theorem. First we need the notion of a convolution.

Definition 6.2. Consider $h, g \in L^1$, then $u = g * h$ denotes the convolution

$$u(t) = \int_{-\infty}^{\infty} g(t-u)h(u)\mathrm{d}u \tag{6.3}$$

for all $t \in \mathbb{R}$.

- Again, the convolution will be defined for a broader class of functions, but for us this is sufficient.
- Notice that convolution at a given point in time is essentially taking a weighted average of one function by the other centred at that point

The convolution theorem (continuous time)

Theorem 6.2. *Consider $h, g \in L^1$, and let $u = g * h$, then*

$$\mathscr{U} = \mathscr{G} \cdot \mathscr{H}. \tag{6.4}$$

In other words, the Fourier transform of a convolution of two functions is the product of their Fourier transforms.

Corollary 6.3. *If we also assume that $\mathscr{G}, \mathscr{H} \in L^1$, and let $v = g \cdot h$, then we also have*

$$\mathscr{V} = \mathscr{G} * \mathscr{H}. \tag{6.5}$$

6.2 Discrete time Fourier transform

The Fourier transform (discrete time)

We have discussed the Fourier transform of the continuous time, however, we are usually working with functions of a discrete domain.

Definition 6.3. Consider a sequence $\{g_t\}_{t\in\Delta\mathbb{Z}}$ in ℓ^1, then the discrete Fourier transform is given by

$$G(f) = \Delta \sum_{t\in\Delta\mathbb{Z}} g_t e^{-2\pi i t f} \tag{6.6}$$

for all $f \in \mathbb{R}$.

Inverse transform

Theorem 6.4. *Consider the case that $g \in \ell^1$, then*

$$g_t = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i t f} \mathrm{d}f \tag{6.7}$$

for all $t \in \Delta\mathbb{Z}$.

- Notice that we really have a Fourier series, just with the domains flipped.

Properties of the Fourier transform (discrete time)

Let $g, h \in \ell^1$. Analogously to the continuous time case, we have similar properties Additionally, we

Time	Frequency
g_t^*	$G(-f)^*$
$g_{\alpha t}$ $(\alpha \neq 0)$	$G(f/\alpha)/\lvert\alpha\rvert$
$g_{t+\tau}$	$G(f)e^{2\pi i \tau f}$
$\alpha g_t + h_t$	$\alpha G(f) + H(f)$

have that for any $k \in \mathbb{Z}$

$$G(f) = G(f + k/\Delta), \tag{6.8}$$

for all $f \in \mathbb{R}$.

Convolution (discrete time)

Definition 6.4 (Discrete time convolution)**.** Again we have the notion of a convolution, so that $u = g * h$ if

$$u_t = \Delta \sum_{u\in\Delta\mathbb{Z}} g_{t-u} h_u \tag{6.9}$$

for all $t \in \Delta\mathbb{Z}$.

Definition 6.5 (Periodic convolution)**.** We can also define the notation of convolution in frequency, i.e. $U = G * H$ taken to mean

$$U(f) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f - f')H(f')\mathrm{d}f' \tag{6.10}$$

The convolution theorem (discrete time)

Theorem 6.5. *Consider $g, h \in \ell^1$, then we have that both*

$$h \cdot g \longleftrightarrow H * G, \tag{6.11}$$

$$h * g \longleftrightarrow H \cdot G. \tag{6.12}$$

- Recall that $\longleftrightarrow$ indicates that two functions/sequences form a Fourier pair.

Relation to continuous time case

We have now discussed both the continuous and discrete time cases. We did this because:

1. We often believe the process of interest is actually evolving over continuous time,
2. but we can only make observations in discrete time.

We now need to understand the relation between the two. In particular, the impact of sampling a continuous function at discrete time points, namely aliasing.

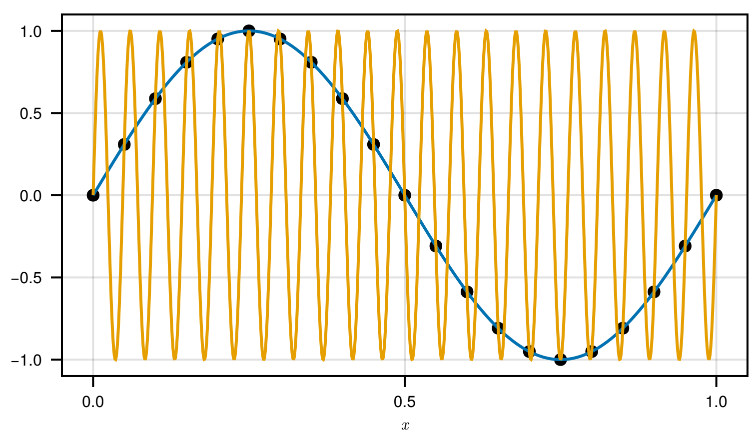


Figure 6.1: A sin function sampled at discrete time points

Aliasing formally

Theorem 6.6. *Consider a continuous function $g \in L^1$. Then writing $\mathscr{G}$ for its Fourier transform and G for the Fourier transform of the sequence $\{g(t)\}_{t\in\Delta\mathbb{Z}}$ and assuming that $\mathscr{G} \in L^1$ and $\{g(t)\}_{t\in\Delta\mathbb{Z}} \in \ell^1$ and the sequence $\{\mathscr{G}(f + k/\Delta)\}_{k\in\mathbb{Z}} \in \ell^1$ for all $f \in \mathbb{R}$, then*

$$G(f) = \sum_{k\in\mathbb{Z}} \mathscr{G}(f + k/\Delta). \tag{6.13}$$

- Therefore, the true Fourier transform will not be recoverable in general.
- This phenomenon is called aliasing, because the value of the function at frequencies $\{f + k/\Delta \mid k \in \mathbb{Z}\}$ are all "given the alias" of frequency f.

6.3 Finite data Fourier transform

Finite data Fourier transform

The discussion so far has assumed that we have access to the function as a whole. Of course, this is often not true. In practice, we will have a finite number of observations, say at times $T = \{0, \Delta, \ldots, (n-1)\Delta\}$.

Definition 6.6. Say we observe $g_0, \ldots, g_{(n-1)\Delta}$, then we define the finite Fourier transform as

$$G_n(f) = \sum_{t\in T} g_t e^{-2\pi i t f} \tag{6.14}$$

for $f \in \mathbb{R}$.

- Whilst this is defined for all frequencies, it is typically evaluated at the Fourier frequencies. We will see them in greater detail later in this lecture.

Relation to discrete time case

- We would like to understand how this relates to the Fourier transform of the original sequence $\{g_t\}_{t\in\Delta\mathbb{Z}}$.
- In other words, the relationship between $G_n(f)$, and the Fourier transform of the whole sequence $G(f)$.

Notice that G_n is equivalent to the Fourier transform of the whole sequence multiplied by an indicator function. In other words,

$$\begin{aligned} G_n(f) &= \sum_{t\in T} g_t e^{-2\pi i t f} \\ &= \Delta \sum_{t\in\Delta\mathbb{Z}} g_t \frac{1}{\Delta} \mathbb{1}_T(t) e^{-2\pi i t f} \\ &= \Delta \sum_{t\in\Delta\mathbb{Z}} g_t d_t e^{-2\pi i t f} \end{aligned}$$

where for $t \in \mathbb{Z}$, $d_t = \frac{1}{\Delta}\mathbb{1}_T(t)$.

Finite data transform as a convolution

Given the Fourier pair we have constructed,

$$G_n(f) \longleftrightarrow g_t d_t \tag{6.15}$$

we can now utilise properties of Fourier transforms to relate G_n and G. In particular, if $g \in \ell^1$

$$G_n(f) = \int_{-1/(2\Delta)}^{1/(2\Delta)} D_T(f - f')G(f')\mathrm{d}f' \tag{6.16}$$

where D_T is the Fourier transform of d.

Blurring

A result of this convolution, is that the true Fourier transform is "blurred". To see why, consider the Fourier transform of the indicator above:

$$\begin{aligned} D_T(f) &= \Delta \sum_{t \in \Delta\mathbb{Z}} \frac{\mathbb{1}_T(t)}{\Delta} e^{-2\pi i f t} \\ &= \sum_{t \in T} e^{-2\pi i f t} \\ &= \frac{\sin(n\pi f\Delta)}{\sin(\pi f \Delta)} e^{-\pi i(n-1)f\Delta}. \end{aligned} \tag{6.17}$$

This is the Dirichlet kernel (or at least one form of it).

Fast Fourier transform

Part of the benefit of working in the Frequency domain is that the finite Fourier transform can be computed very efficiently using the Fast Fourier Transform (FFT).

- Typically one computes the finite Fourier at a set of n frequencies.
- Thus naively we would require $O(n^2)$ computations.
- However, the FFT enables this to be performed in $O(n \log n)$ time.
- The FFT returns the Fourier transform at the Fourier frequencies:

$$\frac{k}{\Delta n} \qquad \text{where} \qquad k \in \mathbb{Z} \text{ and } -\frac{n}{2} \leq k \leq \frac{n-1}{2}.$$

Chapter 7

Frequency domain time series

7.1 Spectral density (simple case)

The spectral density function

For simplicity, for the remainder of the course we consider only $\Delta = 1$.

Definition 7.1. Consider a discrete time process $\{X_t\}_{t\in\mathbb{Z}}$, with autocovariance sequence satisfying $\{\gamma_\tau\} \in \ell^1$. We define the spectral density function to be

$$S\left(f\right) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau e^{-2\pi i\tau f} \tag{7.1}$$

for all $f \in \mathbb{R}$.

- This is simply the discrete Fourier transform of the autocovariance sequence.
- The subscript indicates that this is the spectral density function of the discrete time process.

The spectral density function (continuous time)

Now consider the continuous time case.

Definition 7.2. Consider a continuous time process $\{X(t)\}_{t\in\mathbb{R}}$, with autocovariance function $\gamma \in L^1$, then the spectral density function is

$$\mathscr{S}\left(f\right) = \int_{-\infty}^{\infty} \gamma\left(\tau\right) e^{-2\pi i\tau f} \mathrm{d}\tau \tag{7.2}$$

for $f \in \mathbb{R}$.

7.1.1 Continuous vs discrete

Aliasing

- If we sample a continuous time process at discrete points, then clearly the autocovariance sequence of the sampled process is the autocovariance function of the original process sampled at those same points.

- From the previous lecture, we therefore know that there is an aliasing relation between the spectral density functions of the two respective processes.
- In particular

$$S(f) = \sum_{k \in \mathbb{Z}} \mathscr{S}(f + k). \tag{7.3}$$

- The frequency $\frac{1}{2}$ is called the Nyquist frequency.
- If $\mathscr{S}(f)$ is essentially zero for $|f| \geq \frac{1}{2}$ we can expect a good correspondence between $\mathscr{S}(f)$ and $S(f)$.
- If $\mathscr{S}(f)$ is large for $|f| \geq \frac{1}{2}$ then the correspondence is poor so $S(f)$ tells us nothing about $\mathscr{S}(f)$.

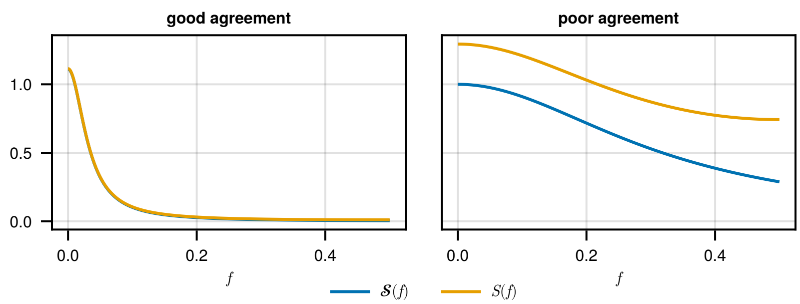


Figure 7.1: Aliasing for two different continuous processes

Harmonic processes

Definition 7.3 (Harmonic (Sinusoidal) process)**.** Consider two independent random variables $A > 0$ and $\Theta \sim \mathrm{Unif}(-\pi, \pi)$. We construct a harmonic process $\{X_t\}$ by setting

$$X_t = A\cos(\nu t + \Theta), \tag{7.4}$$

for all $t \in \mathbb{Z}$ where $\nu \in \mathbb{R}$ determines the frequency of the oscillation.

- We can see that

$$\mathbb{E}[X_t] = 0,$$
$$\mathrm{Cov}(X_{t+\tau}, X_t) = \frac{1}{2}\mathbb{E}\left[A^2\right]\cos(\nu\tau).$$

- This process is stationary, but clearly does not have $\gamma_\tau \in \ell^1$, can we still have a notion of spectra in this case?

7.2 Spectra for general stationary processes

Herglotz's theorem

Theorem 7.1 (Herglotz's theorem)**.** *Given a complex valued sequence $\{g_t\}_{t\in\mathbb{Z}}$, then g is non-negative definite if and only if for all $t \in \mathbb{Z}$*

$$g_t = \int_{-1/2}^{1/2} e^{2\pi i t f} \mathrm{d}G^{(I)}(f) \tag{7.5}$$

where $G^{(I)}(-1/2) = 0$, and $G^{(I)}$ is right continuous, bounded and non-decreasing.

- This can be generalised in a number of ways, but we require only complex valued sequences here.
- Note the properties are essentially the same as a cumulative distribution function.

Integrated spectrum

As a result of Herglotz's theorem, and that the autocovariance sequence is non-negative definite, we have that

$$\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i \tau f} \mathrm{d}S^{(I)}(f) \tag{7.6}$$

- $S^{(I)}(\cdot)$ is referred to as the integrated spectrum, or sometimes the spectral distribution function.
- We have the immediate properties:
  (i) $S^{(I)}\left(-\frac{1}{2}\right) = 0$ and $S^{(I)}\left(\frac{1}{2}\right) = \sigma^2$.
  (ii) $f < f' \Rightarrow S^{(I)}(f) \leq S^{(I)}(f')$.

Relation to the spectral density function

- If there is a function $S(\cdot)$ such that

$$S^{(I)}(f) = \int_{-1/2}^{f} S(\lambda)\,\mathrm{d}\lambda \tag{7.7}$$

  then $S(\cdot)$ is called the spectral density function.
- This occurs if the autocovariance function is absolutely summable, in which case this is equivalent to the definition of the spectral density function seen earlier in the lecture.
- In various books people use $\omega = 2\pi f$ instead of f. This is called angular frequency rather than frequency. I don't like it, because it causes factors of 2π to fly around all over the place, that are easily missed/forgotten, but it is equivalent.
- Intuitively, $S(f)\,\mathrm{d}f$ is the average contribution over all realisations of the process to the power from components with frequencies in a small interval around f. The variance or "power" of $\{X_t\}$ is

$$\mathrm{Var}(X_t) = \gamma_0 = \int_{-\frac{1}{2}}^{\frac{1}{2}} e^{2\pi i f\cdot 0} S(f)\,\mathrm{d}f.$$

Lebesgue decomposition theorem

Theorem 7.2 (Lebesgue decomposition theorem)**.** *Any integrated spectrum $S^{(I)}(\cdot)$ can be written as*

$$S^{(I)}(\cdot) = S_1^{(I)}(\cdot) + S_2^{(I)}(\cdot) + S_3^{(I)}(\cdot),$$

where $S_j^{(I)}(\cdot)$ non-negative, non-decreasing with $S_j^{(I)}\left(-\frac{1}{2}\right) = 0$ for $j = 1, 2, 3$ and where $S_1^{(I)}(\cdot)$ is absolutely continuous, $S_2^{(I)}(\cdot)$ is a step function and $S_3^{(I)}(\cdot)$ is a continuous singular function.

- This is actually a more general result, but we state it here for our specific case.
- Now we will turn to the meaning of the different components $S_j^{(I)}(\cdot)$.

Lebesgue decomposition theorem: further detail

(a) $S_1^{(I)}(\cdot)$ is an absolutely continuous function and satisfies

$$S_1^{(I)}(f) = \int_{-1/2}^{f} S(\lambda)\,\mathrm{d}\lambda \tag{7.8}$$

for all $f \in [-1/2, 1/2]$.

(b) $S_2^{(I)}(\cdot)$ is a step function with jumps of size $\{p_l\}$ at the points $\{f_l\}_l$ where f_l are frequencies of pure sinusoids.

(c) $S_3^{(I)}(\cdot)$ is a continuous singular function (pathological and generally of no practical use).

Lebesgue decomposition theorem: some examples

We can then characterise some common scenarios in terms of this decomposition:

(a) This case corresponds to $S_1^{(I)}(f) \geq 0$ and $S_2^{(I)}(f) \equiv 0$. In this case we say that $\{X_t\}$ has a purely continuous spectrum. Note that as $S_1^{(I)}(f)$ is absolutely continuous and non-decreasing (often increasing). Hence its derivative $S(f)$ is absolutely integrable (Titchmarsh, 1960, for example). But note that if $S(f)$ is absolutely integrable $\int_{-1/2}^{1/2} \cos(2\pi f\tau) S(f)\,\mathrm{d}f \to 0$ $\int_{-1/2}^{1/2} \sin(2\pi f\tau) S(f)\,\mathrm{d}f \to 0$ as $\tau \to \infty$. So as $\tau \to \infty$:

$$\gamma_\tau = \int_{-1/2}^{1/2} \cos(2\pi f\tau) S(f)\,\mathrm{d}f + i\int_{-1/2}^{1/2} \sin(2\pi f\tau) S(f)\,\mathrm{d}f \to 0.$$

(b) This case corresponds to $S_1^{(I)}(f) \equiv 0$ and $S_2^{(I)}(f) \geq 0$. Here the integrated spectrum consists entirely of a step function, and the stationary process is said to have a purely discrete spectrum (or a line spectrum). In this case the ACVS does not damp down to zero.

(c) Mixed spectrum. This case corresponds to $S_1^{(I)}(f) \geq 0$ and $S_2^{(I)}(f) \geq 0$.

- Example case (a): ARMA processes in general.
- Example case (b): sinusoid with random Phase and Amplitude.
- Example case (c): point-wise aggregation of above.

7.3 Spectral representation theorem

Complex random variables

- Note that the covariance of two complex random variables Z_1 and Z_2 is defined as

$$\mathrm{Cov}\left(Z_1, Z_2\right) = \mathbb{E}\left[\left(Z_1 - \mathbb{E}\left[Z_1\right]\right)\left(Z_2 - \mathbb{E}\left[Z_2\right]\right)^*\right].$$

  where a^* denotes the complex conjugate of $a \in \mathbb{C}$.

- The complementary covariance or relation is defined as

$$\mathrm{Rel}\{Z_1, Z_2\} = \mathbb{E}\left[(Z_1 - EZ_1)(Z_2 - EZ_2)\right].$$

Theorem 7.3 (Spectral representation theorem)**.** *Let $\{X_t\}$ be a real-valued discrete time stationary process with mean μ. Then there exists an orthogonal increment process $\{Z(f)\}$ on $\left[-\frac{1}{2}, \frac{1}{2}\right]$ such that*

$$X_t = \mu + \int_{-\frac{1}{2}}^{\frac{1}{2}} e^{2\pi i f t} \mathrm{d}Z(f). \tag{7.9}$$

This equality holds in the mean–square sense. The process $\{Z(f)\}$ has properties for $f, f' \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

1. $\mathbb{E}\left[\mathrm{d}Z(f)\right] = 0$,
2. $\mathrm{Var}\left(\mathrm{d}Z(f)\right) = \mathrm{d}S^{(I)}\left(f\right)$,
3. $\mathrm{Cov}\left(\mathrm{d}Z(f), \mathrm{d}Z(f')\right) = 0$ *if* $f \neq f'$.

Chapter 8

Frequency domain estimation

8.1 The periodogram

Spectral estimation

In this lecture, we will develop methodology to estimate the spectral density function from some observed time series. In other words, say we have a finite set of equally spaced time points

$$T = \{1, \ldots, n\},$$

and we observe X_t for all $t \in T$, then how do we estimate $S(f)$?

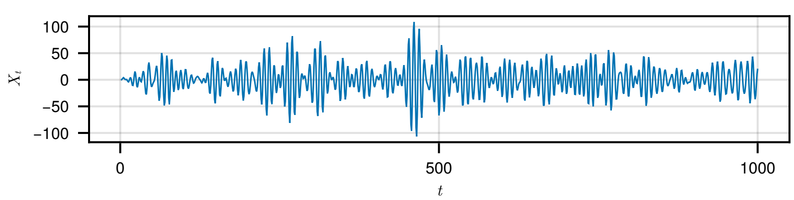


Figure 8.1: Simulated AR(4) time series with $n = 1000$

Recall the relationship

$$S(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau}.$$

We can therefore produce an estimator of $S(f)$ from $\hat{\gamma}_\tau$ by a plug–in estimator.

Definition 8.1 (The periodogram)**.** The periodogram is defined as

$$\widehat{S}^{(p)}(f) = \sum_{\tau \in \mathbb{Z}} \hat{\gamma}_\tau e^{-2\pi i f \tau} \tag{8.1}$$

We set $\sigma^2 = 1$, $\phi = [2.7607, -3.8106, 2.6535, -0.9238]$, replicating an example from Percival and Walden (1993).

where $\{\hat{\gamma}_\tau\}_{\tau\in\mathbb{Z}}$ is given by

$$\hat{\gamma}_\tau = \begin{cases} \frac{1}{n}\sum_{t=1}^{n-|\tau|}(X_t - \bar{X})(X_{t+|\tau|} - \bar{X}) & \text{if } |\tau| \leq n-1, \\ 0 & \text{otherwise.} \end{cases}$$

Relation to the finite Fourier transform of the data

Proposition 8.1. *The periodogram can be written in terms of the finite Fourier transform of the observed data, in that*

$$\widehat{S}^{(p)}(f) = |J(f)|^2, \tag{8.2}$$

where recalling $T = \{1, \ldots, n\}$, we define

$$J(f) = \sqrt{\frac{1}{n}}\sum_{t\in T}\left(X_t - \bar{X}\right)e^{-2\pi itf}. \tag{8.3}$$

- See the exercise sheet for a proof.
- $J(f)$ can be computed efficiently using the Fast Fourier Transform.
- In fact, one can compute the sample autocovariance efficiently by simply inverse transforming the periodogram!

Properties of the periodogram

Ideally as an estimator we would have

1. $\mathbb{E}\left[\widehat{S}^{(p)}(f)\right] = S(f)$,
2. $\mathrm{Var}\left(\widehat{S}^{(p)}(f)\right) \to 0$ as $n \to \infty$,
3. $\mathrm{Cov}\left(\widehat{S}^{(p)}(f), \widehat{S}^{(p)}(f')\right) = 0$ for $f \neq f'$.

However instead we find that

1. is approximately valid.
2. is false.
3. holds approximately if f and f' have a particular form (the Fourier frequencies).

We will deal with each of these in turn.

Expectation

Since the periodogram is a finite sum of the sample autocovariance, (and assuming zero-mean) we see

$$
\begin{aligned}
\mathbb{E}\left[\widehat{S}^{(p)}(f)\right] &= \sum_{\tau\in\mathbb{Z}} \mathbb{E}\left[\hat{\gamma}_\tau\right] e^{-2\pi i\tau f} \\
&= \sum_{\tau\in\mathbb{Z}} \left(1-\frac{|\tau|}{n}\right) \mathbb{1}_{[-n,n]}(\tau)\gamma_\tau e^{-2\pi i\tau f} \\
&= \sum_{\tau\in\mathbb{Z}} w_\tau \gamma_\tau e^{-2\pi i\tau f}
\end{aligned}
$$

where

$$
w_\tau = \left(1-\frac{|\tau|}{n}\right) \mathbb{1}_{[-n,n]}(\tau)
$$

is a weight sequence, which results in bias.

Example

Consider an example of an AR(4) model. Below is a comparison between the true spectral density and the periodogram from a simulated series with $n = 1000$ points. We see a substantial discrepancy here.

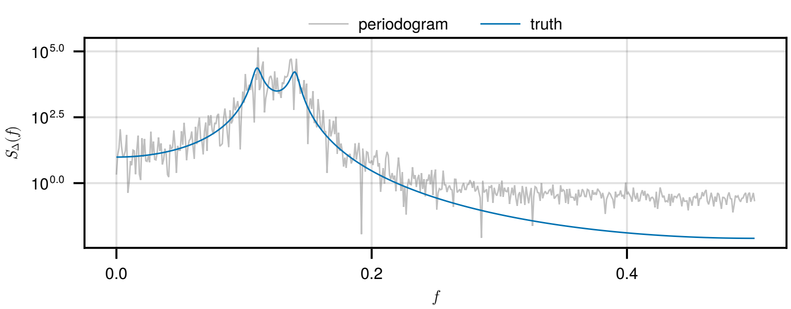


Figure 8.2: The periodogram of a simulated AR(4) model vs the true spectral density function, recreated from Percival and Walden (1993).

Expectation as a convolution

Since we are interested in the spectral density function, it may be useful to rewrite this expectation in the frequency domain. Since we have a multiplication in time, this becomes a convolution in frequency so that

$$
\mathbb{E}\left[\widehat{S}^{(p)}(f)\right] = \int_{-\frac{1}{2}}^{\frac{1}{2}} S(f')\,\mathcal{F}_n(f-f')\,\mathrm{d}f' \tag{8.4}
$$

where

$$\mathcal{F}_n(f) = \sum_{\tau \in \mathbb{Z}} w_\tau e^{-2\pi i \tau f}$$
$$= \begin{cases} \frac{1}{n} \frac{\sin^2(n\pi f)}{\sin^2(\pi f)} & \text{if } f \neq 0, \\ n & \text{if } f = 0. \end{cases}$$

The function $\mathcal{F}_n$ called the Fejér kernel.

Blurring

Examining the Fejér kernel, we see that it does not decay quickly in frequency. This causes the bias seen earlier, an effect typically called blurring.

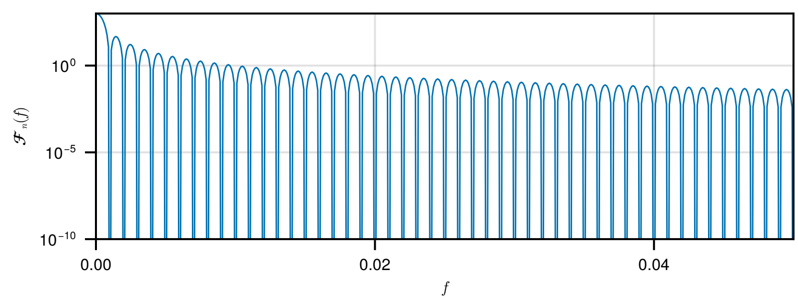


Figure 8.3: The Fejér kernel for $n = 1000$.

Variance

The second issue is the variance. In particular, one can show that (for $f \neq 0 \mod \pi$)

$$\mathrm{Var}\left(\widehat{S}^{(p)}(f)\right) \to S(f)^2 \tag{8.5}$$

as $n \to \infty$.

- So the variance does not decrease with increasing sample size.
- We cover solutions to both the bias and variance problems in the following sections.

8.2 Tapering

Tapering

One way to fix blurring is known as tapering. Here we replace the discrete Fourier transform by a tapered version

$$J_h(f) = \sum_{t \in T} h_t \cdot \left(X_t - \bar{X}\right) e^{-2\pi i t f} \tag{8.6}$$

where $\{h_t\}_{t\in T}$ is called a data taper, and satisfies

$$\|h\|_2^2 = 1.$$

Clearly setting $h_t = \sqrt{1/n}$ we recover the standard discrete Fourier transform. Now the tapered periodogram is

$$\widehat{S}_h^{(p)}(f) = |J_h(f)|^2 . \tag{8.7}$$

Effect of tapering

One can show that (see exercises)

$$\mathbb{E}\left[\widehat{S}_h^{(p)}(f)\right] = \int_{-1/2}^{1/2} S(f')\,|H(f-f')|^2\,\mathrm{d}f' \tag{8.8}$$

where H is the discrete Fourier transform of h. We therefore want to choose h so that $|H(f)|^2$ is focused around zero frequency.

One choice of such tapers are the dpss tapers, which try to optimally achieve this. This is beyond the scope of this course, but see Percival and Walden (1993) for more details.

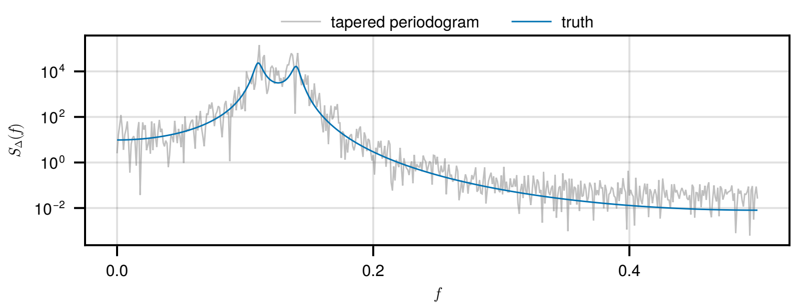


Figure 8.4: The same AR(4) example but applying a taper.

Taper kernel

In Figure 8.5, the $|H|^2$ decays faster than Fejér kernel, resulting in the decrease in bias we previously noted.

8.3 Multitapering

A family of tapers

In order to reduce variance, we need to average. Define a family of tapers for $k = 1, \ldots, K$

$$h_k = \{h_{t,k}\}_{t\in T} \tag{8.9}$$

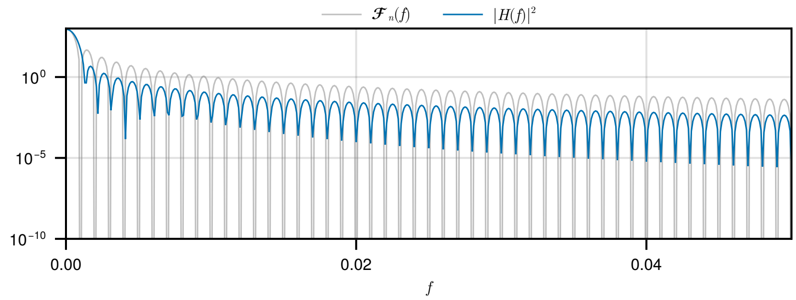


Figure 8.5: A comparison of the Fejér kernel and $|H|^2$ for a choice of taper.

which are orthogonal, i.e. for $1 \leq k, k' \leq K$

$$\sum_{t \in T} h_{t,k} h_{t,k'} = \delta_{k,k'}.$$

Note that these tapers do depend on n, but we suppress this for notational convenience.

Multitaper

The simplest multitaper estimate is

$$\widehat{S}^{(mt)}(f) = \frac{1}{K} \sum_{k=1}^{K} \widehat{S}^{(p)}_{h_k}(f). \tag{8.10}$$

We can determine its expectation:

$$\begin{aligned}
\mathbb{E}\left[\widehat{S}^{(mt)}(f)\right] &= \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}\left[\widehat{S}^{(p)}_{h_k}(f)\right] \\
&= \int_{-\frac{1}{2}}^{\frac{1}{2}} \overline{\mathcal{H}}(f - f') S(f') \, \mathrm{d}f'.
\end{aligned}$$

We call $\overline{\mathcal{H}}(f) = \frac{1}{K} \sum_{k=1}^{K} |H(f)|^2$ the average kernel.

One can show that asymptotically (see exercises)

$$\mathrm{Cov}\left(\widehat{S}^{(p)}_{h_k}(f), \widehat{S}^{(p)}_{h_{k'}}(f)\right) \to S(f)^2 \delta_{k,k'}$$

as $n \to \infty$ and therefore

$$\mathrm{Var}\left(\widehat{S}^{(mt)}(f)\right) \approx \frac{S(f)^2}{K}.$$

Chapter 9

Multivariate Time Series

9.1 Multivariate Time Series

Multivariate time series

A multivariate time series is a time series which takes values in $\mathbb{R}^d$ instead of $\mathbb{R}$. In other words, $\{\mathbf{X}_t\}$ denotes a real d-vector-valued discrete time stochastic process with

$$\mathbf{X}_t = \begin{pmatrix} X_t^{(1)} \\ X_t^{(2)} \\ \vdots \\ X_t^{(d)} \end{pmatrix}, \quad t \in \mathbb{Z},$$

where each of the marginal processes $\left\{X_t^{(1)}\right\}, \ldots, \left\{X_t^{(d)}\right\}$ are themselves univariate time series.

Stocks example

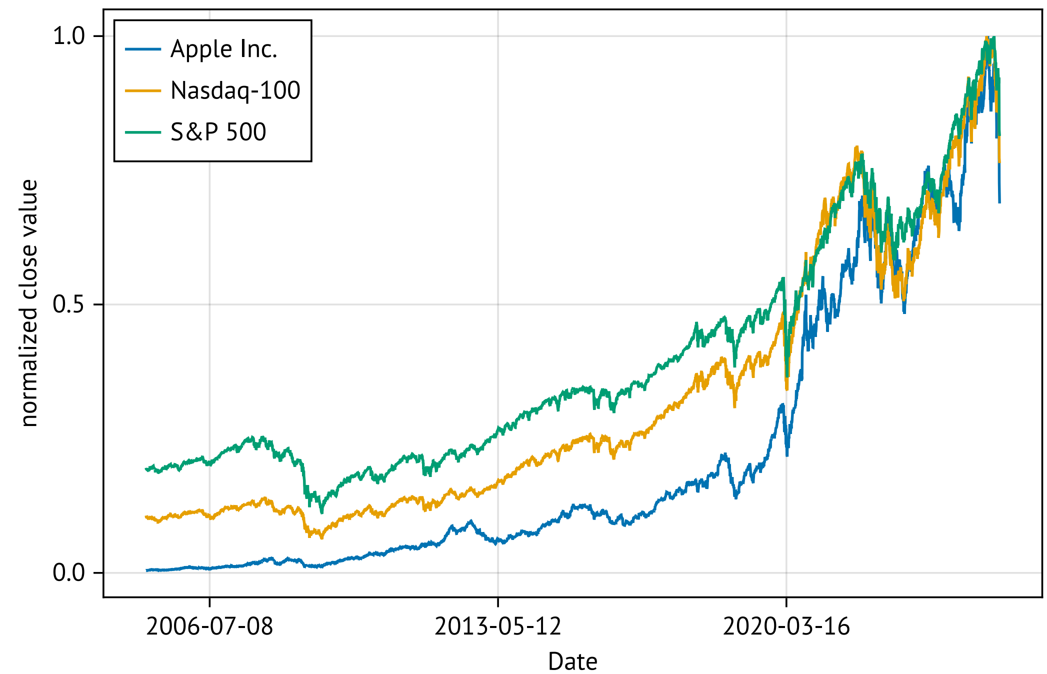
Drifter example

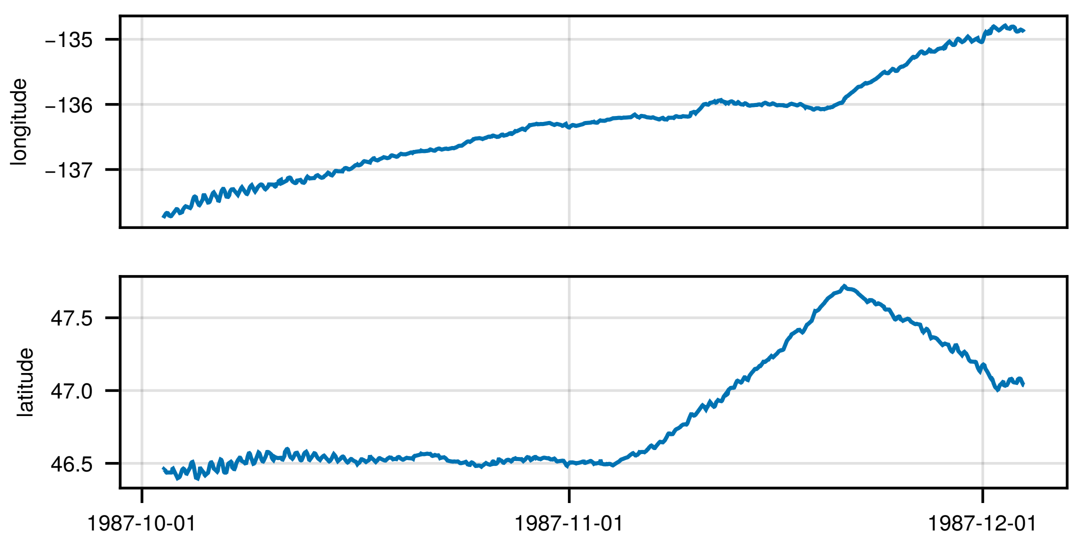
Second order stationarity

Definition 9.1 (Multivariate second-order stationary)**.** For all $t, s, \tau \in \mathbb{Z}$

$$\begin{aligned} \mathbb{E}\left[\mathbf{X}_t\right] &= \mathbb{E}\left[\mathbf{X}_s\right], \\ \operatorname{Cov}\left(\mathbf{X}_{t+\tau}, \mathbf{X}_t\right) &= \operatorname{Cov}\left(\mathbf{X}_{s+\tau}, \mathbf{X}_s\right), \\ \operatorname{trace}\left(\operatorname{Var}\left(\mathbf{X}_t\right)\right) &< \infty. \end{aligned}$$

NOAA - hourly position, current, and sea surface temperature from drifters was accessed on 8/4/25 from https://registry.opendata.aws/noaa-oar-hourly-gdp.





- Equivalently, we require that each of the univariate processes $\left\{X_t^{(1)}\right\}, \ldots, \left\{X_t^{(d)}\right\}$ are second-order stationary, and

$$\mathrm{Cov}\left(X_{t+\tau}^{(j)}, X_t^{(k)}\right) = \mathrm{Cov}\left(X_{s+\tau}^{(j)}, X_s^{(k)}\right),$$

for all $1 \leq j, k \leq d$ and for all $t, s, \tau \in \mathbb{Z}$.

Joint autocovariance sequence structure

Definition 9.2. The autocovariance sequence of a second-order stationary time series is the matrix valued sequence

$$\Gamma_\tau = \mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \mathbf{X}_t\right), \quad \tau \in \mathbb{Z}.$$

- Notice that the lag is in the first argument. This does not matter in the univariate (real-valued) case, but does matter in the multivariate case!
- The jk^{th} element of Γ_τ is $\gamma_\tau^{(j,k)} = \mathrm{Cov}\left(X_{t+\tau}^{(j)}, X_t^{(k)}\right)$.
- When $j = k$, this is the usual autocovariance sequence of the univariate process.
- When $j \neq k$, this is called the cross-covariance sequence.

Properties of the autocovariance sequence (multivariate)

- From stationarity, we can see that

$$\begin{aligned}\gamma_\tau^{(j,k)} = \mathrm{Cov}\left(X_{t+\tau}^{(j)}, X_t^{(k)}\right) &= \mathrm{Cov}\left(X_t^{(k)}, X_{t+\tau}^{(j)}\right) && \text{(symmetry)}\\ &= \mathrm{Cov}\left(X_{s-\tau}^{(k)}, X_s^{(j)}\right) && \text{(stationarity)}\\ &= \gamma_{-\tau}^{(k,j)}.\end{aligned}$$

- Therefore we have $\Gamma_{-\tau} = \Gamma_\tau^T$.
- Notice that the cross-covariance sequences need not be symmetric in τ, unlike the univariate autocovariance sequences.
- $\{\Gamma_\tau\}$ is positive semi-definite, i.e. $\forall n \in \mathbb{N}$, $\forall \boldsymbol{a}_1, \ldots, \boldsymbol{a}_n \in \mathbb{R}^d$

$$\sum_{j=1}^{n} \sum_{k=1}^{n} \boldsymbol{a}_j^T \Gamma_{k-j} \boldsymbol{a}_k \geq 0.$$

Cross-correlation sequence

Definition 9.3 (Cross-correlation sequence)**.** We define the cross-correlation sequence (CCS) $\rho_\tau^{(j,k)}$ as

$$\rho_\tau^{(j,k)} = \frac{\gamma_\tau^{(j,k)}}{\sqrt{\gamma_0^{(j,j)} \cdot \gamma_0^{(k,k)}}}$$

- For $j \neq k$, we have that

$$\rho_\tau^{(j,k)} = \rho_{-\tau}^{(k,j)}.$$

Multivariate White Noise

Definition 9.4 (Multivariate White Noise)**.** The d-variate series $\{\boldsymbol{\varepsilon}_t\}_{t\in\mathbb{Z}}$ is said to be white noise with zero-mean and covariance matrix Σ, written

$$\{\boldsymbol{\varepsilon}_t\} \sim \mathrm{WN}(\mathbf{0}, \Sigma)$$

if and only if $\{\boldsymbol{\varepsilon}_t\}$ is stationary with mean vector $\mathbf{0}$ and

$$\Gamma_\tau^{(\boldsymbol{\varepsilon})} = \begin{cases} \Sigma & \text{if } \tau = 0 \\ 0 & \text{otherwise.} \end{cases}$$

- Notice that there can be correlation between the different processes at lag 0.
- However, sometimes it is useful to assume that Σ is a diagonal matrix.

Example 9.1 (Uncorrelated processes)*.* Two jointly second-order processes $\left\{X_t^{(1)}\right\}$ and $\left\{X_t^{(2)}\right\}$ are said to be uncorrelated if

$$\gamma_\tau^{(1,2)} = 0$$

for all $\tau \in \mathbb{Z}$.

Example 9.2 (Contemporaneous correlation)*.* Say that $\mathbf{X}_t = (X_t^{(1)}, X_t^{(2)})^T$ is second-order stationary and $\{\epsilon_t\}$ is a univariate mean-zero white noise process which is independent of $\{\mathbf{X}_t\}$. Define

$$\boldsymbol{Y}_t = \mathbf{X}_t + \epsilon_t = \begin{pmatrix} X_t^{(1)} + \epsilon_t \\ X_t^{(2)} + \epsilon_t \end{pmatrix}$$

Then we have (letting $\mathbf{1}$ be the 2×2 matrix of all ones)

$$\begin{aligned} \Gamma_\tau^{(\boldsymbol{Y})} &= \mathrm{Cov}\left(\boldsymbol{Y}_{t+\tau}, \boldsymbol{Y}_t\right) \\ &= \mathrm{Cov}\left(\boldsymbol{X}_{t+\tau}, \boldsymbol{X}_t\right) + \sigma_\epsilon^2 \delta_{\tau,0}\mathbf{1} \\ &= \Gamma_\tau^{(\mathbf{X})} + \sigma_\epsilon^2 \delta_{\tau,0}\mathbf{1} \end{aligned}$$

- Note that there is therefore always lag zero correlation between $Y_t^{(1)}$ and $Y_t^{(2)}$, irrespective of the correlation structure in $\mathbf{X}_t$.

9.2 Multivariate spectra

Spectral density matrix function

Definition 9.5 (Spectral density matrix function)**.** Assume that $\left\{\gamma_\tau^{(j,k)}\right\}_{\tau\in\mathbb{Z}} \in \ell^1$ for all $1 \leq j, k \leq d$. Then the spectral density matrix function is given by

$$\boldsymbol{S}(f) = \sum_{\tau\in\mathbb{Z}} \Gamma_\tau e^{-2\pi i\tau f}, \quad f \in [-1/2, 1/2].$$

- The jk^{th} element of the spectral matrix at frequency f is denoted $S_{j,k}(f)$.
- When $j = k$ it is the usual spectral density function.
- When $j \neq k$, it is called the cross-spectral density function.
- Note that $S_{j,k}(f)$ is in $\mathbb{C}$ in general, but if $j = k$ it is in $\mathbb{R}$.

Properties of the spectral density matrix function

- We have for any $f \in [-1/2, 1/2]$, for all $1 \leq j, k \leq d$,

$$S_{j,k}(f) = S_{k,j}(f)^*, \quad S_{j,k}(f) = S_{j,k}(-f)^*.$$

- As a consequence, we get the matrix results

$$\boldsymbol{S}(f) = \boldsymbol{S}(f)^H, \quad \boldsymbol{S}(f) = \boldsymbol{S}(-f)^T$$

  for any $f \in [-1/2, 1/2]$.
- Furthermore, we have that the matrix $\boldsymbol{S}(f)$ should be positive semi-definite.
- We also have the inverse relations

$$\Gamma_\tau = \int_{-1/2}^{1/2} \boldsymbol{S}(f) e^{2\pi i f \tau} \mathrm{d}f.$$

Theorem 9.1 (Multivariate spectral representation theorem)**.** *Let $\{\mathbf{X}_t\}$ be a vector-valued discrete time stationary process with mean $\boldsymbol{\mu}$. Then there exists a vector-valued orthogonal increment process $\{\boldsymbol{Z}(f)\}$ on $\left[-\frac{1}{2}, \frac{1}{2}\right]$ such that*

$$\mathbf{X}_t = \mu + \int_{-\frac{1}{2}}^{\frac{1}{2}} e^{2\pi i f t} \mathrm{d}\boldsymbol{Z}(f). \tag{9.1}$$

Assuming that $\left\{\gamma_\tau^{(j,k)}\right\}_{\tau \in \mathbb{Z}} \in \ell^1$, then for $f, f' \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

1. $\mathbb{E}\left[\mathrm{d}\boldsymbol{Z}(f)\right] = 0$,
2. $\mathrm{Var}\left(\mathrm{d}\boldsymbol{Z}(f)\right) = \boldsymbol{S}(f)\,\mathrm{d}f$,
3. $\mathrm{Cov}\left(\mathrm{d}\boldsymbol{Z}(f), \mathrm{d}\boldsymbol{Z}(f')\right) = 0$ *if $f \neq f'$.*

- Note that $\boldsymbol{S}(f)$ here is a matrix,
- and that 0 in the third property refers to the zero matrix.

Coherence

Definition 9.6. The coherence between the j^{th} and k^{th} processes is given by

$$r_{j,k}(f) = \frac{S_{j,k}\left(f\right)}{\sqrt{S_{j,j}\left(f\right)S_{k,k}\left(f\right)}},$$

and is in essence the correlation of $\mathrm{d}Z_j$ and $\mathrm{d}Z_k$

- Coherence is a complex-valued quantity.
- Typically, we consider its magnitude or magnitude square as one statistic.
- We then look at its argument as an notion of the "sign" of the correlation (in the complex plane), which is called phase.

Example 9.3. Consider a stationary process $\{Y_t\}$, and the bivariate process $\{\mathbf{X}_t\}$ s.t.

$$\mathbf{X}_t = \begin{pmatrix} Y_t, & Y_{t-\nu} \end{pmatrix}^T, \quad t \in \mathbb{Z}$$

for some $\nu \in \mathbb{Z}$, then we have that

$$S_{1,1}\left(f\right) = S_{2,2}\left(f\right) = S_Y\left(f\right).$$

Furthermore, we have $\gamma_\tau^{(1,2)} = \mathrm{Cov}\left(Y_{t+\tau}, Y_{t-\nu}\right) = \gamma_{\tau+\nu}^{(Y)}$ and therefore

$$S_{1,2}\left(f\right) = S_Y\left(f\right)e^{2\pi i f \nu}.$$

So the coherence is

$$r_{1,2}(f) = e^{2\pi i f \nu}.$$

This has magnitude one and phase $2\pi\nu f$.

9.3 Vector Autoregression: VAR

From univariate to multivariate models

- We treated vector-valued time series in terms of pair-wise relationships.
- But how do we propose models for them?
- The simplest framework is to start from AR processes

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$$

- How would we generalize this to a vector-valued process?

Vector Autoregressive Process, VAR(p)

Definition 9.7 (Vector autoregression)**.** Let $\{\boldsymbol{\varepsilon}_t\}$ be a d-dimensional multivariate white noise process with zero-mean. A process $\{\mathbf{X}_t\}$ is called a vector auto-regressive process of order p, VAR(p), if

$$\underset{d\times 1}{\mathbf{X}_t} = \underset{d\times d}{\Phi_1}\,\underset{d\times 1}{\mathbf{X}_{t-1}} + \underset{d\times d}{\Phi_2}\,\underset{d\times 1}{\mathbf{X}_{t-2}} + \cdots + \underset{d\times d}{\Phi_p}\,\underset{d\times 1}{\mathbf{X}_{t-p}} + \underset{d\times 1}{\boldsymbol{\varepsilon}_t}$$

where the $\Phi_j \in \mathbb{R}^{d\times d}$ are matrices such that $\Phi_p \neq 0$.

- Note that we have the same regressors in the equation, namely past values of $\mathbf{X}_t$.
- Defining the polynomial $\Phi(z) = 1 - \sum_{j=1}^{p} \Phi_j z^j$, then we write

$$\Phi(\mathrm{B})\mathbf{X}_t = \boldsymbol{\varepsilon}_t.$$

Everything is a VAR(1)

Consider a VAR(p), $\{\mathbf{X}_t\}$, as before. Now we want to find a way of writing this as

$$\underset{dp\times 1}{\mathbf{Y}_t} = \underset{dp\times dp}{F}\,\underset{dp\times 1}{\mathbf{Y}_{t-1}} + \underset{dp\times 1}{\mathbf{U}_t}$$

for some White noise process $\{\mathbf{U}_t\}$ and where

$$\mathbf{Y}_t = \begin{pmatrix} \mathbf{X}_t \\ \mathbf{X}_{t-1} \\ \vdots \\ \mathbf{X}_{t-p+1} \end{pmatrix}$$

- If we can do this, then we can study a VAR(1) process, and then recover the VAR(p) processes properties from it.

The companion form for a VAR(p)

Notice that

$$\begin{aligned}
\mathbf{Y}_t = \begin{pmatrix} \mathbf{X}_t \\ \mathbf{X}_{t-1} \\ \vdots \\ \mathbf{X}_{t-p+1} \end{pmatrix} &= \begin{pmatrix} \Phi_1\mathbf{X}_{t-1} + \Phi_2\mathbf{X}_{t-2} + \cdots + \Phi_p\mathbf{X}_{t-p} + \boldsymbol{\varepsilon}_t \\ \mathbf{X}_{t-1} \\ \vdots \\ \mathbf{X}_{t-p+1} \end{pmatrix} \\
&= \begin{pmatrix} \Phi_1 & \Phi_2 & \dots & \Phi_{p-1} & \Phi_p \\ I_d & 0 & \dots & 0 & 0 \\ 0 & I_d & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & I_d & 0 \end{pmatrix} \begin{pmatrix} \mathbf{X}_{t-1} \\ \mathbf{X}_{t-2} \\ \vdots \\ \mathbf{X}_{t-p} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\varepsilon}_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}
\end{aligned}$$

Therefore, we have the companion form for a VAR(p) process as

$$\underset{dp\times 1}{\mathbf{Y}_t} = \underset{dp\times dp}{F}\ \underset{dp\times 1}{\mathbf{Y}_{t-1}} + \underset{dp\times 1}{\mathbf{U}_t}$$

where

$$\mathbf{Y}_t = \begin{pmatrix} \mathbf{X}_t \\ \mathbf{X}_{t-1} \\ \vdots \\ \mathbf{X}_{t-p+1} \end{pmatrix}, \quad F = \begin{pmatrix} \Phi_1 & \Phi_2 & \dots & \Phi_{p-1} & \Phi_p \\ I_d & 0 & \dots & 0 & 0 \\ 0 & I_d & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & I_d & 0 \end{pmatrix}, \quad \mathbf{U}_t = \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Conditions for stationarity

The VAR(p) model is called stable if the roots of

$$\det\{I_d - \Phi_1 z - \cdots - \Phi_p z^p\} = 0$$

all lie outside the complex unit circle or equivalently we can require

$$F = \begin{pmatrix} \Phi_1 & \Phi_2 & \dots & \Phi_{p-1} & \Phi_p \\ I_d & 0 & \dots & 0 & 0 \\ 0 & I_d & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & I_d & 0 \end{pmatrix}$$

has eigenvalues with modulus less than one. Stability implies stationarity, but not the other way around in general.

Infinite moving average representation

Assume that the VAR(1) is stable. We can write the VAR(1) model as

$$\begin{aligned} \mathbf{X}_t &= \Phi_1 \mathbf{X}_{t-1} + \varepsilon_t \\ &= \Phi_1 (\Phi_1 \mathbf{X}_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\ &= \Phi_1^2 \mathbf{X}_{t-2} + \Phi_1 \varepsilon_{t-1} + \varepsilon_t \\ &= \cdots \\ &= \sum_{j=0}^{\infty} \Phi_1^j \varepsilon_{t-j} \end{aligned}$$

This manipulation can be shown to be formally correct, but is beyond the scope of the course.

Covariance

The covariance of a VAR(1) process is then given by

$$\begin{aligned}
\Gamma_\tau &= \mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \mathbf{X}_t\right) \\
&= \mathrm{Cov}\left(\sum_{j=0}^{\infty} \Phi_1^j \boldsymbol{\varepsilon}_{t-j+\tau}, \sum_{k=0}^{\infty} \Phi_1^k \boldsymbol{\varepsilon}_{t-k}\right) \\
&= \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} \Phi_1^j \,\mathrm{Cov}\left(\boldsymbol{\varepsilon}_{t-j+\tau}, \boldsymbol{\varepsilon}_{t-k}\right)(\Phi_1^k)^T \\
&= \sum_{k=0}^{\infty} \Phi_1^{k+\tau}\Sigma(\Phi_1^k)^T
\end{aligned}$$

Recovering the VAR(p)

To recover the VAR(p) process from its companion form, we can note that

$$\underset{d\times 1}{\mathbf{X}_t} = \underset{d\times dp}{G}\ \underset{dp\times 1}{\mathbf{Y}_t}$$

where

$$G = \begin{pmatrix} I_d & 0 & \cdots & 0 \end{pmatrix}.$$

Therefore, we have that

$$\begin{aligned}
\Gamma_\tau^{(\mathbf{X})} &= \mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \mathbf{X}_t\right) \\
&= \mathrm{Cov}\left(G\mathbf{Y}_{t+\tau}, G\mathbf{Y}_t\right) \\
&= G\Gamma_\tau^{(\mathbf{Y})}G^T \\
&= G\left(\sum_{k=0}^{\infty} F^{k+\tau}\Sigma(F^k)^T\right)G^T
\end{aligned}$$

Example 9.4 (bivariate-VAR(1)). We consider the bivariate or (two-dimensional) process of

$$\mathbf{X}_t = \boldsymbol{\nu} + \begin{pmatrix} 0.5 & 0.1 \\ 0.4 & 0.5 \end{pmatrix}\mathbf{X}_{t-1} + \begin{pmatrix} 0 & 0 \\ 0.25 & 0 \end{pmatrix}\mathbf{X}_{t-2} + \varepsilon_t.$$

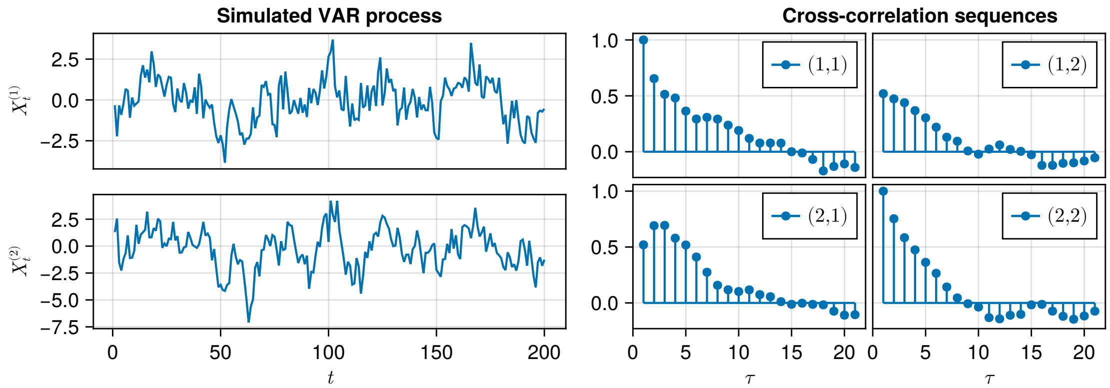
For this process the reverse characteristic polynomial is

$$\begin{aligned}
&\det\left\{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.5 & 0.1 \\ 0.4 & 0.5 \end{pmatrix}z - \begin{pmatrix} 0 & 0 \\ 0.25 & 0 \end{pmatrix}z^2\right\} \\
&= 1 - z + 0.21z^2 - 0.025z^3.
\end{aligned}$$

The roots are now instead

$$z_1 = 1.3, \quad z_2 = 3.55 + 4.26i, \quad z_3 = 3.55 - 4.26i.$$

Clearly $|z_1| > 1$ and also $|z_2|^2 = |z_3|^2 = 3.55^2 + 4.26^2$. Taking the squareroot of the latter quantity we get $|z_2| = |z_3| = 5.545$



$$\mathbf{X}_t = \boldsymbol{\nu} + \begin{pmatrix} 0.5 & 0.1 \\ 0.4 & 0.5 \end{pmatrix} \mathbf{X}_{t-1} + \begin{pmatrix} 0 & 0 \\ 0.25 & 0 \end{pmatrix} \mathbf{X}_{t-2} + \varepsilon_t.$$

Example 9.5 (trivariate-VAR(1)).

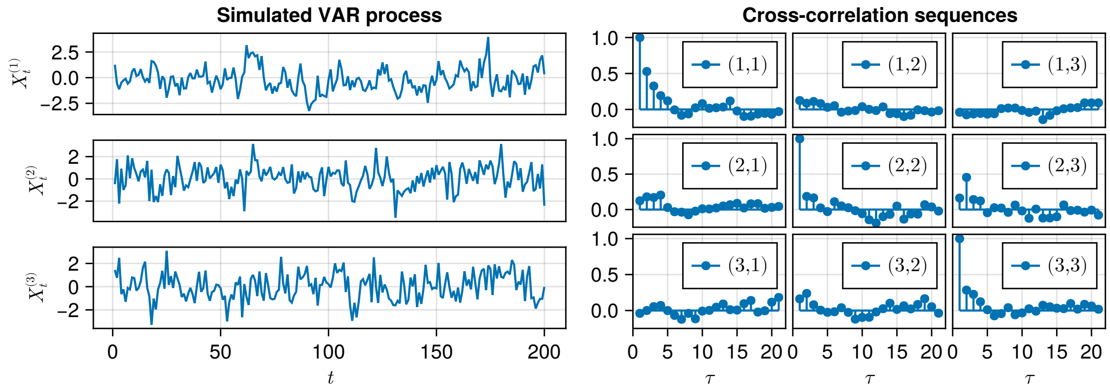
$$\mathbf{X}_t = \boldsymbol{\nu} + \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \mathbf{X}_{t-1} + \varepsilon_t$$

For this process the reverse characteristic polynomial is

$$\begin{aligned} &\det\left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} z \right\} \\ &= (1 - 0.5z)\left(1 - 0.4z - 0.03z^2\right). \end{aligned}$$

The roots are $z_1 = 2$, $z_2 \approx 2.1525$, $z_3 \approx -15.4858$. These are obviously greater than unity in magnitude, and so the VAR(1) is stable.

$$\mathbf{X}_t = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \mathbf{X}_{t-1} + \varepsilon_t$$



Yule-Walker equations

We can derive the Yule-Walker equations also for the VAR process.

$$\mathbf{X}_t = \Phi_1 \mathbf{X}_{t-1} + \varepsilon_t.$$

It therefore follows that

$$\begin{aligned}\mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \mathbf{X}_t\right) &= \mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \Phi_1 \mathbf{X}_{t-1} + \varepsilon_t\right) \\ &= \Phi_1 \Gamma_{\tau-1} + \Sigma \delta_{\tau,0}.\end{aligned}$$

We therefore get

$$\Gamma_\tau = \Phi_1 \Gamma_{\tau-1} + \Sigma \delta_{\tau,0}.$$

If we know Σ and Γ_0 then we obtain the form of the other auto-covariance matrices $\{\Gamma_\tau\}$ iteratively.

Vector Autoregression

For a higher order VAR(p) we have

$$\mathbf{X}_t = \Phi_1 \mathbf{X}_{t-1} + \cdots + \Phi_p \mathbf{X}_{t-p} + \varepsilon_t.$$

Therefore for $\tau \geq 0$ we have

$$\begin{aligned}\Gamma_\tau &= \mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \mathbf{X}_t\right) \\ &= \mathrm{Cov}\left(\mathbf{X}_{t+\tau}, \Phi_1 \mathbf{X}_{t-1} + \cdots + \Phi_p \mathbf{X}_{t-p} + \varepsilon_t\right) \\ &= \Phi_1 \Gamma_{\tau-1} + \cdots + \Phi_p \Gamma_{\tau-p} + \delta_{\tau,0} \Sigma.\end{aligned}$$

We can solve the first $p+1$ of these to obtain $\Gamma_0, \ldots, \Gamma_p$ in terms of Σ and the Φ_j matrices. The remaining Γ_τ for $\tau > p$ can be obtained recursively from the above equation.

Chapter 10

Forecasting

10.1 Forecasting

What is forecasting

The purpose of time series analysis is often to forecast future values based on the data collected up to the present. To do this we

1. assume the form of the model;

2. estimate the parameters of the model;

3. use the assumed model structure and the estimated parameters to form predictions.

For now, we assume that the model structure and the parameters are known, and predict h lags ahead, i.e. X_{n+h} based on $X_1, \ldots, X_n$.

- $h > 0$ is called the lead or the horizon
- n is called the origin

Conditional expectation

Let $g(X_1, \ldots, X_n)$ be an estimator of X_{n+h} based on $X_1, \ldots, X_n$. We assess its performance with the prediction mean squared error

$$P_{n+h}^n := \mathbb{E}\left[(X_{n+h} - g(X_1, \ldots, X_n))^2\right]. \tag{10.1}$$

Lemma 10.1. *The conditional expectation*

$$g(X_1, \ldots, X_n) = \mathbb{E}\left[X_{n+h} \mid X_1, \ldots, X_n\right]$$

minimizes the prediction mean square error for X_{n+h} based on $X_1, \ldots, X_n$.

Proof of Lemma 10.1. For Y a real-valued random variable with $\mu = \mathbb{E}[Y]$, we have

$$\begin{aligned}\mathrm{MSE}(c) &= \mathbb{E}\left[(Y-c)^2\right]\\ &= \mathbb{E}\left[(Y-\mu+\mu-c)^2\right]\\ &= \mathbb{E}\left[(Y-\mu)^2\right] + (\mu-c)^2\\ &= \mathrm{Var}(Y) + (\mu-c)^2.\end{aligned}$$

Thus $\mu = \arg\min_c \mathrm{MSE}(c)$. Then,

$$\begin{aligned}\mathrm{MSE}(g(X_1,\dots,X_n)) &= \mathbb{E}\left[(X_{n+h} - g(X_1,\dots,X_n))^2\right]\\ &= \mathbb{E}\left[\mathbb{E}\left[(X_{n+h} - g(X_1,\dots,X_n))^2 \mid X_1,\dots,X_n\right]\right]\end{aligned}$$

will be minimised at $g(X_1,\dots,X_n) = \mathbb{E}[X_{n+h}|X_1,\dots,X_n]$. □

Best linear predictor

Recall the definition of linear predictor:

Definition 10.1. The best linear predictor of X_{n+h} based on $X_1,\dots,X_n$ is the linear function

$$X_{n+h}^n = \mathcal{P}_{X_1,\dots,X_n}(X_{n+h}) = \sum_{j=1}^n \beta_j X_j$$

that minimizes the prediction mean square error.

- For Gaussian time series, the conditional expectation is linear in the data.
- But in general, the conditional expectation will be a nonlinear function of the data.

For simplicity, we restrict attention to linear predictors.

Example: Gaussian AR(1)

Example 10.1. Suppose X_t is such that $X_t = \phi X_{t-1} + \varepsilon_t$, with $\varepsilon_t \overset{iid}{\sim} \mathcal{N}(0,\sigma^2)$. For $h > 0$, the best linear predictor of X_{n+h} based on $X_1,\dots,X_n$ is

$$\begin{aligned}X_{n+h}^n &= \mathbb{E}[X_{n+h} \mid X_1,\dots,X_n]\\ &= \mathbb{E}[\phi X_{n+h-1} + \varepsilon_{n+h} \mid X_1,\dots,X_n]\\ &= \phi^h X_n\end{aligned}$$

Since X_t is stationary, $|\phi| < 1$ and $X_{n+h}^n \to 0$ as $h \to \infty$.

- X_{n+h}^n reverts to the mean of the process.
- This is a general property of stationary ARMA models.

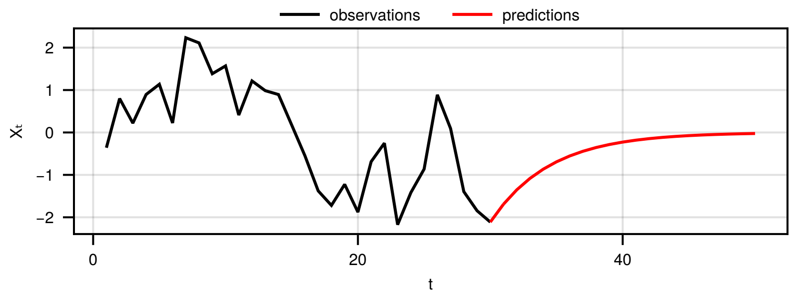


Figure 10.1: Predictions of an AR(1) process with $\phi = 0.8$ and $\sigma^2 = 1$.

Example: Gaussian MA(1)

Example 10.2. Suppose X_t is such that $X_t = \mu + \varepsilon_t - \theta\varepsilon_{t-1}$, with $\varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$. For $h > 0$, we get

$$\begin{aligned} X_{n+h}^n &= \mu + \mathbb{E}\left[\varepsilon_{n+h} \mid X_1, \dots, X_n\right] - \theta\mathbb{E}\left[\varepsilon_{n+h-1} \mid X_1, \dots, X_n\right] \\ &= \mu - \theta\mathbb{E}\left[\varepsilon_{n+h-1} \mid X_1, \dots, X_n\right], \end{aligned}$$

as ε_t are strictly uncorrelated with $X_1, \dots, X_n$ for $t > n$. For $h > 1$, this reduces to

$$X_{n+h}^n = \mu.$$

Prediction equations

The best linear predictor depends only on the second-order properties.

Theorem 10.2. *For a zero-mean stationary process $\{X_t\}$, X_{n+h}^n is found by solving for $\beta_1, \dots, \beta_n$ the prediction equations*

$$\mathbb{E}\left[\left(X_{n+h} - X_{n+h}^n\right) X_k\right] = 0, \quad k = 1, \dots, n.$$

The β_j are then given by the solution of the system of equations

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1} & \gamma_{n-2} & \cdots & \gamma_0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} = \begin{pmatrix} \gamma_{n+h-1} \\ \gamma_{n+h-2} \\ \vdots \\ \gamma_h \end{pmatrix}. \tag{10.2}$$

- The proof for the non-zero mean processes is an exercise.

Application to ARMA processes

Rewriting the prediction equations (10.2) in matrix form

$$\mathbf{\Gamma}_n \boldsymbol{\beta} = \boldsymbol{\gamma}_{[h]},$$

we get that if $\mathbf{\Gamma}_n$ is invertible

$$X_{n+h}^n = \boldsymbol{\gamma}_{[h]}^{\mathrm{T}} \mathbf{\Gamma}_n^{-1} \boldsymbol{X} \text{ and } P_{n+h}^h = \gamma_0 - \boldsymbol{\gamma}_{[h]}^{\mathrm{T}} \mathbf{\Gamma}_n^{-1} \boldsymbol{\gamma}_{[h]}$$

with $\boldsymbol{X} = (X_1, \dots, X_n)^{\mathrm{T}}$.

- $\mathbf{\Gamma}_n$ is invertible for invertible ARMA models, but not in all other cases. However the best prediction X_{n+h}^n is always unique.

Comments on prediction equations

- The calculation of the solution $X_{n+h}^n = \boldsymbol{\gamma}_{[h]}^{\mathrm{T}} \mathbf{\Gamma}_n^{-1} \boldsymbol{X}$ is inefficient when n is large because the $n \times n$ matrix $\mathbf{\Gamma}_n$ must be inverted.
- Recursive algorithms that do not require any matrix inversion have been proposed: the Durbin-Levinson and the innovations algorithms are discussed in Shumway and Stoffer (2000, §3.5).
- Theorem 10.2 is valid for any stationary process. In the next section, we focus on predictions for causal ARMA models.

10.2 Prediction for causal ARMA models

Prediction based on linear representation

We use the linear process representation and the implicit definition from an ARMA equation.

Theorem 10.3. *The best linear predictor X_{n+h}^n for X_{n+h} in a causal ARMA process with general linear representation $\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ is*

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j} = \psi_h \varepsilon_n + \psi_{h+1} \varepsilon_{n-1} + \cdots .$$

The corresponding prediction mean square error is $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$.

- X_{n+h}^n can be evaluated from its linear representation, with unrealised innovations $\varepsilon_{n+1}, \dots, \varepsilon_{n+h} \equiv 0$.
- The proof is an exercise for this week

Truncated predictions for ARMA models

In practice it is simpler to predict based on $\Phi(\mathrm{B})X_{t+h} = \Theta(\mathrm{B})\varepsilon_{t+h}$ than to obtain the general linear representation. The residuals $\widehat{\varepsilon}_t$ and fitted values/predictions $\tilde{X}_t^n$ are obtained recursively

$$\begin{aligned}
\widehat{\varepsilon}_t &= \begin{cases} \sum_{i=1}^{p} \phi_i \tilde{X}_{t-i}^n + \left(\theta_1 \widehat{\varepsilon}_{t-1} + \cdots + \theta_q \widehat{\varepsilon}_{t-q}\right), & t = 1, \ldots, n, \\ 0, & \text{otherwise}. \end{cases} \\
\tilde{X}_{t+h}^n &= \begin{cases} \sum_{i=1}^{p} \phi_i \tilde{X}_{t+h-i}^n - \sum_{j=1}^{q} \theta_j \widehat{\varepsilon}_{t+h-j}, & t+h > n, \\ X_{t+h}, & t+h = 1, \ldots, n, \\ 0, & t+h \leq 0. \end{cases}
\end{aligned} \tag{10.3}$$

- The forecast error is estimated by the expression in Theorem 10.3.
- Note that $\tilde{X}_t^n$ is an approximation as we had to truncate.

Comments

The best linear predictor for a causal process with mean μ has form

$$X_{n+h}^n = \mu + \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j}.$$

For a causal ARMA model $\psi_j \to 0$ exponentially fast:

$$X_{n+h} \to \mu \text{ as } h \to \infty.$$

Likewise for large h, the prediction error

$$\mathbb{E}\left[\left(X_{n+h} - X_{n+h}^n\right)^2\right] = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \to \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 = \gamma_0, \quad h \to \infty.$$

10.3 Prediction for integrated processes

Truncated point forecasts example for ARIMA(3,1,1)

Example 10.3 (Example: ARIMA(3,1,1)). We have

$$\left(1 - \phi_1 \mathrm{B} - \phi_2 \mathrm{B}^2 - \phi_3 \mathrm{B}^3\right)(1 - \mathrm{B})X_t = (1 - \theta_1 \mathrm{B})\, \varepsilon_t,$$

which we can rewrite as

$$\begin{aligned}
X_t = &\underbrace{(1 + \phi_1)\, X_{t-1} - (\phi_1 - \phi_2)\, X_{t-2} - (\phi_2 - \phi_3)\, X_{t-3} - \phi_3 X_{t-4}}_{\text{AR}} \\
&+ \underbrace{\varepsilon_t - \theta_1 \varepsilon_{t-1}}_{\text{MA}}.
\end{aligned}$$

We will deal with the AR and MA part as we did for ARMA models:

- AR part: replace $X_{t-1}, \dots, X_{t-4}$ with $X_{t-1}^n, \dots, X_{t-4}^n$.
- MA part: replace ε_t with the residuals if $t \leq n$, and otherwise set to 0.

Example 10.4 (Example: ARIMA(3,1,1) (cont.)). Let $\hat{\varepsilon}_n$ be the last observed residual (similar to eq. (10.3)). For the one-step ahead forecast we get

$$X_{n+1}^n = (1+\phi_1)X_n - (\phi_1 - \phi_2)X_{n-1} - (\phi_2 - \phi_3)X_{n-2} - \phi_3 X_{n-3} - \theta_1 \hat{\varepsilon}_n.$$

For $t > n+1$, the best linear predictor for X_{n+h}^n based on $X_1, \dots, X_n$ is then

$$X_t^n = (1+\phi_1)X_{t-1}^n - (\phi_1 - \phi_2)X_{t-2}^n - (\phi_2 - \phi_3)X_{t-3}^n - \phi_3 X_{t-4}^n.$$

- We can generalize this example to any ARIMA(p, d, q) model.

General case: falling back to ARMA

Let $\{X_t\}$ be ARIMA(p, d, q), i.e.

$$\Phi(\mathrm{B})(1-\mathrm{B})^d X_t = \Theta(\mathrm{B})\varepsilon_t.$$

Then $Z_t = \nabla^d X_t$ is ARMA(p, q)

$$Z_t = \sum_{j=1}^{p} \phi_j Z_{t-j} + \varepsilon_t - \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}.$$

We know how to predict Z_{n+h} based on $Z_1, \dots, Z_n$, and we can use them to predict X_{n+h} based on $X_1, \dots, X_n$:

$$\nabla^d X_{n+h}^n = Z_{n+h}^n.$$

Example 10.5 (predicting X based on Z). Suppose that X_t ARIMA$(p, 1, q)$. Then we have $X_{n+h} - X_{n+h-1} = Z_{n+h}$ and

$$X_{n+h} = X_{n+h-1} + Z_{n+h} = \cdots = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}, \quad h > 1,$$

giving

$$X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n, \quad h > 1.$$

If $\{Z_t\}$ has mean $\alpha \neq 0, Z_{n+h}^n \to \alpha$ as $h \to \infty$, and $X_{n+h}^n \approx X_n + \alpha h$ for h large.

- In general for ARIMA models $P_{n+h}^n \to \infty$ as $h \to \infty$.
- For SARIMA models with $d = D = 1$ and $\alpha = 0$, it can be shown that the forecast will be linear and seasonal in h.

10.3.1 Details on truncated forecasts

1. Expand the ARIMA equation so that y_t is on the left-hand side and all other terms are on the right.
2. Rewrite the equation by replacing t with $n+h$.
3. On the right-hand side of the equation, replace future observations with their forecasts, future errors with zero, and past errors with the corresponding residuals.

Beginning with $h=1$, these steps are then repeated for $h=2,3,\ldots$ until all forecasts have been calculated. The procedure is most easily understood via an example. We will illustrate it using the ARIMA(3,1,1) model fitted in the previous section. The model can be written as follows:

$$\left(1-\hat{\phi}_1 B-\hat{\phi}_2 B^2-\hat{\phi}_3 B^3\right)(1-B)y_t=\left(1+\hat{\theta}_1 B\right)\varepsilon_t$$

where $\hat{\phi}_1=0.0044, \hat{\phi}_2=0.0916, \hat{\phi}_3=0.3698$ and $\hat{\theta}_1=-0.3921$. Then we expand the left hand side to obtain

$$\left[1-\left(1+\hat{\phi}_1\right)B+\left(\hat{\phi}_1-\hat{\phi}_2\right)B^2+\left(\hat{\phi}_2-\hat{\phi}_3\right)B^3+\hat{\phi}_3 B^4\right]y_t=\left(1+\hat{\theta}_1 B\right)\varepsilon_t$$

and applying the backshift operator gives

$$y_t-\left(1+\hat{\phi}_1\right)y_{t-1}+\left(\hat{\phi}_1-\hat{\phi}_2\right)y_{t-2}+\left(\hat{\phi}_2-\hat{\phi}_3\right)y_{t-3}+\hat{\phi}_3 y_{t-4}=\varepsilon_t+\hat{\theta}_1\varepsilon_{t-1}$$

Finally, we move all terms other than y_t to the right hand side:

$$y_t=\left(1+\hat{\phi}_1\right)y_{t-1}-\left(\hat{\phi}_1-\hat{\phi}_2\right)y_{t-2}-\left(\hat{\phi}_2-\hat{\phi}_3\right)y_{t-3}-\hat{\phi}_3 y_{t-4}+\varepsilon_t+\hat{\theta}_1\varepsilon_{t-1}.$$

This completes the first step. While the equation now looks like an ARIMA($4,0,1$), it is still the same ARIMA $(3,1,1)$ model we started with. It cannot be considered an ARIMA$(4,0,1)$ because the coefficients do not satisfy the stationarity conditions. For the second step, we replace t with $T+1$ in (8.6):

$$y_{T+1}=\left(1+\hat{\phi}_1\right)y_T-\left(\hat{\phi}_1-\hat{\phi}_2\right)y_{T-1}-\left(\hat{\phi}_2-\hat{\phi}_3\right)y_{T-2}-\hat{\phi}_3 y_{T-3}+\varepsilon_{T+1}+\hat{\theta}_1\varepsilon_T.$$

Assuming we have observations up to time T, all values on the right hand side are known except for ε_{T+1}, which we replace with zero, and ε_T, which we replace with the last observed residual e_T :

$$\hat{y}_{T+1|T}=\left(1+\hat{\phi}_1\right)y_T-\left(\hat{\phi}_1-\hat{\phi}_2\right)y_{T-1}-\left(\hat{\phi}_2-\hat{\phi}_3\right)y_{T-2}-\hat{\phi}_3 y_{T-3}+\hat{\theta}_1 e_T.$$

A forecast of y_{T+2} is obtained by replacing t with $T+2$ in (8.6). All values on the right hand side will be known at time T except y_{T+1} which we replace with $\hat{y}_{T+1|T}$, and ε_{T+2} and ε_{T+1}, both of which we replace with zero:

$$\hat{y}_{T+2|T}=\left(1+\hat{\phi}_1\right)\hat{y}_{T+1|T}-\left(\hat{\phi}_1-\hat{\phi}_2\right)y_T-\left(\hat{\phi}_2-\hat{\phi}_3\right)y_{T-1}-\hat{\phi}_3 y_{T-2}.$$

The process continues in this manner for all future time periods. In this way, any number of point forecasts can be obtained.

10.4 Prediction uncertainty

Source of uncertainty

There are three components of uncertainty in predictions:

1. model uncertainty,
2. estimation uncertainty,
3. innovation uncertainty.

- We assume that the model is known and that the parameters are estimated without error. This is equivalent to removing the first two sources of uncertainty.
- There are techniques (bootstrap) to account for the model and estimation uncertainties, but we will not cover them in this course.

Innovation uncertainty for ARMA models

Let $e_n(h)$ be the h-step ahead forecast error for X_{n+h} based on $X_1, \ldots, X_n$. Then using theorem 10.3 we have

$$e_n(h) = X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}.$$

We have $\mathbb{E}\left[e_n(h)\right] = 0$ and

$$\operatorname{Var}\left(e_n(h)\right) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \to \gamma_0, h \to \infty.$$

Prediction intervals

If we consider a Gaussian process, we can construct a $1-\alpha$ prediction interval for X_{n+h} based on $X_1, \ldots, X_n$ as

$$X_{n+h}^n \pm z_{1-\alpha/2} \operatorname{Var}\left(e_n(h)\right)^{1/2},$$

where z_α is the αth percentile of the Gaussian distribution. See Figure 10.2.

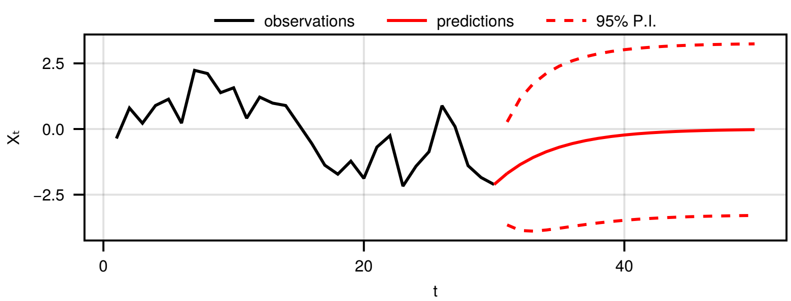


Figure 10.2: Prediction intervals for a Gaussian process.

Chapter 11

Diagnostics, model selection and long memory

11.1 Diagnostics for general ARIMA models

Residuals

Model checking is usually based on residuals. Informally they are the difference between the observed and the fitted values.

Definition 11.1 (Standardized residual)**.** Consider observations of a time series $\{X_t\}$. Say that for a given model we have a fitted value $\hat{X}_t$ at time t. The residuals are

$$e_t = X_t - \hat{X}_t$$

and the standardized residuals are

$$\tilde{e}_t = e_t / \sqrt{\operatorname{Var}(e_t)}.$$

- Do the residuals have a constant zero mean?
- Is their variance constant wrt t (homoscedasticity)?
- Are they uncorrelated in t?
- Are they Gaussian?

Residuals for an AR(1)

Example 11.1. For an AR(1) model with parameters ϕ and σ, the fitted value at time t is ϕX_{t-1}. Therefore, the residual (if we know the true model) is given by

$$e_t = X_t - \phi X_{t-1} = \varepsilon_t.$$

This is the original noise process, meaning that the residuals have constant zero mean, constant variance and are uncorrelated. If the noise process was Gaussian, then the residuals are Gaussian.

Checking the mean

We make a plot of $\{(t, \tilde{e}_t)\}$. There should be no trends, and they should be close to zero like in the left-hand side of Figure 11.1.

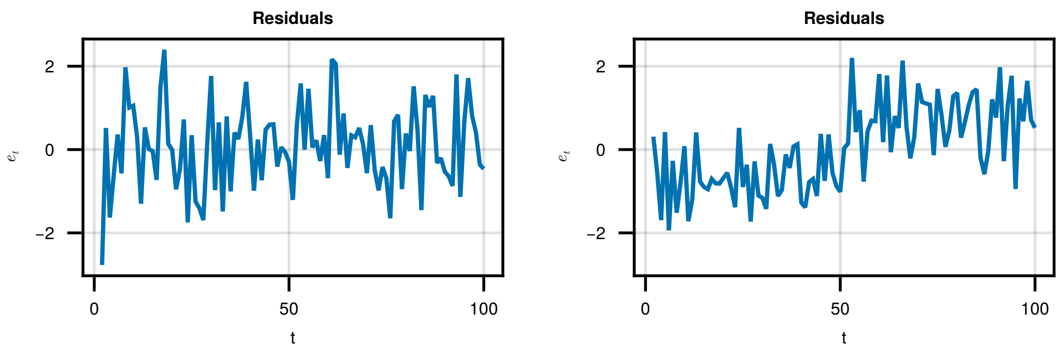


Figure 11.1: Left: centered residuals. Right: residuals with a trend.

Checking the variance

We make a plot of $\{(t, \tilde{e}_t)\}$. The variance should be constant like in the left-hand side of Figure 11.2.

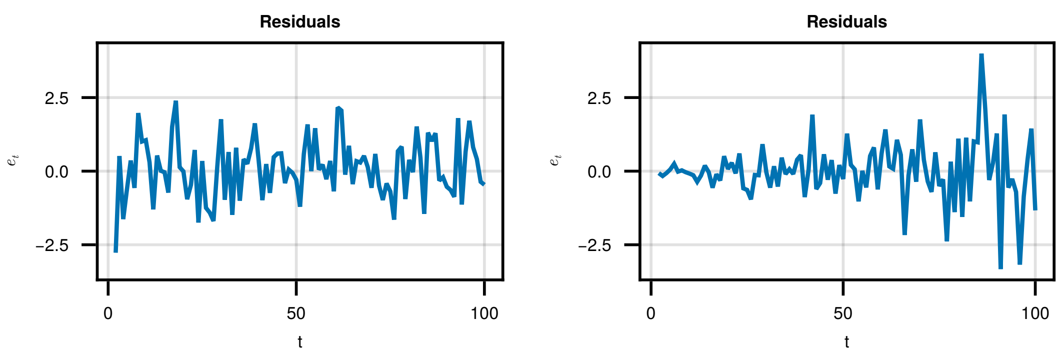


Figure 11.2: Left: constant variance. Right: non-constant variance.

Checking the Gaussianity

To check the Gaussianity of the residuals we can use a Q-Q plot. This is a plot of the quantiles of the residuals against the quantiles of a normal distribution. If the residuals are Gaussian then the points should lie on a straight line as in the left-hand side of Figure 11.3. Note that this only checks marginal Gaussianity.

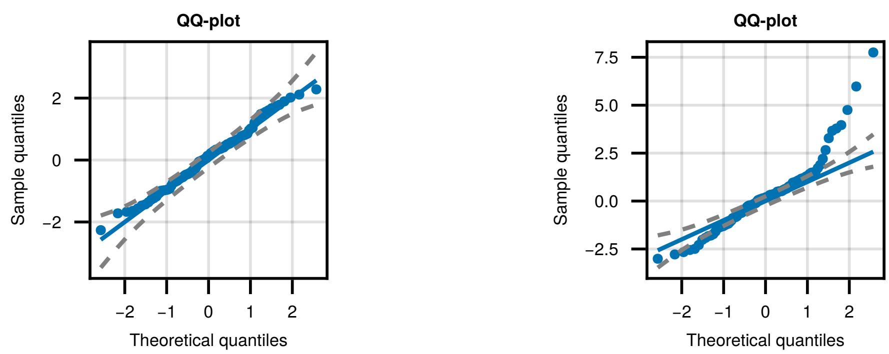


Figure 11.3: Left: Gaussian residuals. Right: student residuals.

Checking the autocorrelation

- To check the correlation we can consider the correlation between the residuals at different lags.
- Recall that the autocorrelation at lag k is given by

$$\hat{r}_\tau = \hat{\rho}_\tau^{(e)} = \frac{\sum_{t=1}^{n-\tau} (e_{t+\tau} - \bar{e})(e_t - \bar{e})}{\sum_{t=1}^{n} (e_t - \bar{e})^2}.$$

- We plot the corellelogram for the residuals. Under the assumption of white noise we compare to $(-1.96/\sqrt{n}, 1.96/\sqrt{n})$.

Testing the autocorrelation

- We can also, more realistically, try to test that a long range of correlations are zero:

$$H_0: \quad \rho_1 = \rho_2 = \cdots = \rho_m = 0$$

for some choice of m, against the alternative that at least one autocorrelation in that range is non-zero.

Proposition 11.1 (Box-Pierce)**.** *We introduce the Box-Pierce statistic*

$$Q_m = n \sum_{\tau=1}^{m} \hat{r}_\tau^2$$

Under H_0 this is approximately χ^2_{m-p-q} for an ARMA(p,q).

Improving the Box-Pierce test

In practice this approximation is inaccurate. Thus we chose to use the modified Box-Pierce statistic, usually called the Ljung–Box statistic.

Proposition 11.2 (Ljung-Box)**.**

$$Q_m = n(n+2)\sum_{\tau=1}^{m} \frac{\hat{r}_\tau^2}{n-\tau}$$

Under H_0 this is a χ^2_{m-p-q} for an ARMA(p,q).

- m has to be chosen by the practitioner.
- Often a series of m values are tested and displayed in a plot.

Figure 11.4 shows example of diagnostic plots for checking the autocorrelation of residuals and the Ljung-Box test.

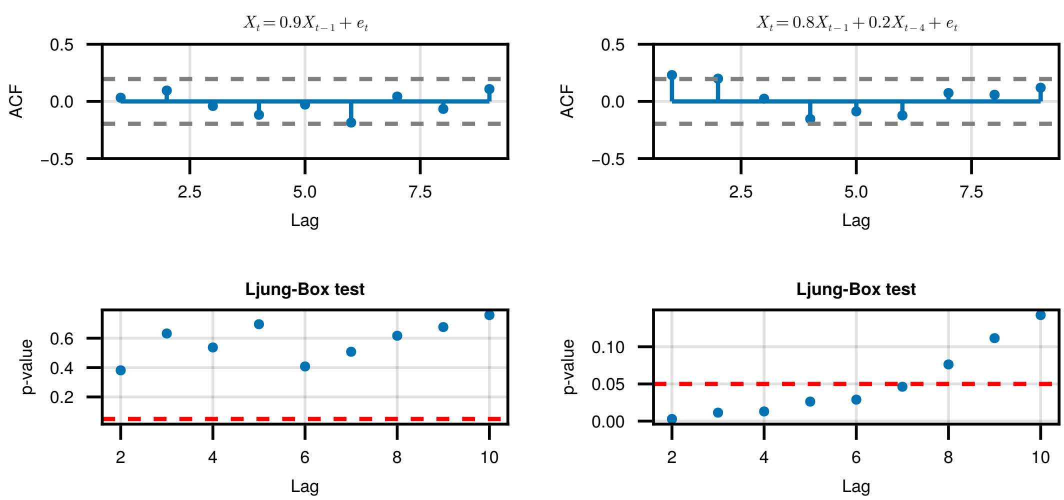


Figure 11.4: Left are the diagnostic plots for the true model; right are the diagnostic plots for an AR(2) model fitted to the same data.

Evaluating model fit: the dangers of $\mathbf{R}^2$

- We might think that residual variance can help with model selection. We define the R^2 statistic to be
$$R^2 = 1 - \frac{s_e^2}{s_y^2}$$

- Some care is needed; more parameters $\Rightarrow R^2$ improves.
- Stochastic processes are not perfectly predictable. Therefore there is a limit. Consider an AR(1) model.
- The process variance is $\sigma^2 / \left(1-\phi^2\right)$, thus

$$R^2 = 1 - \frac{\sigma^2\left(1-\phi^2\right)}{\sigma^2} = \phi^2$$

As $|\phi| < 1$ say with $\phi = 0.4$ we get $R^2 = 0.16$. Does not look good.

11.2 Model selection

Model Comparison

There are formal methods of comparing models.

- Most come from information theory, and correspond to information criteria.

These criteria are defined for models with k parameters

- For ARMA $k = p + q + 1$, AR and MA plus noise variance.

Information criterion are regularly used in any statistics context to compare models.
A model with smaller AIC is deemed better

$$\mathrm{AIC}(\boldsymbol{\theta}) = -2\log L(\boldsymbol{\theta} \mid \mathbf{y}) + 2k$$

where L is the likelihood function.

- AIC overestimates p in the ARMA model.

Hurvich and Tsai (1989) suggested a corrected Aikake Information Criterion that works better in practice:

$$\mathrm{AICC}(\boldsymbol{\theta}) = -2\log L(\boldsymbol{\theta} \mid \mathbf{y}) + 2k\frac{n}{n-k-1}$$

- AICC and AIC become equivalent as n diverges.

The Bayesian Information Criterion (BIC) is given by

$$\mathrm{BIC}(\boldsymbol{\theta}) = -2\log L(\boldsymbol{\theta} \mid \mathbf{y}) + k\log(n)$$

Box–Jenkins method

The Box-Jenkins methodology is a framework for building models:

- starts by identifying reasonable values for p, d and q,
- then estimates the parameters of the proposed ARIMA model,
- checks diagnostics to verify that the model fitting is appropriate,
- the subsequent step might be forecasting or some other inference.

Model identification

For choosing d

- Plot the data and look for non-stationarity.
- If the data looks non-stationary, plot differences of the data.
- Hopefully low orders of differencing are enough.

For choosing p and q (using the differenced data)

- Look at the acf and pacf.
- Sharp drop in the ACF at lag q suggests an MA(q) model.
- Sharp drop in the PACF at lag p suggests an AR(p) model.
- No sharp drops suggests an ARMA model.
- Very slow decay suggests non-stationary or some other issue.

11.3 Long Memory

Long memory

- Assume that $\{X_t\}$ is a stationary process; it is characterised by its mean $\mathbb{E}[X_t] = \mu$ and its auto-covariance sequence

$$\gamma_\tau = \mathrm{Cov}\left(X_t, X_{t+\tau}\right)$$

- We have looked at ARMA models.
- These have very rapidly decaying autocovariances, and in fact

$$S(0) = \sum_{\tau=-\infty}^{\infty} \gamma_\tau$$

- If we assume the spectrum is absolutely continuous, then $S(0)$ is finite.
- But what if we don't?

Long memory

- A consequence of the absolutely continuous spectrum is that

$$\mathrm{Var}\left(\bar{X}\right) \sim \frac{S(0)}{n}$$

- But in practice Smith (1938) found for many practical processes that this scales like $1/n^\alpha$ where $0 < \alpha < 1$.
- For example if $\gamma_j \sim j^{2d-1}$ then we find that if we aggregate

$$\sum_{j=-n}^{n} \gamma_j$$

then this will recover different powers of n.

Long memory

- The term "long-memory" was coined for processes where the covariance decayed like a polynomial in j rather than being compact or exponential.
- The simplest such process is a fractionally differenced process. This is defined in terms of the Backshift operator B as

$$(1-B)^d X_t = \epsilon_t$$

- We can expand the power to be

$$(1-B)^d = \sum_{j=0}^{\infty} \frac{\Gamma(j-d)}{\Gamma(-d)\Gamma(j+1)} B^j$$

- This will depend on the power of d, we have treated positive integer values as ARIMA processes.

Long memory

- ARMA models $\rightarrow$ covariance is compact or exponentially decaying.
- A second-order stationary process X_t with autocovariance $\{\gamma_\tau\}$ is said to exhibit long-memory in terms of its correlations if

$$\gamma_\tau \sim L_\gamma(\tau)|\tau|^{-\alpha}$$

  as $|\tau| \rightarrow \infty$. We assume $0 < \alpha < 1$ and L_γ is a function that varies slowly as $|\tau|$ becomes large.
- Before $\gamma_\tau = 0$ if $|\tau| > q$ or $\gamma_\tau \sim |\phi|^\tau$.

Long memory

- Long memory can also be defined in terms of the spectral behaviour of the process:

$$S(f) \sim L_S(f)|f|^{\alpha-1}$$

- Referring back to γ_τ it can be shown it takes the form

$$L_S(f) = 2\gamma(0)\Gamma(1-\alpha)\sin\frac{\pi\alpha}{2}$$
$$L_\gamma(\tau) = 2\Gamma(\alpha)\sin\frac{\pi(1-\alpha)}{2}$$

- This gives an interesting range of behaviour. The range of α is $0 < \alpha < 2$: this yields three different sorts of stationary processes: (a) long memory ($0 < \alpha < 1$), (b) short-range dependence ($\alpha = 1$), and (c) anti-persistence ($1 < \alpha < 2$).

Long memory

- We note that (c) implies that

$$\sum_k \gamma_k = 0$$

- We can achieve (a-c) using the FARIMA (fractional ARIMA) class defined as the solution of

$$(1-B)^d \varphi(B) X_t = \psi(B)\epsilon_t$$

and $d = (1-\alpha)/2 \in (-0.5, 0.5), \epsilon_t$ are IID zero mean variables, with variance $\sigma_\epsilon^2 > 0, B$ is the back-shift operator, and $\varphi(z)$ and $\psi(z)$ are polynomials with order p and q, respectively, and where we can assume all roots are outside the unit circle.

Long memory

- We recall that

$$\begin{aligned}
\{1-B\}^d &= \sum_{k=0}^{\infty} \begin{pmatrix} d \\ k \end{pmatrix} (-1)^k B^k \\
&= \sum_{k=0}^{\infty} \frac{\Gamma(d+1)}{\Gamma(k+1)\Gamma(d-k+1)} (-1)^k B^k
\end{aligned}$$

is the so-called fractional differencing operator. The spectral density is then equal to

$$S(f) = \sigma^2 \frac{\left|\psi\left(e^{-2i\pi f}\right)\right|^2}{\left|\varphi\left(e^{-2i\pi f}\right)\right|^2} \left|1 - e^{-2i\pi f}\right|^{-2d} \overset{|f|\to 0}{\sim} c_f |f|^{-2d}$$

- Here $c_f = \sigma^2 \frac{|\psi(1)|^2}{|\varphi(1)|^2}$

Long memory

- Long-memory is closely related to self-similar processes.
- A process Y_t is "self-similar" with self-similarity parameter H, if for any $c > 0, Y_{ct}$ is equal in distribution to $c^H Y_t$.
- Self-similarity implies that we have the covariance structure

$$\operatorname{Cov}(Y_t, Y_s) = \frac{\sigma^2}{2} \left\{ |t|^{2H} + |s|^{2H} - |t-s|^{2H} \right\} \tag{11.1}$$

- If Y_t has stationary increments then $X_t = Y_t - Y_{t-1}$ has covariance

$$\begin{aligned}
\gamma_X(\tau) &= \frac{\sigma^2}{2} \left\{ |\tau+1|^{2H} - 2|\tau|^{2H} + |\tau = 1|^{2H} \right\} \\
&\sim H(2H-1)|\tau|^{2H-2}, \quad |\tau| \to \infty
\end{aligned}$$

Long memory

- A Gaussian process with covariance (11.1) is called fractional Brownian motion, written as $B_H(t)$.
- The increment process $X_t = B_H(t) - B_H(t-1)$ is fractional Gaussian noise.
- It can be shown that fractional Brownian motion is the only self-similar Gaussian process.
- For instance, for one-dimensional fractional Brownian motion, the Hausdorff dimension of the graph $(t, B_H(t))$ is equal to $D = 2 - H$.
- Fractal dimension is a local property, whereas long-range dependence characterizes global behavior.
- It is possible to construct processes where the long-memory parameter and the fractal dimension are completely independent of each other.

Long memory

- How do we estimate Long-Memory?
- By calculating the periodogram we can arrive at estimators for d.
- If we take the logarithm of $I(f)$ then its expectation for a Gaussian process takes the form

$$\mathbb{E}I(f) = L_S(f)|f|^{\alpha-1},$$

however to determine α we need to take the logarithm of $I(f)$ before taking expectations.

Chapter 12

Non-Linear Time Series

12.1 Motivation

Financial returns

- We begin with a financial time series X_t, such as a stock index.
- Rather than modelling prices directly, we usually study log-returns:

$$r_t = \log\left(\frac{X_t}{X_{t-1}}\right) = \log(X_t) - \log(X_{t-1}).$$

- These measure relative changes in the price.

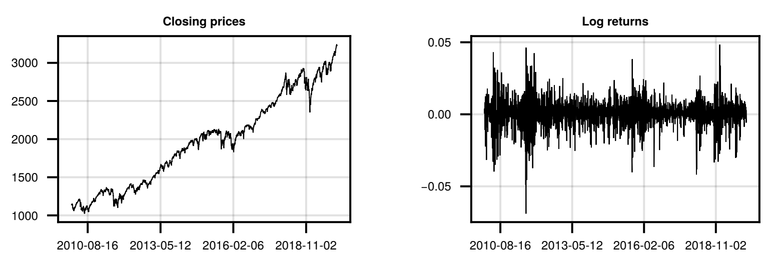


Figure 12.1: S&P 500 stock index prices from 01/01/2010 to 01/01/2020.

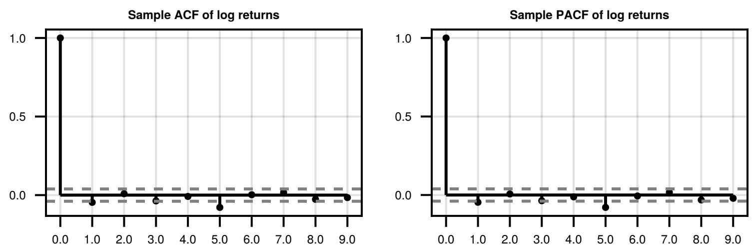


Figure 12.2: ACF and PACF of the log-returns from 01/01/2010 to 01/01/2020.

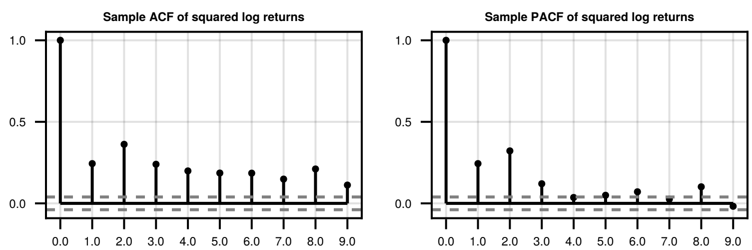


Figure 12.3: ACF and PACF of the square of the log-returns from 01/01/2010 to 01/01/2020.

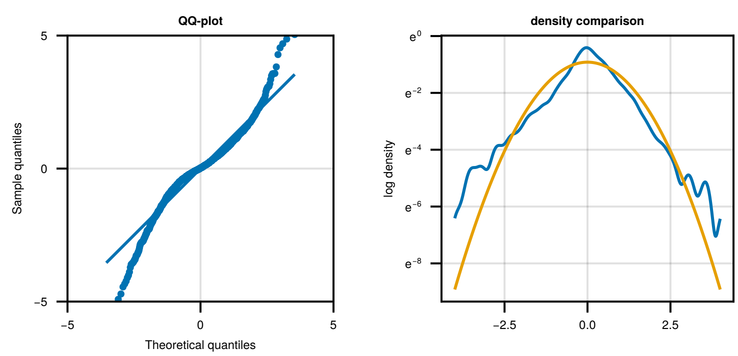


Figure 12.4: qq plot of standardised log returns (left) and log density (right) both compared to a standard normal.

ACF and PACF

ACF and PACF of the squared log-returns

Marginal distribution of the log-returns

Why non-linear time series?

- The log-returns have little linear dependence.

- So a linear model for the conditional mean may not be very useful.
- But the squared log-returns do show dependence.
- This suggests that the conditional variance changes over time.
- This phenomenon is called volatility clustering:
  - large changes tend to be followed by large changes;
  - small changes tend to be followed by small changes.
- ARCH and GARCH models are designed to model this changing conditional variance.

12.2 ARCH

Definition 12.1 (ARCH models)**.** We will say that a time series $\{r_t\}$ is an AutoRegressive Conditionally Heteroscedastic (ARCH) model if it takes the form

$$r_t = \sigma_t \varepsilon_t, \qquad t \in \mathbb{Z}$$

where $\{\varepsilon_t\}$ is a white noise process with mean zero and variance one, and

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2$$

the parameters $\alpha_0, \alpha_1 > 0$.

- So we are doing the same thing as before, but for the variance of the noise process, not the process itself.
- The idea is that this will allow the conditional variance to change over time.

ARCH

- Informally, we write the information available at time t as $\mathcal{F}_t$.
- We are interested in the properties of r_t given $\mathcal{F}_{t-1}$.
- The conditional mean is given by

$$\mathbb{E}\left[r_t \mid \mathcal{F}_{t-1}\right] = \mathbb{E}\left[\sigma_t\varepsilon_t \mid \mathcal{F}_{t-1}\right] = \sigma_t \mathbb{E}\left[\varepsilon_t \mid \mathcal{F}_{t-1}\right] = 0$$

- We now compute the conditional variance

$$\begin{aligned}
\operatorname{Var}\left(r_t \mid \mathcal{F}_{t-1}\right) &= \mathbb{E}\left[r_t^2 \mid \mathcal{F}_{t-1}\right] - \mathbb{E}\left[r_t \mid \mathcal{F}_{t-1}\right]^2 \\
&= \mathbb{E}\left[\left(\sigma_t\varepsilon_t\right)^2 \mid \mathcal{F}_{t-1}\right] \\
&= \sigma_t^2 \mathbb{E}\left[\varepsilon_t^2 \mid \mathcal{F}_{t-1}\right] \\
&= \sigma_t^2 \\
&= \alpha_0 + \alpha_1 r_{t-1}^2.
\end{aligned}$$

ARCH: mean and variance

- We can also use this to compute unconditional expectations and variances.
- We can compute the unconditional expectation using the law of iterated expectation:

$$\begin{aligned}
\mathbb{E}[r_t] &= \mathbb{E}[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]] \\
&= \mathbb{E}[0] \\
&= 0
\end{aligned}$$

ARCH: variance

- We can also compute a recursion for the marginal variance:

$$\begin{aligned}
\operatorname{Var}(r_t) &= \mathbb{E}\left[r_t^2\right] - \mathbb{E}[r_t]^2 \\
&= \mathbb{E}\left[\mathbb{E}\left[r_t^2 \mid \mathcal{F}_{t-1}\right]\right] - 0^2 \\
&= \mathbb{E}\left[\alpha_0 + \alpha_1 r_{t-1}^2\right] \\
&= \alpha_0 + \alpha_1 \operatorname{Var}(r_{t-1}).
\end{aligned}$$

- If a weakly stationary solution exists, then $\operatorname{Var}(r_t) = \operatorname{Var}(r_{t-1})$.
- Hence

$$\operatorname{Var}(r_t) = \frac{\alpha_0}{1-\alpha_1}.$$

- So $\alpha_1 < 1$ is necessary for finite variance.
- In fact, for ARCH(1), this condition is also sufficient.

ARCH: autocovariance

- The autocovariance for $\tau > 0$ is given by

$$\begin{aligned}
\operatorname{Cov}(r_t, r_{t+\tau}) &= \mathbb{E}[r_{t+\tau} r_t] - \mathbb{E}[r_{t+\tau}]\,\mathbb{E}[r_t] \\
&= \mathbb{E}[\sigma_{t+\tau}\varepsilon_{t+\tau} r_t] - 0^2 \\
&= \mathbb{E}[\mathbb{E}[\sigma_{t+\tau}\varepsilon_{t+\tau} r_t \mid \mathcal{F}_{t+\tau-1}]] \\
&= \mathbb{E}[r_t \sigma_{t+\tau}\mathbb{E}[\varepsilon_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]] \\
&= 0
\end{aligned}$$

- So ARCH is an uncorrelated process despite having dependent volatility!

Isserlis' Theorem

Theorem 12.1 (Isserlis' Theorem)**.** *If* $(X_1, \dots X_N)$ *is a zero-mean multivariate normal random vector and* N *is even, then*

$$\mathbb{E}\{X_1 \dots X_N\} = \sum_{p \in P_N} \prod_{\{i,j\} \in p} \mathrm{Cov}\{X_i, X_j\}$$

where P_N *is the set of all partitions of* $\{1, \dots, N\}$ *into pairs.*

A special case of this theorem for $N = 4$ is

$$\begin{aligned}\mathbb{E}\{X_1 X_2 X_3 X_4\} &= \mathbb{E}\{X_1 X_2\} \mathbb{E}\{X_3 X_4\} \\ &+ \mathbb{E}\{X_1 X_3\} \mathbb{E}\{X_2 X_4\} \\ &+ \mathbb{E}\{X_1 X_4\} \mathbb{E}\{X_2 X_3\}\end{aligned}$$

ARCH: higher order moments

- Often, financial time series exhibit "fat tails", which means a kurtosis greater than 3.
- If the innovations are Gaussian, we can compute the fourth-order moments of the ARCH process

$$\begin{aligned}\mathbb{E}\left[r_t^4 \mid \mathcal{F}_{t-1}\right] &= \mathbb{E}\left[\sigma_t^4 \varepsilon_t^4 \mid \mathcal{F}_{t-1}\right] \\ &= \sigma_t^4 \mathbb{E}\left[\varepsilon_t^4 \mid \mathcal{F}_{t-1}\right] \\ &= 3\sigma_t^4 \mathbb{E}\left[\varepsilon_t^2 \mid \mathcal{F}_{t-1}\right]^2 \\ &= 3\sigma_t^4 \\ &= 3\left(\alpha_0 + \alpha_1 r_{t-1}^2\right)^2\end{aligned}$$

  where Isserlis' Theorem gives $\mathbb{E}\left[\varepsilon_t^4\right] = 3$.

ARCH: higher order moments

Again assuming shift invariance, the unconditional fourth moment is

$$\begin{aligned}\mathbb{E}\left[r_t^4\right] &= \mathbb{E}\left[\mathbb{E}\left[r_t^4 \mid \mathcal{F}_{t-1}\right]\right] \\ &= 3\mathbb{E}\left[\left(\alpha_0 + \alpha_1 r_{t-1}^2\right)^2\right] \\ &= 3\left[\alpha_0^2 + 2\alpha_0\alpha_1 \mathbb{E}\left[r_t^2\right] + \alpha_1^2 \mathbb{E}\left[r_t^4\right]\right] \\ &= 3\left[\alpha_0^2 + 2\alpha_0\alpha_1 \frac{\alpha_0}{1-\alpha_1} + \alpha_1^2 \mathbb{E}\left[r_t^4\right]\right]\end{aligned}$$

Re-arrangement gives

$$\mathbb{E}\left[r_t^4\right] = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)}$$

Which is finite if we require $0 \leq \alpha_1^2 < 1/3$.

Then the kurtosis is given by

$$\begin{aligned}\frac{\mathbb{E}\left[r_t^4\right]}{\operatorname{Var}\left(r_t\right)^2} &= \frac{3\alpha_0^2\left(1+\alpha_1\right)}{\left(1-\alpha_1\right)\left(1-3\alpha_1^2\right)} \frac{\left(1-\alpha_1\right)^2}{\alpha_0^2} \\ &= 3\frac{1-\alpha_1^2}{1-3\alpha_1^2}\end{aligned}$$

which is greater than 3 if $\alpha_1 > 0$.

The autocorrelation of the squares

- One can also show (see the exercises) that

$$\operatorname{Corr}\left(r_{t+\tau}^2, r_t^2\right) = \alpha_1^{|\tau|}$$

- This means that we have a geometric decay in the autocovariance.
- This is sometimes fine, but we might want more persistent volatility than this.

Parameter estimation

- The parameters of the ARCH model can be estimated using maximum likelihood estimation (MLE).
- The idea is to construct a likelihood function from the conditional distribution of each observation given the past.
- We usually use the conditional likelihood, conditioning on the initial observation r_1:

$$L_c(\theta \mid r_1, \ldots, r_n) = \prod_{t=2}^{n} f(r_t \mid r_{t-1}, \ldots, r_1; \theta).$$

- This avoids specifying a marginal distribution for the initial value.

ARCH parameter estimation

- Under Gaussian innovations, the ARCH(1) model implies that

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}\left(0, \alpha_0 + \alpha_1 r_{t-1}^2\right).$$

- Therefore, conditional on r_1, the likelihood is

$$\begin{aligned}L_c(\alpha_0, \alpha_1) &:= L_c(\alpha_0, \alpha_1 \mid r_1, \ldots, r_n) \\ &= \prod_{t=2}^{n} \frac{1}{\sqrt{2\pi(\alpha_0 + \alpha_1 r_{t-1}^2)}} \exp\left\{-\frac{r_t^2}{2(\alpha_0 + \alpha_1 r_{t-1}^2)}\right\}.\end{aligned}$$

- We then maximise this over $\alpha_0 > 0$ and $\alpha_1 \geq 0$.

Does the fitted ARCH model look like the data?

- Assuming Gaussian innovations, we get $\alpha_0 = 6.663 \times 10^{-5}$ and $\alpha_1 = 0.2384$.

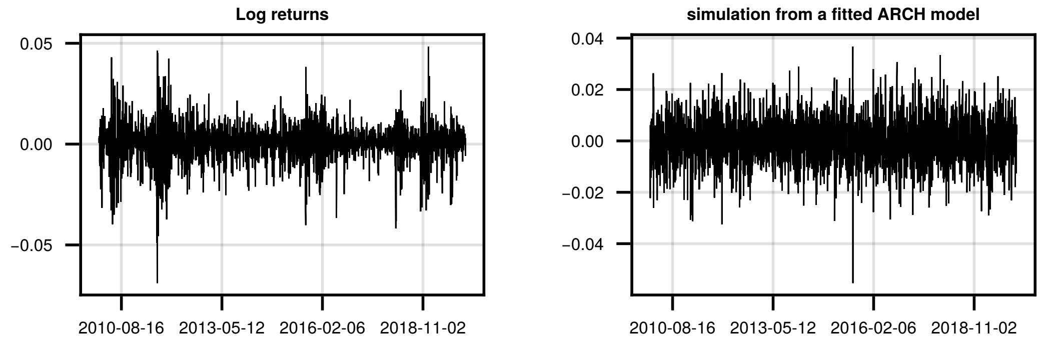


Figure 12.5: Observed log-returns (left) and a simulated ARCH(1) process using the fitted parameters (right).

- This is not ideal, the simulated data does not look like the real data.

What about the ACF and PACF?

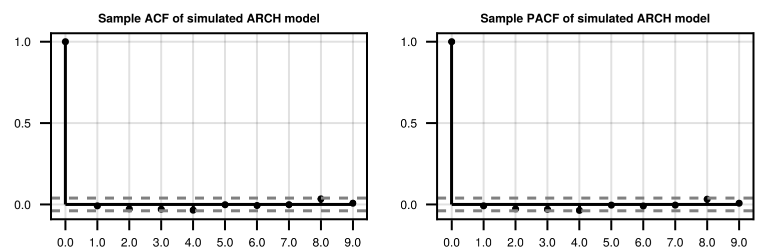


Figure 12.6: Simulated realisation of the ARCH process.

And the square?

Recall the ACF and PACF of the squared log-returns

- They do not look similar to the ACF and PACF of the simulated ARCH(1) process.

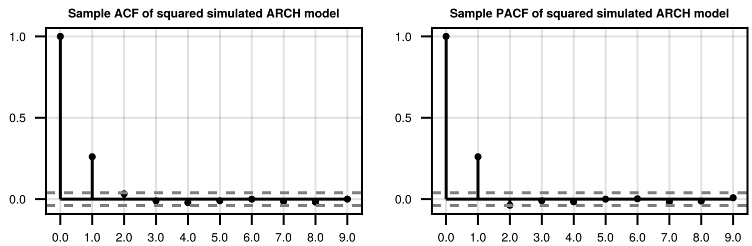


Figure 12.7: Simulated realisation of the ARCH process.

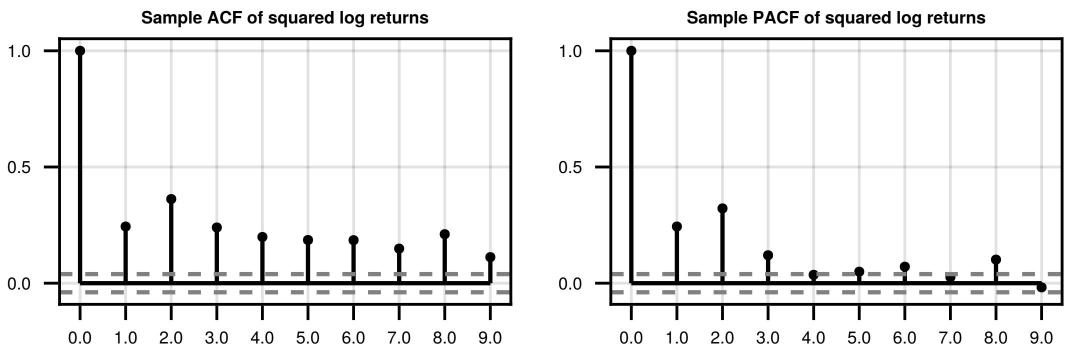


Figure 12.8: acf and pacf of the square of the log-returns from 01/01/2010 to 01/01/2020.

What happened?

- Recall that the ARCH(1) model is

$$\begin{aligned} r_t &= \sigma_t \varepsilon_t, \\ \sigma_t^2 &= \alpha_0 + \alpha_1 r_{t-1}^2. \end{aligned}$$

- A large shock r_{t-1}^2 increases the next conditional variance σ_t^2.
- A larger conditional variance makes another large shock more likely.
- This should create clusters of high volatility.

So what went wrong?

How could we fix this?

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2.$$

- In ARCH(1), the conditional variance only depends on the most recent squared return.
- This means that a single small return can quickly reduce the conditional variance, even after a period of high volatility.
- As a result, it is difficult for the model to sustain long periods of high volatility (”volatility clustering”).

What modification to the model could make volatility more persistent?

Solution 1: ARCH(p)

Definition 12.2 (ARCH(p) models)**.** We will say that a time series $\{r_t\}$ is an ARCH(p) model if it takes the form

$$r_t = \sigma_t \varepsilon_t, \qquad t \in \mathbb{Z}$$

where $\{\varepsilon_t\}$ is a white noise process with mean zero and variance one, and

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \ldots + \alpha_p r_{t-p}^2$$

The parameters satisfy $\alpha_0, \alpha_p > 0$ and $\alpha_j \geq 0$ for $1 \leq j < p$.

- The idea now is that volatility can depend on several past squared shocks.
- As before, one can show that $\mathrm{Var}\left(r_t \mid \mathcal{F}_{t-1}\right) = \sigma_t^2$.
- But, this might require a lot of parameters.

Returning to the ACF and PACF of the squared log returns

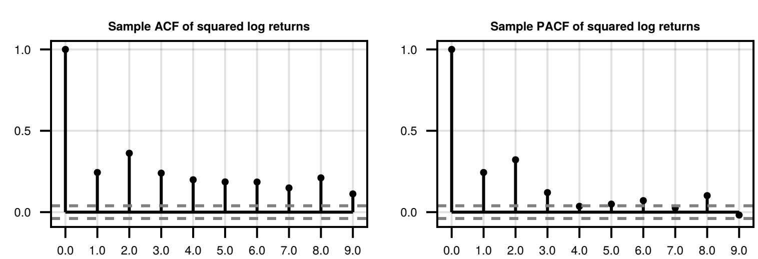


Figure 12.9: acf and pacf of the square of the log-returns from 01/01/2010 to 01/01/2020.

Can we do this more parsimoniously?

Instead of adding many lagged squared returns, is there another quantity we could feed back into the model to achieve persistent volatility with fewer parameters?

Solution 2: GARCH

Definition 12.3 (GARCH(p,q) models)**.** We will say that a time series $\{r_t\}$ is a GARCH(p,q) model if it takes the form

$$r_t = \sigma_t \varepsilon_t, \qquad t \in \mathbb{Z}$$

where $\{\varepsilon_t\}$ is a white noise process with mean zero and variance one, and

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \ldots + \alpha_p r_{t-p}^2 + \beta_1 \sigma_{t-1}^2 + \ldots + \beta_q \sigma_{t-q}^2$$

The parameters satisfy $\alpha_0, \alpha_p, \beta_q > 0$ and $\alpha_j, \beta_j \geq 0$ for $1 \leq j < p$.

- Often low order GARCH models are used, such as GARCH(1,1).
- This is far more parsimonious than the ARCH(p) model.
- The key idea: feed past volatility estimates (previous σ_t^2 values) back into the model.
- This allows volatility to persist over time, even after a sequence of large or small shocks.

Properties of GARCH

- GARCH processes are also mean-zero and white noise (see the exercises)
- GARCH processes enable us to model more dependent volatility
- As before, one can show that $\mathrm{Var}\,(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$
- For a GARCH(1,1), $\alpha + \beta < 1$ implies stationarity
- In financial applications, fitted GARCH(1,1) models often have $\alpha_1 + \beta_1$ close to one.
- This produces highly persistent volatility, although the dependence still decays geometrically.
- For a GARCH(1,1) with Gaussian noise, the fourth moment exists if and only if

$$3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$$

Proposition 12.2. *A GARCH(p,q) has a weakly stationary solution if and only if*

$$\sum_{j=1}^{p} \alpha_j + \sum_{j=1}^{q} \beta_j < 1.$$

In this case, its variance is given by

$$\frac{\alpha_0}{1 - \sum_{j=1}^{p} \alpha_j - \sum_{j=1}^{q} \beta_j}$$

- In general, $\sum_{j=1}^{p} \alpha_j + \sum_{j=1}^{q} \beta_j < 1$ is sufficient for strict stationarity
- For certain classes of noise (including Gaussian) $\sum_{j=1}^{p} \alpha_j + \sum_{j=1}^{q} \beta_j = 1$ also has a strictly stationary solution, just not a weakly stationary one

Let's fit a GARCH(1,1) model

- Assuming Gaussian innovations, we get

$$\alpha_0 = 3.686 \times 10^{-6},\ \alpha_1 = 0.1713,\ \beta_1 = 0.7897$$

- Look at a simulation, this is much better!

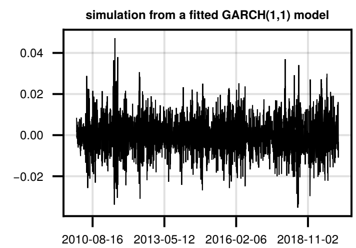


Figure 12.10: Simulated realisation of the GARCH(1,1) process.

Now let's check the ACF and PACF

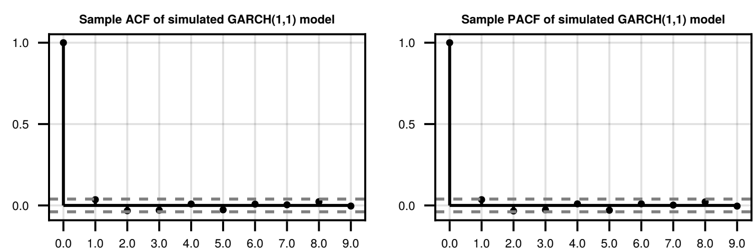


Figure 12.11: Simulated realisation of the GARCH(1,1) process.

And the square?

- This looks much better!
- The ACF and PACF of the squared log-returns are now much more similar to the simulated GARCH(1,1) process.

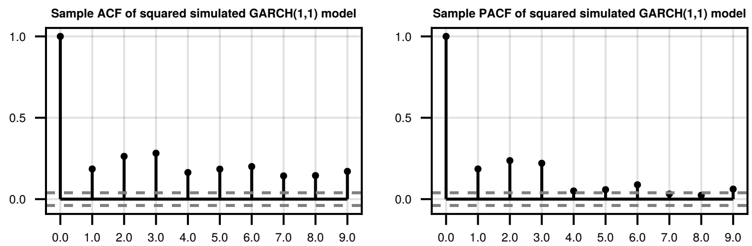


Figure 12.12: Simulated realisation of the GARCH(1,1) process.

12.3 Diagnostics

In sample prediction

- One thing we can do is look at whether the model can predict the variance of the data well.
- Assuming Gaussian innovations, our model at a given time t is

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}\left(0, \sigma_t^2\right).$$

- Therefore, we look at the prediction confidence intervals $\pm 1.96\hat{\sigma}_t$.

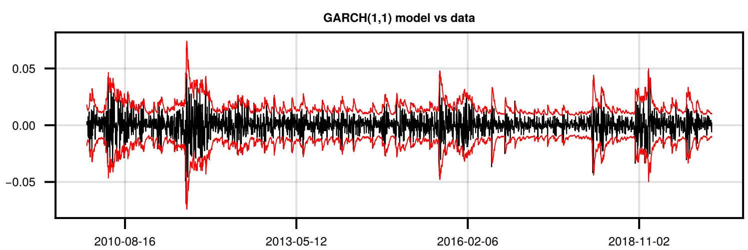


Figure 12.13: The log-returns with confidence intervals from the GARCH process.

Standardised residuals

- We can also look at the standardised residuals

$$\tilde{e}_t = \frac{r_t}{\sigma_t}.$$

ACF and PACF of the residuals

- We can see that the residuals are probably white noise.
- This is a good sign!

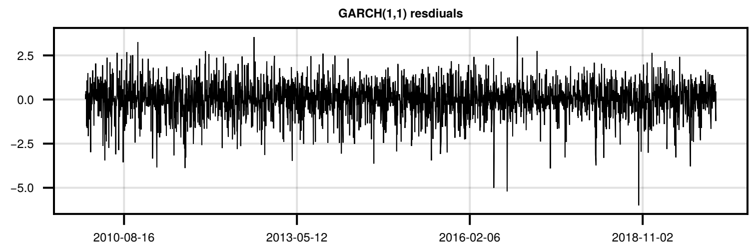


Figure 12.14: Residuals of the log-returns.

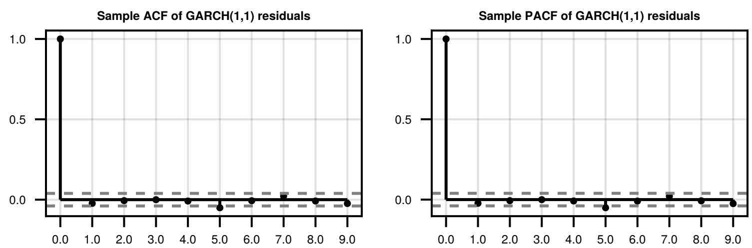


Figure 12.15: ACF and PACF of the residuals.

ACF and PACF of the squared residuals

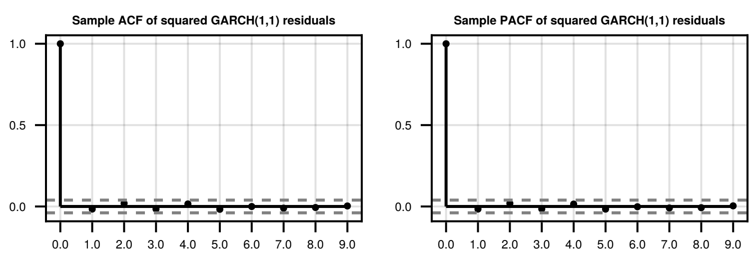


Figure 12.16: ACF and PACF of the squared residuals.

- We can see that the squared residuals are now behaving like white noise!

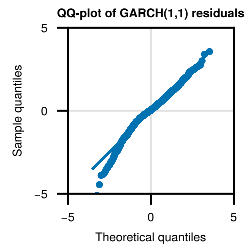


Figure 12.17: QQ plot of the residuals.

Residual distribution

Combining GARCH and ARMA

- Since GARCH is a white noise process, we can simply make

$$X_t = \sum_{j=1}^{p} \phi_j X_{t-j} - \sum_{k=0}^{q} \theta_k r_{t-k}$$

  where r_t is a GARCH$(\tilde{p}, \tilde{q})$ process and the θs and ϕs satisfy the usual conditions for an ARMA(p, q).

- We just need to be careful with the likelihood, since we cannot assume iid Gaussian errors any more!

Summary

- ARCH and GARCH models can capture changes over time and volatility "clusters".
- We can model "outlier" behaviour and large shocks due to its large kurtosis.
- The GARCH model does not affect the correlation structure.
- GARCH allows us to model more persistent volatility in a parsimonious way.

But:

- Positive and negative shocks affect the system in the same way.
- The dependence in the volatility still decays geometrically, which may not be realistic for financial data.
- Extensions to the GARCH model can address these issues, but we will not cover them in this course.

Chapter 13

Linear time-invariant filters

13.1 Linear filters

Complex-valued time series

In this lecture, we sometimes allow time series to take values in $\mathbb{C}$. A complex-valued time series can be written as

$$X_t = U_t + iV_t,$$

where $\{U_t\}$ and $\{V_t\}$ are real-valued time series. Its mean is

$$\mathbb{E}[X_t] = \mathbb{E}[U_t] + i\mathbb{E}[V_t].$$

For complex-valued time series, we define the autocovariance function by

$$\gamma_\tau^{(X)} = \mathrm{Cov}(X_{t+\tau}, X_t) = \mathbb{E}[(X_{t+\tau} - \mu)(X_t - \mu)^*].$$

This convention reduces to the usual autocovariance when X_t is real-valued.

Digital filters

A digital filter maps some input sequence to another output sequence. In this lecture, these sequences will have the index set $\mathbb{Z}$. Other filters exist, but in this course, filter will mean digital filter.

- If a filter is given by L, then the filter of a sequence $\{X_t\}$ will be another sequence, say $\{Y_t\}$, and we write

$$\{Y_t\} = L[\{X_t\}]. \tag{13.1}$$

- When we refer to an element of the output of a filter, we will write a subscript after the bracket, e.g.

$$Y_u = L[\{X_t\}]_u. \tag{13.2}$$

To be clear, the t in $\{X_t\}$ is a dummy variable, whilst $u \in \mathbb{Z}$ has meaning.

For stationarity, we require the corresponding real and imaginary parts to be jointly second-order stationary.
We have dropped the index set from the sequences for notational convenience, i.e. in this lecture $\{X_t\} = \{X_t\}_{t\in\mathbb{Z}}$.

Linear time-invariant filters

Recall the backshift operator, denoted B. It has the property that

$$\mathrm{B}\left[\{X_t\}\right]_u = X_{u-1} \tag{13.3}$$

for any $u \in \mathbb{Z}$.

Definition 13.1. A digital filter L is called a linear time-invariant (LTI) filter if for any sequences $\{X_t\}$, $\{Y_t\}$ and any $\alpha \in \mathbb{C}$ we have

1. Linearity:

$$L\left[\{\alpha X_t + Y_t\}\right] = \alpha L\left[\{X_t\}\right] + L\left[\{Y_t\}\right]. \tag{13.4}$$

2. Time invariance:

$$L\left[\mathrm{B}\left[\{X_t\}\right]\right] = \mathrm{B}\left[L\left[\{X_t\}\right]\right]. \tag{13.5}$$

Some examples

Example 13.1. The backshift operator is clearly a linear time-invariant filter.

Example 13.2. Consider a filter L such that for any $u \in \mathbb{Z}$

$$L\left[\{X_t\}\right]_u = X_u + \phi X_{u-1},$$

where $\phi \in \mathbb{R}$. This is linear because for any $u \in \mathbb{Z}$

$$\begin{aligned} L\left[\{\alpha X_t + Y_t\}\right]_u &= \alpha(X_u + \phi X_{u-1}) + Y_u + \phi Y_{u-1} \\ &= \alpha L\left[\{X_t\}\right]_u + L\left[\{Y_t\}\right]_u. \end{aligned}$$

Time invariance is left to the interested reader, but is easily verified.

Arbitrary time shifts

Say we want to shift by some arbitrary $u \in \mathbb{Z}$, then we need only apply B u times, i.e. for any $s \in \mathbb{Z}$

$$\mathrm{B}^u\left[\{X_t\}\right]_s = X_{s-u}. \tag{13.6}$$

(If $u < 0$, this means applying the inverse operator, B^{-1}, $-u$ times).

If L is an LTI filter, then for any $u \in \mathbb{Z}$,

$$L\left[\mathrm{B}^u\{X_t\}\right] = \mathrm{B}^u L\left[\{X_t\}\right]. \tag{13.7}$$

Often, (13.7) is stated as the condition for time-invariance (as this parallels the continuous-time version), however, one only needs to show the weaker condition given in (13.5).

Linear combinations of linear filters

Proposition 13.1. *Consider two LTI filters L_1 and L_2, and let $\alpha \in \mathbb{C}$ then the filter*

$$L = \alpha L_1 + L_2 \tag{13.8}$$

is also an LTI filter.

Proposition 13.2. *Consider two LTI filters L_1 and L_2. The filter $L = L_1 L_2$, i.e. so that*

$$L\left[\{X_t\}\right] = L_1\left[L_2\left[\{X_t\}\right]\right] \tag{13.9}$$

is also an LTI filter. (This is called a cascaded filter.)

- See the exercises for proofs of these two results.

13.2 Impulse response and transfer function

Properties of a filter

In order to explore the properties of a given LTI filter, we will consider the effect of applying the filter to some test sequences. The sequences in question will be:

1. the impulse sequence,
2. a complex wave.

Formal definitions will be given on the following slides, however, informally they tell us about:

1. the effect of the filter if we input a single shock to the system at a given time,
2. the effect the filter has on a signal made up of one specific frequency.

13.2.1 Impulse response

The impulse response sequence

For a given $m \in \mathbb{Z}$, define the impulse sequence $\{\delta_{t,m}\}$ so that for $t \in \mathbb{Z}$

$$\delta_{t,m} = \begin{cases} 1 & \text{if } m = t, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 13.2. For some LTI filter L, let the impulse response sequence be $\{h_m\}$ such that for any $m \in \mathbb{Z}$

$$h_m = L\left[\{\delta_{t,-m}\}\right]_0 . \tag{13.10}$$

- Note: the zero here is the time index.

A note on convergence

For finite linear filters, such as

$$L\left[\{X_t\}\right]_u = X_u + \phi X_{u-1},$$

there are no convergence issues.

For infinite filters, such as

$$L\left[\{X_t\}\right]_u = \sum_{m\in\mathbb{Z}} h_{u-m} X_m,$$

we need the sum to be well-defined.

In this lecture, we will assume the required sums converge. A common sufficient condition is

$$h \in \ell^1.$$

Mean-square equality

For random variables $U, V \in L^2$, we write

$$U \overset{ms}{=} V$$

to mean that U and V are equal in mean square, i.e.

$$\mathbb{E}\left[|U - V|^2\right] = 0.$$

This is the natural notion of equality for the stochastic infinite sums used below.

Linear filters as a convolution

Theorem 13.3. *Under suitable convergence assumptions, an LTI filter with impulse response h can be written as*

$$L\left[\{X_t\}\right]_u \overset{ms}{=} \sum_{m\in\mathbb{Z}} h_{u-m} X_m. \tag{13.11}$$

Conversely, any filter defined by such a convolution is LTI.

- The impulse response tells us the weights in the convolution.
- For infinite filters, the convergence assumptions are technical; throughout this lecture we assume the required sums are well-defined.
- In the exercise session, we will prove the deterministic version: assuming $\{X_t\} \in \ell^1$ and replacing $\overset{ms}{=}$ by $=$.

13.2.2 Transfer function

The transfer function

Recall, the second kind of test sequence of interest is a complex wave.
For $f \in \mathbb{R}$, let $\{\xi_{t,f}\}$ be such that for all $t \in \mathbb{Z}$,

$$\xi_{t,f} = e^{2\pi i f t}.$$

In this section, we assume that the filter can be applied to these complex waves. For convolution filters with $h \in \ell^1$, this is guaranteed because the relevant sums converge absolutely.

Definition 13.3. The transfer function of an LTI filter L is

$$H(f) = L\left[\{\xi_{t,f}\}\right]_0, \qquad f \in \mathbb{R}. \tag{13.12}$$

- The transfer function is useful for understanding the frequency-domain properties of a linear filter.

Theorem 13.4. *Consider an LTI filter L that can be applied to the complex waves $\{\xi_{t,f}\}$. These waves are eigensequences of L, and $H(f)$ gives the corresponding eigenvalue. In other words,*

$$L\left[\{\xi_{t,f}\}\right] = H(f)\{\xi_{t,f}\} \tag{13.13}$$

for any $f \in \mathbb{R}$.

- So we see that applying a linear filter to a complex wave just modifies its phase and amplitude, whilst retaining the same frequency.
- The shape of $H(f)$ determines how this weighting occurs.

Proof of Theorem 13.4. For any $u \in \mathbb{Z}$ and $f \in \mathbb{R}$, begin by noting that for any $\tau \in \mathbb{Z}$

$$\mathrm{B}^{-u}\left[\{\xi_{t,f}\}\right]_\tau = \xi_{\tau+u,f} = e^{2\pi i(\tau+u)f} = e^{2\pi i u f}\xi_{\tau,f},$$

and so

$$\mathrm{B}^{-u}\left[\{\xi_{t,f}\}\right] = e^{2\pi i u f}\{\xi_{t,f}\}. \tag{13.14}$$

Therefore

$$\begin{aligned}
L\left[\{\xi_{t,f}\}\right]_u &= \mathrm{B}^u\left[L\left[\mathrm{B}^{-u}\left[\{\xi_{t,f}\}\right]\right]\right]_u && \text{(time invariance)}\\
&= L\left[e^{2\pi i f u}\{\xi_{t,f}\}\right]_0 && \text{(by (13.14))}\\
&= e^{2\pi i f u} L\left[\{\xi_{t,f}\}\right]_0 && \text{(linearity)}\\
&= \xi_{u,f} H(f). && \text{(by definition)}
\end{aligned}$$

Therefore since this holds for any $u \in \mathbb{Z}$, the result holds. $\square$

Why did we do this?

Informally, there are two useful ways to understand a time series:

$$X_t = \sum_{m\in\mathbb{Z}} X_m \delta_{m,t}, \qquad \text{(as a sum of impulses)}$$

$$X_t \overset{ms}{=} \mu + \int_{-1/2}^{1/2} e^{2\pi i f t} \mathrm{d}Z(f), \qquad \text{(as a sum of waves)}$$

Under the convergence assumptions above, an LTI filter acts simply on both building blocks:

$$L\left[\{X_t\}\right]_u \overset{ms}{=} \sum_{m\in\mathbb{Z}} X_m L\left[\{\delta_{m,t}\}\right]_u,$$

$$L\left[\{X_t\}\right]_u \overset{ms}{=} \mu H(0) + \int_{-1/2}^{1/2} H(f) e^{2\pi i f u} \mathrm{d}Z(f).$$

The first line explains the impulse response; the second explains the transfer function.

Relating the impulse response and transfer function

Theorem 13.5. *Assume the impulse response satisfies $h \in \ell^1$. Then the transfer function is the Fourier transform of the impulse response: for $f \in \mathbb{R}$,*

$$H(f) = \sum_{m\in\mathbb{Z}} h_m e^{-2\pi i f m}. \tag{13.15}$$

- This provides a convenient way to compute the transfer function.
- In the exercises, you will prove this using the convolution representation.

13.3 The spectral density function and linear filters

Spectra of the output process

In what follows, it will be convenient to augment our usual notation. For the stationary discrete-time time series $\{X_t\}_{t\in\mathbb{Z}}$, we will write

- μ_X for the mean,
- $\gamma_\tau^{(X)}$ for the autocovariance,
- $S_X(f)$ for the spectral density function.

Theorem 13.6. *Consider a stationary time series $\{X_t\}_{t\in\mathbb{Z}}$. If L is an LTI filter with impulse response $h \in \ell^1$, and*

$$\{Y_t\} = L\left[\{X_t\}\right]$$

then $\{Y_t\}$ is a stationary process.

Proof. From Theorem 13.3, for each $t \in \mathbb{Z}$

$$Y_t \overset{ms}{=} \sum_{m \in \mathbb{Z}} h_m X_{t-m}.$$

Hence Y_t has the same mean and covariance as the right-hand side. Thus for any $t \in \mathbb{Z}$, by Fubini's theorem

$$\mathbb{E}[Y_t] = \sum_{m \in \mathbb{Z}} h_m \mathbb{E}[X_{t-m}] = \mu_X \sum_{m \in \mathbb{Z}} h_m$$

which does not depend on t.

Now for $t, \tau \in \mathbb{Z}$ we have

$$\begin{aligned}
\gamma_\tau^{(Y)} &= \operatorname{Cov}(Y_{t+\tau}, Y_t) \\
&= \sum_{m \in \mathbb{Z}} \sum_{s \in \mathbb{Z}} h_m h_s^* \operatorname{Cov}(X_{t+\tau-m}, X_{t-s}) && \text{(Fubini)} \\
&= \sum_{m \in \mathbb{Z}} \sum_{s \in \mathbb{Z}} h_m h_s^* \gamma_{\tau-m+s}^{(X)} && \text{(stationarity)}
\end{aligned}$$

which does not depend on t. Finally by stationarity and $h \in \ell^1$

$$\gamma_0^{(Y)} \leq \gamma_0^{(X)} \|h\|_1^2 < \infty.$$

□

Theorem 13.7. *Consider a stationary time series* $\{X_t\}_{t \in \mathbb{Z}}$*, with* $\gamma^{(X)} \in \ell^1$*. If* L *is an LTI filter with impulse response* $h \in \ell^1$*, and* $\{Y_t\} = L[\{X_t\}]$ *then*

$$S_Y(f) = |H(f)|^2 S_X(f) \tag{13.16}$$

where H *is the transfer function of* L*.*

- From this result, we can conveniently compute the spectral density function of different processes.

Proof. Firstly from the proof of Theorem 13.6

$$\begin{aligned}
\gamma_\tau^{(Y)} &= \sum_{m \in \mathbb{Z}} \sum_{s \in \mathbb{Z}} h_m h_s^* \gamma_{\tau-m+s}^{(X)} \\
&= \sum_{s \in \mathbb{Z}} \sum_{u \in \mathbb{Z}} h_{u+s} h_s^* \gamma_{\tau-u}^{(X)} && (u = m - s) \\
&= \sum_{u \in \mathbb{Z}} \sum_{s \in \mathbb{Z}} h_{u+s} h_s^* \gamma_{\tau-u}^{(X)} && \text{(Fubini)} \\
&= \sum_{u \in \mathbb{Z}} w_u \gamma_{\tau-u}^{(X)}
\end{aligned}$$

where

$$w_u = \sum_{s\in\mathbb{Z}} h_{u+s} h_s^*.$$

Since $h \in \ell^1$, we also have $w \in \ell^1$.

Therefore, since $w \in \ell^1$ and $\gamma^{(X)} \in \ell^1$, we can apply the convolution theorem, obtaining for $f \in \mathbb{R}$

$$S_Y(f) = W(f) S_X(f).$$

All that remains is to find W, the Fourier transform of w. Letting $\tilde{h}_t = h^*_{-t}$ for all $t \in \mathbb{Z}$, we see

$$\begin{aligned} w_u &= \sum_{s\in\mathbb{Z}} h_{u+s} h_s^* \\ &= \sum_{m\in\mathbb{Z}} h_m \tilde{h}_{u-m} && \text{(setting } m = u+s\text{)} \end{aligned}$$

so $w = h * \tilde{h}$ and thus applying the convolution theorem again

$$W(f) = H(f)\tilde{H}(f) = |H(f)|^2,$$

because $\tilde{H}(f) = H(f)^*$. This gives the required result. □

Lemma 13.8. *The filter given by $\Phi(\mathrm{B})$ for some p^{th} order polynomial Φ is an LTI filter with transfer function*

$$H(f) = \Phi(e^{-2\pi i f})$$

for $f \in \mathbb{R}$.

Proof. A linear combination of LTI filters is an LTI filter. The impulse response is

$$h_m = \begin{cases} \phi_m & \text{if } 0 \le m \le p, \\ 0 & \text{otherwise.} \end{cases}$$

So its transfer function is

$$H(f) = \sum_{m\in\mathbb{Z}} h_m e^{-2\pi i f m} = \sum_{j=0}^{p} \phi_j e^{-2\pi i f j} = \Phi(e^{-2\pi i f}).$$

□

Theorem 13.9. *Consider a stationary ARMA(p, q) process*

$$\Phi(\mathrm{B}) X_t = \Theta(\mathrm{B})\epsilon_t,$$

the spectral density function of X is given by

$$\begin{aligned} S_X(f) &= \sigma^2 \frac{\left|\Theta\left(e^{-2\pi i f}\right)\right|^2}{\left|\Phi\left(e^{-2\pi i f}\right)\right|^2} \\ &= \sigma^2 \left| \frac{\sum_{j=0}^{q} \theta_j e^{-2\pi i j f}}{\sum_{j=0}^{p} \phi_j e^{-2\pi i j f}} \right|^2 \end{aligned}$$

- We do not have a nice closed form for the autocovariance function of an ARMA process.
- But, we do have a nice form for the spectral density function.

Proof. If we have a stationary ARMA process, then we can write

$$\Phi(\mathrm{B})\left[\{X_t\}\right] = \Theta(\mathrm{B})\left[\{\epsilon_t\}\right].$$

Therefore we have from Lemma 13.8

$$\left|\Phi(e^{-2\pi i f})\right|^2 S_X\left(f\right) = \left|\Theta(e^{-2\pi i f})\right|^2 S_\epsilon\left(f\right)$$

and so

$$S_X\left(f\right) = \sigma^2 \frac{\left|\Theta\left(e^{-2\pi i f}\right)\right|^2}{\left|\Phi\left(e^{-2\pi i f}\right)\right|^2}$$

because the stationarity condition implies that $\Phi(z)$ has no zeros on the unit circle, so $\Phi(e^{-2\pi i f}) \neq 0$. $\square$

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